Laboratory Manual

Electronic System Design

EEE383

Department of Electrical and Electronic Engineering
BSRM School of Engineering
BRAC University



Revision: August 2026

A. Course Objectives:

The objective of this course are to:

I. Provide students with the basics of single and double layered Printed Circuit Board (PCB) design and efficient component selection and bill of material preparation
II. Enable students to integrate different electrical and electronic subsystems to form a full-fledged electronic system
III. Introduce basic performance requirements of some common electronic subsystems and provide hands-on experience in their design.

B. Course Outcomes, CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
EEE383_Electronic System Design					
CO1	**Design** and **Implement** fully functional electronic system by integrating different sub-systems	c	Cognitive/ Create	Lab Class	Project review
CO2	**Demonstrate** engineering project management and economic decision-making skills in electronic system design project	k	Cognitive/ Apply	Lab Class	Project review
CO3	**Perform** effectively as an individual as well as a member of a team to develop electronic system design project	i	Affective/ Valuing	Lab Class	Project Review, Peer-evaluation

C. Mark Distribution

Assessment Tools	Weightage (%)
Attendance & Lab Performance	10%
Assignment	10%
Quiz	10%
Final Assessment	20%
Project	40%
Team Work	10%

Prepared By:

Version	Experiments	Prepared By & Designation
Version 01	Exp. 2, 3, 4, 5	**Atib Mohammad Oni** Lecturer, Department of EEE, BRAC University
Version 02	Exp. 0, 1, 2, 6, 7	**Saad Mahmud Chowdhury** Lecturer, Department of EEE, BRAC University **Abu Sayed Chowdhury** Lecturer, Department of EEE, BRAC University

Acknowledgement:

Version	Contributor & Affiliation	Contribution
Version 01	**Fasiul Abedin Khan** Student Tutor, Fall 2024, EEE, BRAC University	Drafting (Version 01)
Version 01	**Tajrean Tamas** Student Tutor, Fall 2025, EEE, BRAC University	Contribution to Version 01
Version 02	**MD Shazzad Ahammed Shuvo** Student Tutor, Summer 2026, EEE, BRAC University	Contribution to Version 02

Lab Safety and Security Issues

Laboratory Safety Rules (General Guidelines):

The Department of EEE maintains general safety rules for laboratories. The guideline is attached in front of the door in each of the laboratories. The written rules are as follows.

a. Closed shoes must be worn that will provide full coverage of the feet and appropriate personnel clothing must be worn.
b. Always check if the power switch is off before plugging into the outlet. Also, turn the instrument or equipment OFF before unplugging from the outlet.
c. Before supplying power to the circuit, the connections and layouts must be checked by the teacher.
d. Voltage equal to or above 50V is always dangerous. Therefore, extra precautions must be taken when the voltage level is increased.
e. Extension cords should be used only when necessary and only on a temporary basis.
f. Once the lab exercise is done, all equipment must be powered down and all probes, cords, and other instruments must be returned to their proper position.
g. In case of fire, disconnect the electrical main power source if possible.
h. Students must be familiar with the locations and operations of safety and emergency equipment like Emergency power off, Fire alarm switches, and so on.
i. Eating, drinking, and chewing gum inside electrical laboratories are strictly prohibited.
j. Do not use damaged cords or cords that become too hot or cords with exposed wiring and if something like that is found, inform the teacher/LTO right away.
k. No laboratory equipment can be removed from its fixed places without the teacher/LTO's authorization.
l. No lab work must be performed without the laboratory teacher/lab technical officer being present.

2. Electrical Safety:

To prevent electrical hazards, there are symbols in front of the Electrical Distribution Board, High voltage three-phase lines in the lab, Backup generator, and substation. Symbols related to Arc Flash and Shock Hazard, Danger: High Voltage, Authorized personnel Only, no smoking, etc. are posted in required places. Only authorized personnel are allowed to open the distribution boxes.

3. Electrical Fire:

If an electrical fire occurs, try to disconnect the electrical power source, if possible. If the fire is small, you are not in immediate danger, use any type of fire extinguisher except water to extinguish the fire. When in doubt, push the Emergency Power Off button.

4. IMPORTANT:

Do not use water on an electrical fire.

Laboratory Plan

Exp. NO.	Activity / Experiment name	Page	Tentative Schedule
0	Ice-breaking session: Introducing necessary software, discussion on course outcomes & project, assessment plan, group formation, Altium Setup, etc.	5–13	1st Week
1	Introduction to PCB fundamentals, Project Initialization	14–25	2nd–3rd Week
2	Introduction to Altium Designer Environment, Submission of Project Proposal & BOM (Bill of Materials)	26–29	4th Week
3	Component Selection, Sourcing, Downloading and Integration from SnapEDA and External Libraries	30–38	5th Week
4	Schematic Design and Component Integration Using Altium Designer	39–46	6th Week
MID Week			**7th Week**
5	Board Shape Configuration, Component Placement and Routing Optimization for PCB Layout	47–57	8th Week
6	Implementation of Copper Planes, Design Finalization and Manufacturing Output Generation	58–63	9th Week
7	Custom Library Making	64–78	10th Week
8	Soldering Basics	79–84	11th Week
9	Final Assessment & Project Presentation	–	12th Week

BRAC University
Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-0: Introduction to Electronic System Design and Altium Setup

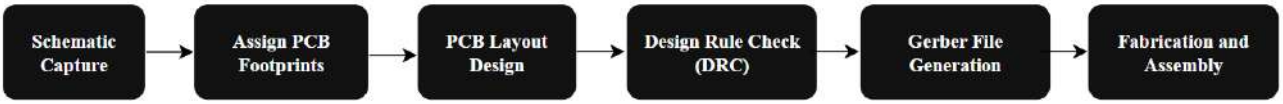
1. **Objective:**
   - Understand the complete Electronic System Design (ESD) workflow from concept to fabrication.
   - Navigate the Altium Designer interface.
   - Create and save a properly structured Altium project workspace.

2. **Introduction:**

Electronic System Design (ESD) is a multi-stage engineering discipline that transforms a functional concept into a manufacturable electronic product. It is a complete arrangement of electronic blocks that work together to perform a specific function. The typical design flow consists of: (1) Requirement analysis (2) Circuit design (3) Schematic capture (4) Simulation & verification (5) PCB layout (6) Design Rule Check (7) Gerber file generation and (8) PCB fabrication.

Altium Designer is an industry-standard EDA (Electronic Design Automation) tool widely used in academia and industry. It integrates schematic editing, PCB layout, component libraries, simulation and manufacturing output generation in a single environment. Key workspace elements include the Projects panel, Inspector, Properties panel and the schematic/PCB editors.

Designing a PCB is an engineering process, not just drawing lines. The flow below ensures your circuit idea turns into a manufacturable and functional product:



A typical Altium project (.PrjPcb) contains schematic sheets (.SchDoc), PCB documents (.PcbDoc), component library links and output job configurations. Proper project organization from the start prevents errors during hand-off to manufacturing.

3. **Software and Device requirement:**

Altium Designer

Minimum PC specifications:

Windows/Mac: Microsoft® Windows® 7 Professional, Enterprise, Ultimate or Home Premium (64-bit); Windows 8 (64-bit) (All Service Packs); Windows 10 (64-bit); Windows 2008 R2 Server; Windows 2012 *Server* (All Service Packs).
Ram: 2 GB, *Processor*: Intel® Pentium® 4 or AMD Athlon XP 2000 with multi-core CPU
Display resolutions: 1,024 x 768 display resolution with true color (16-bit color)

4. Altium Designer Setup:

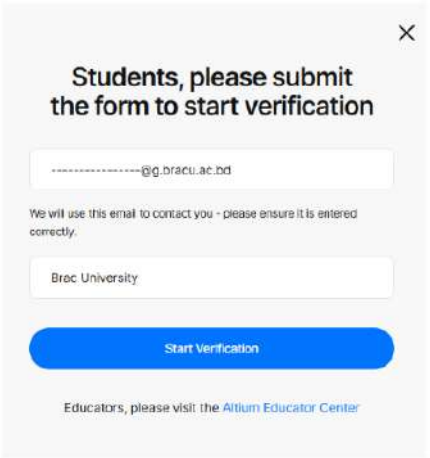
- Go to the official Altium Education portal: https://www.altium.com/education/



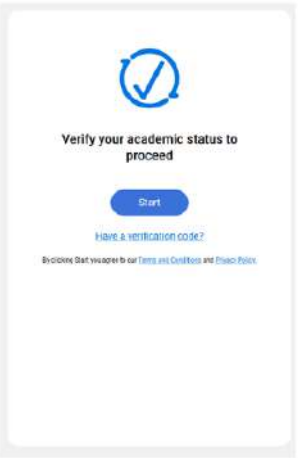
Step-01: Click “See Student Programs” button



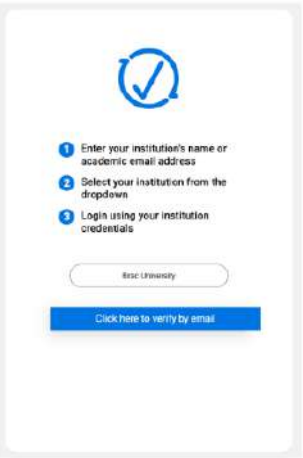
Step-02: Click “Enroll for free” button



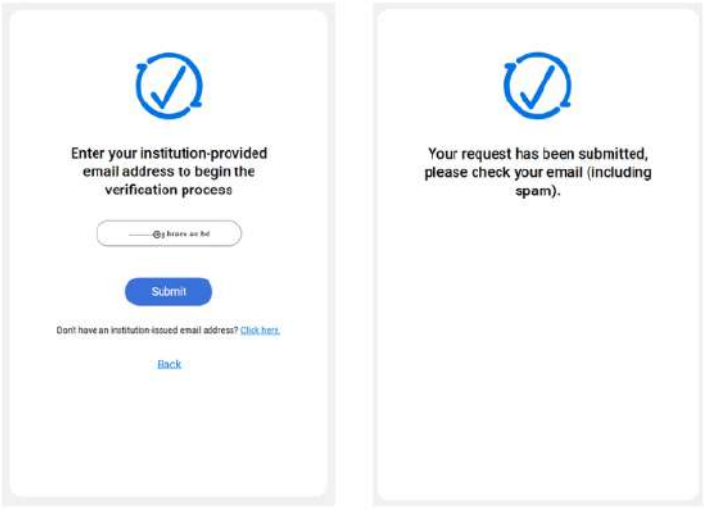
Step-03: Fill up the form



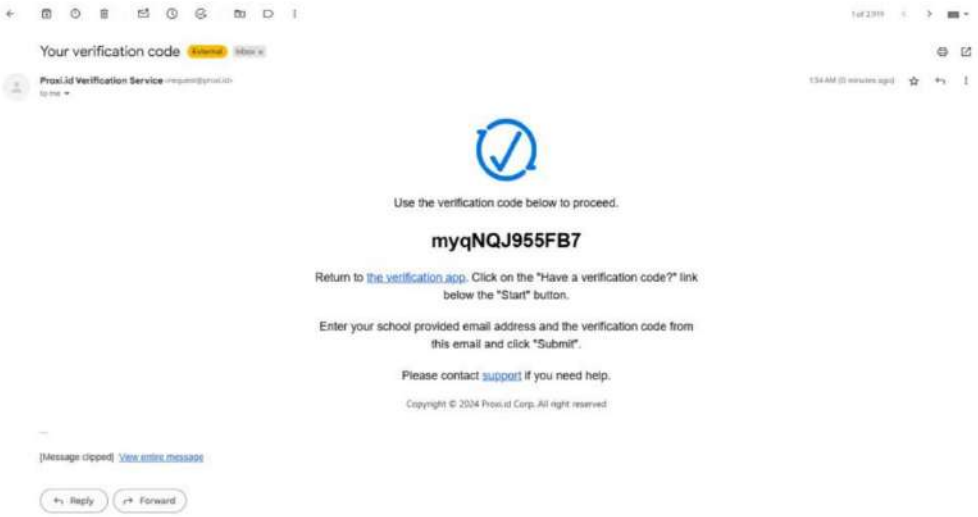
Step-04: Click *Start*



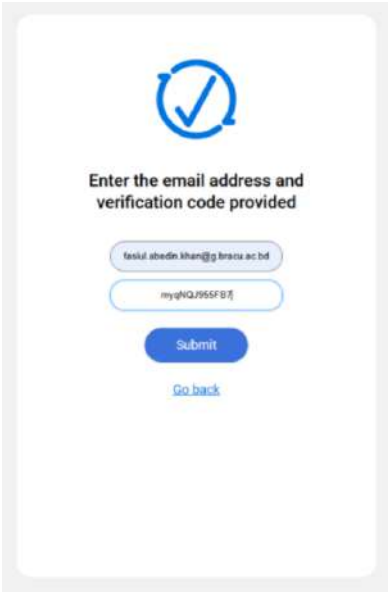
Step-05: Click *Verify*



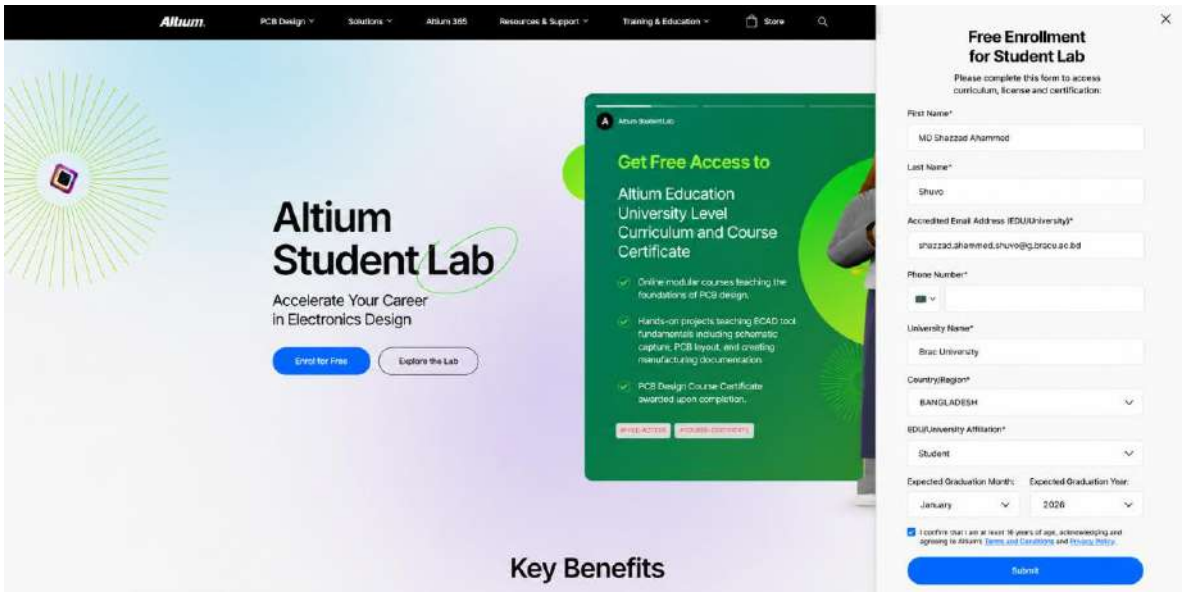
Step-06: Fill up the form & submit



Step-07: You will get a verification code

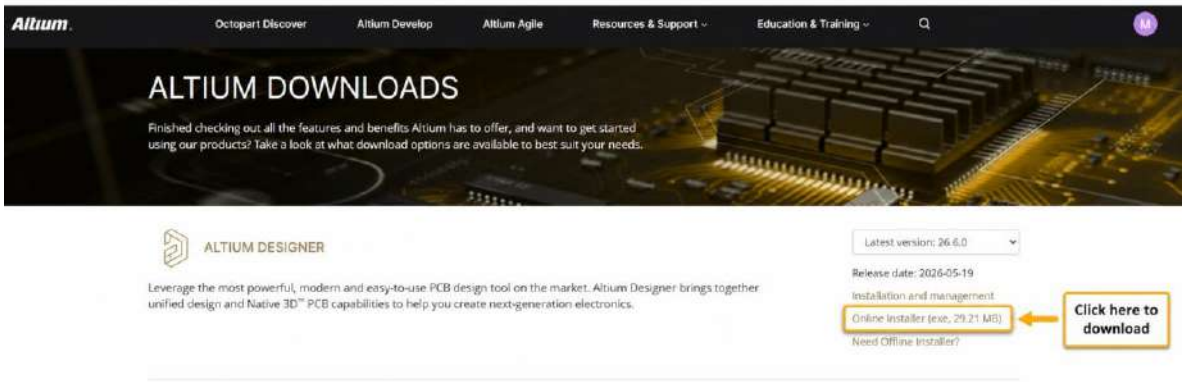


Step-08: Verify yourself using your gsuite ID and verification code

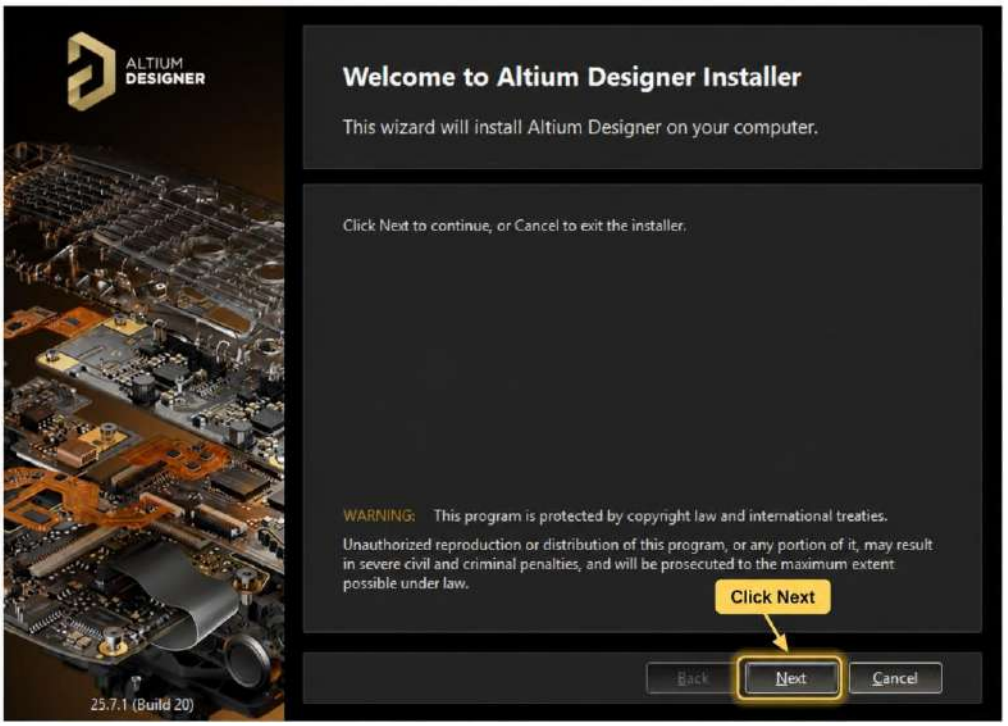


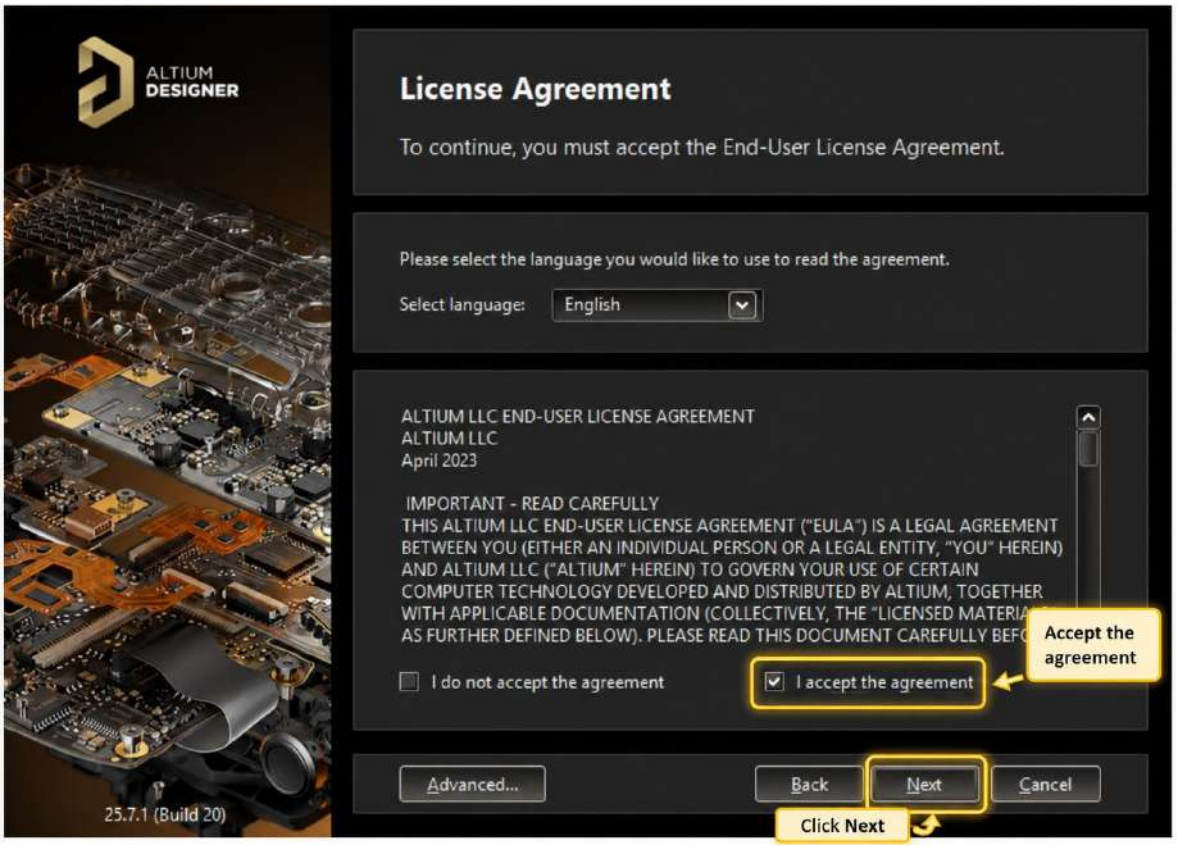
Step-09: Fill up the form and hit *Submit*. You will get a confirmation mail from Altium.

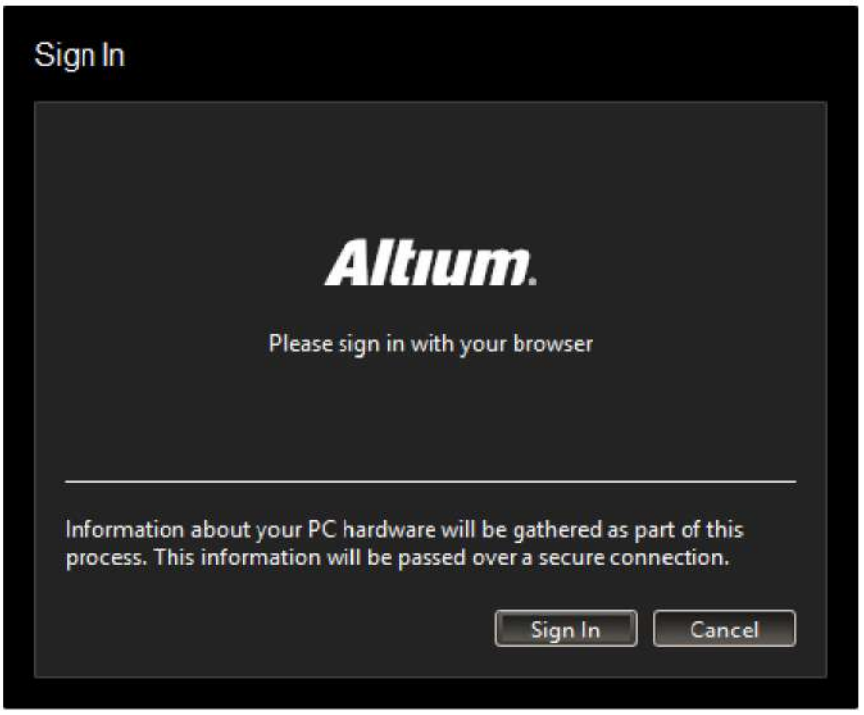
- Log into your Altium account at: https://www.altium.com/
- Go to the link: https://www.altium.com/products/downloads
- Click *Online Installer(.exe)* under the most recent version (26.6.0 or latest).

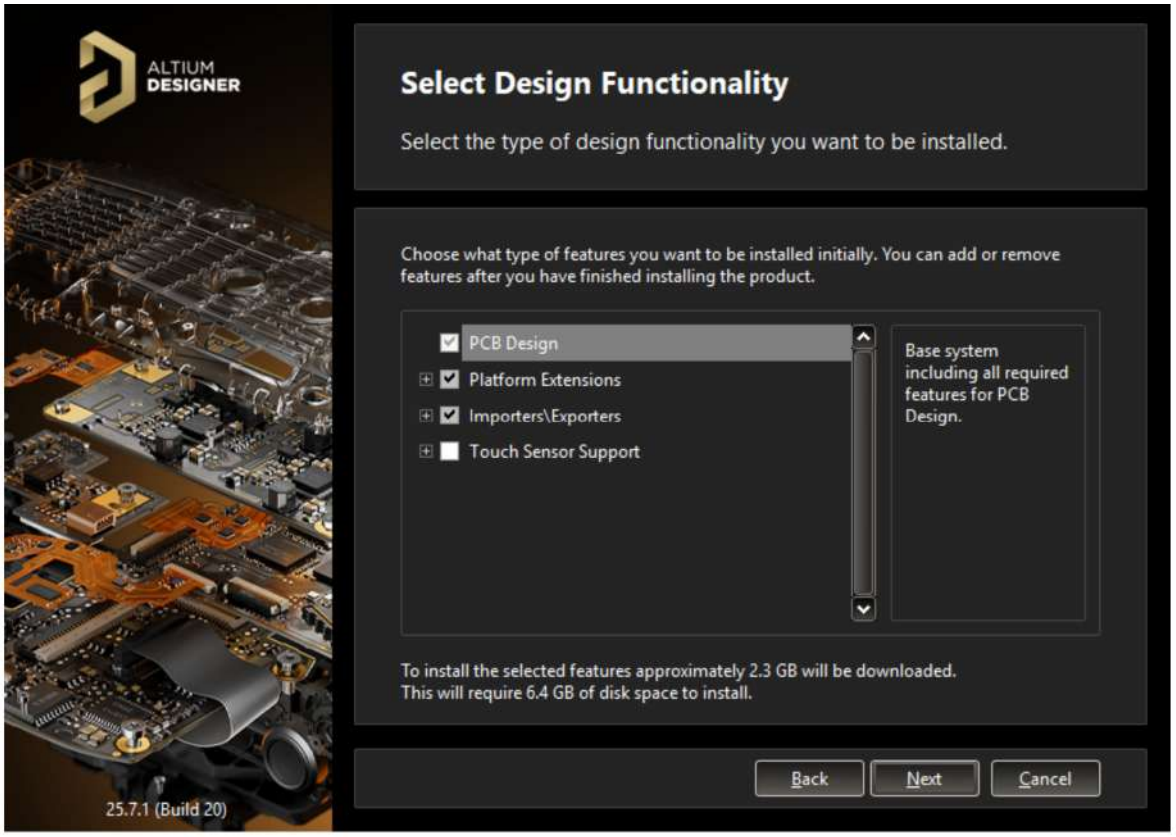


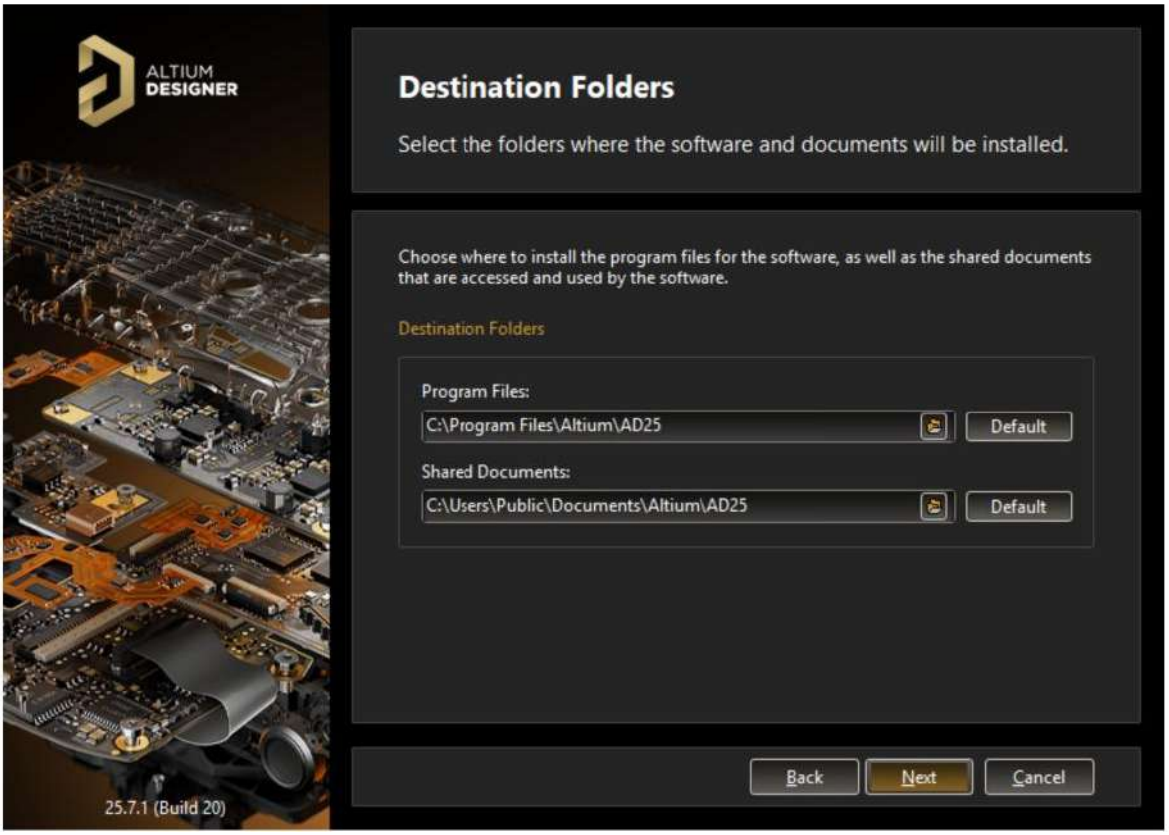
- Run the downloaded *.exe installer file*.
- Choose the *Installation Directory* (default is fine for most users).
- Select the design tools and extensions you want to install:
    - Schematic Capture
    - PCB Editor
    - Draftsman
    - Simulation Tools
    - Vault / Altium 365 (optional for cloud sync)
- Click *Next* and let the installer complete the process (this may take 10–15 minutes).
- When prompted, sign in with your student Altium account credentials.

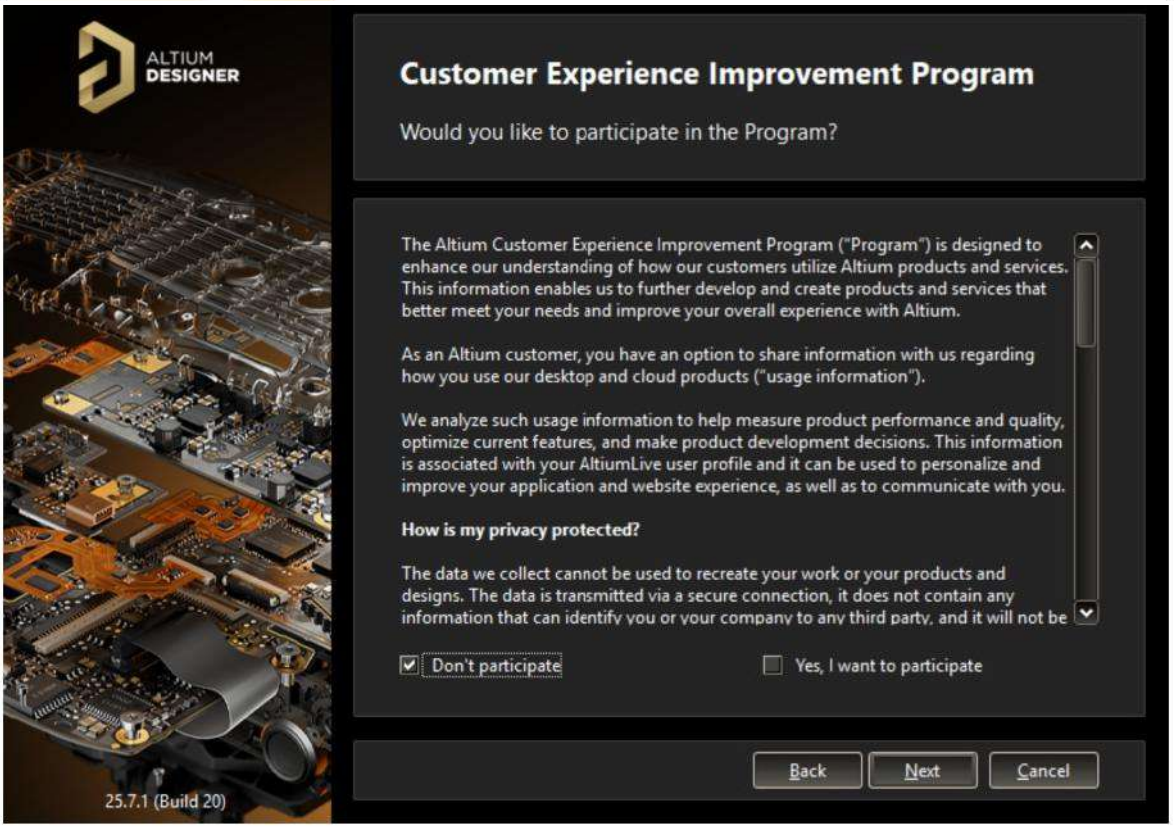

ALTIUM DESIGNER
Welcome to Altium Designer Installer
This wizard will install Altium Designer on your computer.
Click Next to continue, or Cancel to exit the installer.
WARNING: This program is protected by copyright law and international treatie
Unauthorized reproduction or distribution of this program, or any portion of it, may result in severe civil and criminal penalties, and will be prosecuted to the maximum extent possible under law.
Click Next
Back
Next
Cancel
25.7.1 (Build 20)

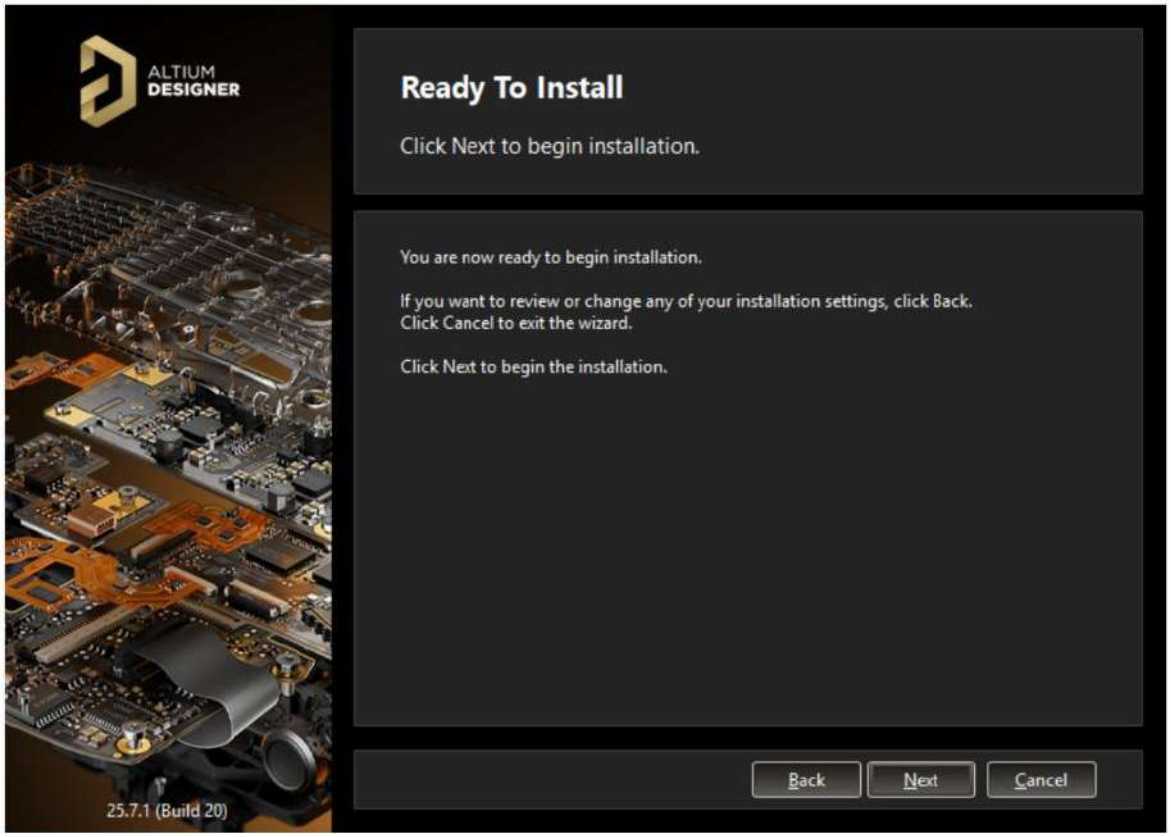

ALTIUM DESIGNER
License Agreement
To continue, you must accept the End-User License Agreement.
Please select the language you would like to use to read the agreement.
Select language: English
ALTIUM LLC END-USER LICENSE AGREEMENT
ALTIUM LLC
April 2023
IMPORTANT - READ CAREFULLY
THIS ALTIUM LLC END-USER LICENSE AGREEMENT ("EULA") IS A LEGAL AGREEMENT BETWEEN YOU (EITHER AN INDIVIDUAL PERSON OR A LEGAL ENTITY, "YOU" HEREIN) AND ALTIUM LLC ("ALTIUM" HEREIN) TO GOVERN YOUR USE OF CERTAIN COMPUTER TECHNOLOGY DEVELOPED AND DISTRIBUTED BY ALTIUM, TOGETHER WITH APPLICABLE DOCUMENTATION (COLLECTIVELY, THE "LICENSED MATERIA
AS FURTHER DEFINED BELOW). PLEASE READ THIS DOCUMENT CAREFULLY BEF
I do not accept the agreement
I accept the agreement
Accept the agreement
Advanced...
Back
Next
Cancel
Click Next
25.7.1 (Build 20)

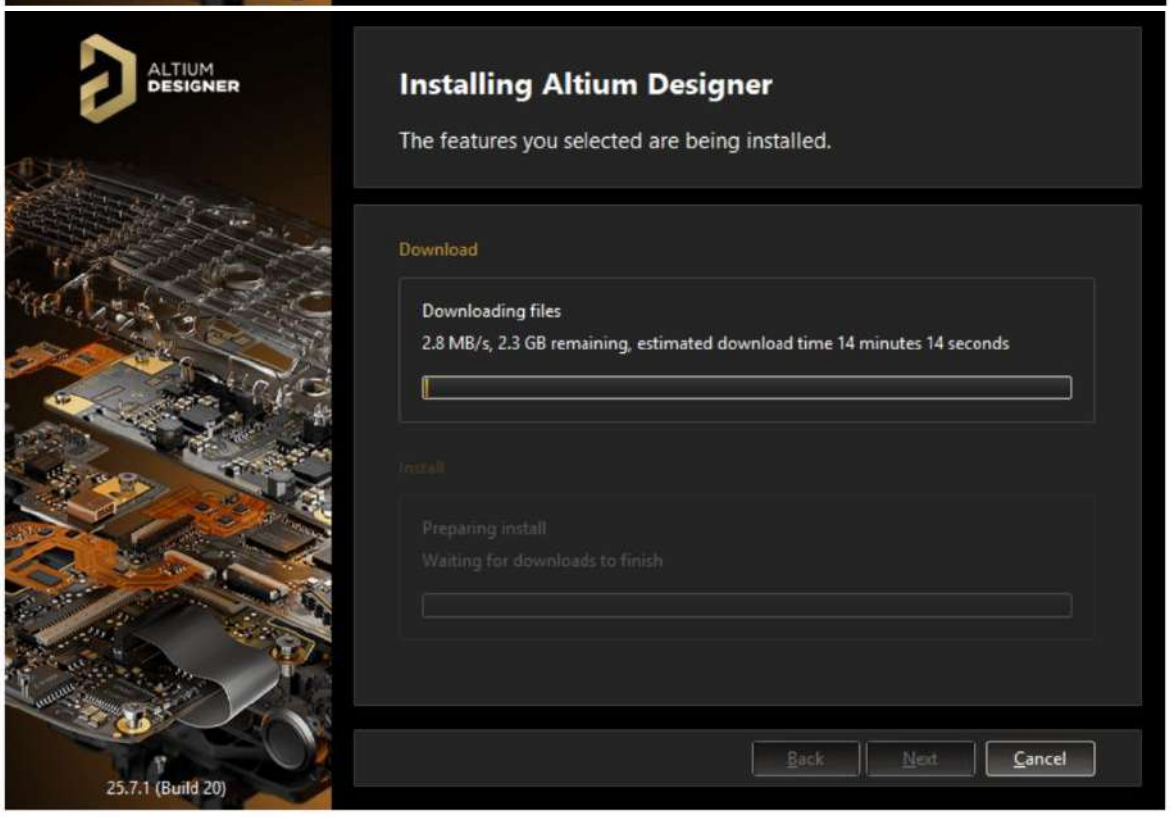

Sign In
Altium
Please sign in with your browser
Information about your PC hardware will be gathered as part of this process. This information will be passed over a secure connection.
Sign In
Cancel


ALTIUM DESIGNER
25.7.1 (Build 20)
Select Design Functionality
Select the type of design functionality you want to be installed.
Choose what type of features you want to be installed initially. You can add or remove features after you have finished installing the product.
PCB Design
Platform Extensions
Importers\Exporters
Touch Sensor Support
Base system including all required features for PCB Design.
To install the selected features approximately 2.3 GB will be downloaded.
This will require 6.4 GB of disk space to install.
Back
Next
Cancel


ALTIUM DESIGNER
25.7.1 (Build 20)
Destination Folders
Select the folders where the software and documents will be installed.
Choose where to install the program files for the software, as well as the shared documents that are accessed and used by the software.
Destination Folders
Program Files:
C:\Program Files\Altium\AD25
Default
Shared Documents:
C:\Users\Public\Documents\Altium\AD25
Default
Back
Next
Cancel

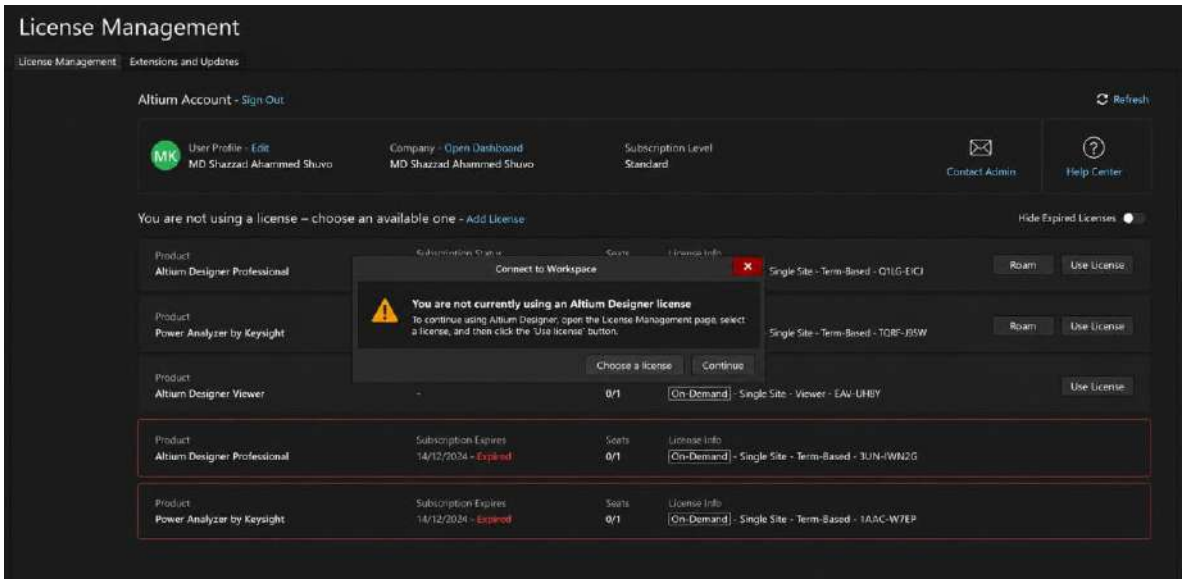

ALTIUM DESIGNER
25.7.1 (Build 20)
Customer Experience Improvement Program
Would you like to participate in the Program?
The Altium Customer Experience Improvement Program ("Program") is designed to enhance our understanding of how our customers utilize Altium products and services. This information enables us to further develop and create products and services that better meet your needs and improve your overall experience with Altium.
As an Altium customer, you have an option to share information with us regarding how you use our desktop and cloud products ("usage information").
We analyze such usage information to help measure product performance and quality, optimize current features, and make product development decisions. This information is associated with your AltiumLive user profile and it can be used to personalize and improve your application and website experience, as well as to communicate with you.
How is my privacy protected?
The data we collect cannot be used to recreate your work or your products and designs. The data is transmitted via a secure connection, it does not contain any information that can identify you or your company to any third party, and it will not be
Don't participate
Yes, I want to participate
Back
Next
Cancel

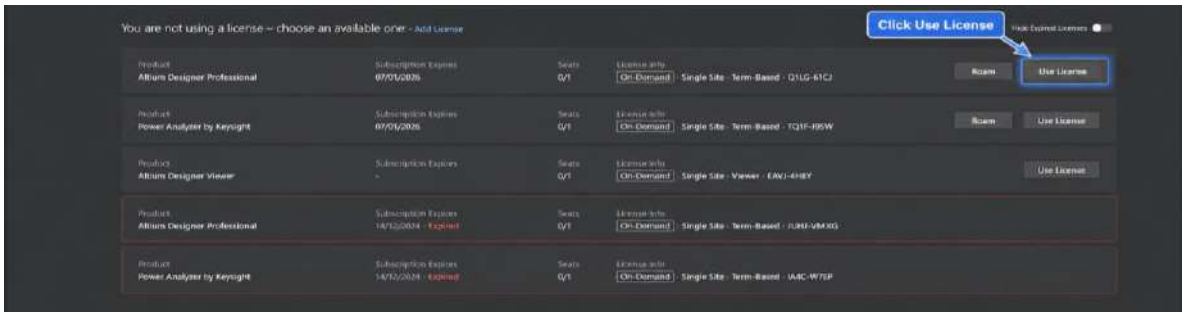


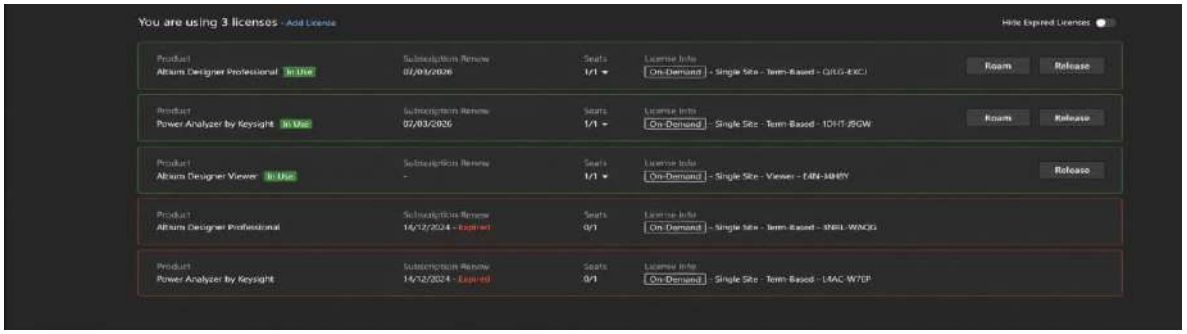


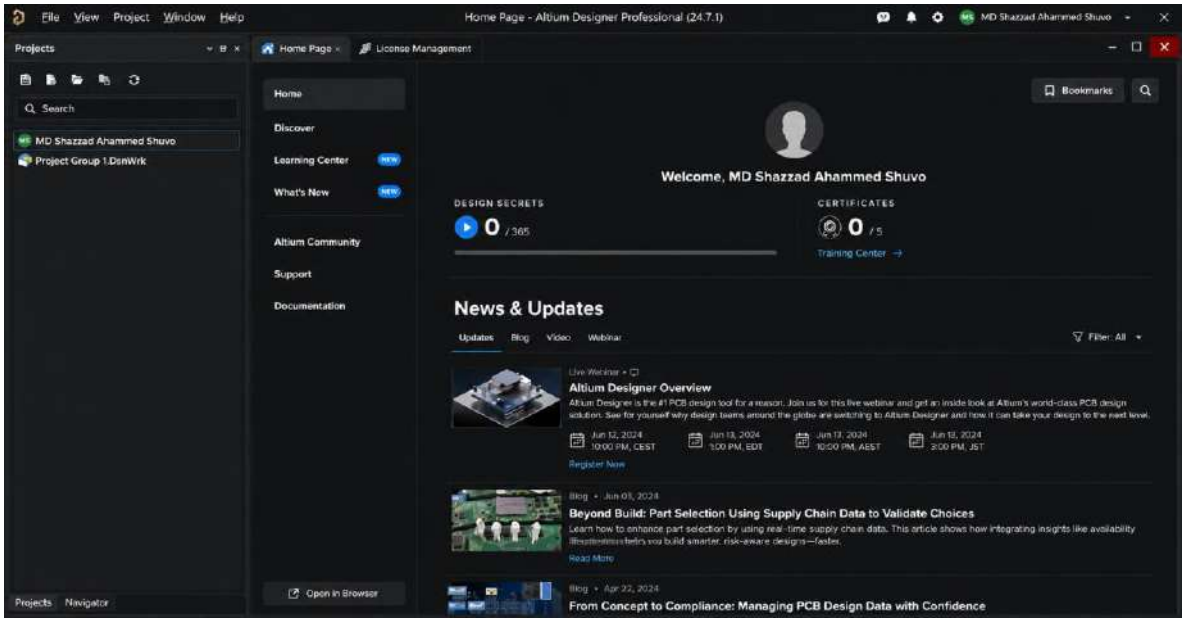
After installation, you can launch Altium from the Start Menu or desktop shortcut.

- The first-time launch may take a few extra seconds while setting up your environment.
- Sign in again if prompted to activate your student license.


License Management
License Management
Extensions and Updates
Altium Account - Sign Out
Refresh
User Profile - Edit
MD Shazzad Ahammed Shuvo
Company - Open Dashboard
MD Shazzad Ahammed Shuvo
Subscription Level
Standard
Contact Admin
Help Center
You are not using a license – choose an available one - Add License
Hide Expired Licenses
Connect to Workspace
You are not currently using an Altium Designer license
To continue using Altium Designer, open the License Management page, select a license, and then click the 'Use license' button.
Choose a license
Continue
Product
Altium Designer Professional
Single Site - Term-Based - Q1LG-EICJ
Roam
Use License
Product
Power Analyzer by Keysight
Single Site - Term-Based - TQ1F-J9SW
Roam
Use License
Product
Altium Designer Viewer
0/1
On-Demand - Single Site - Viewer - EAV-UHBY
Use License
Product
Altium Designer Professional
Subscription Expires
14/12/2024 - Expired
Seats
0/1
License Info
On-Demand - Single Site - Term-Based - 3UN-IWN2G
Product
Power Analyzer by Keysight
Subscription Expires
14/12/2024 - Expired
Seats
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License Info
On-Demand - Single Site - Term-Based - 1AAC-W7EP


You are not using a license – choose an available one - Add License
Click Use License
Use License


You are using 3 licenses - Add License
Hide Expired Licenses
Roam
Release


File View Project Window Help
Home Page - Altium Designer Professional (24.7.1)
MD Shazzad Ahammed Shuvo
Projects
Home Page
License Management
Search
MD Shazzad Ahammed Shuvo
Project Group 1.DsnWrk
Home
Discover
Learning Center
What's New
Altium Community
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DESIGN SECRETS
0 /365
CERTIFICATES
0 /5
Training Center
News & Updates
Updates
Blog
Video
Webinar
Filter: All
Altium Designer Overview
Register Now
Beyond Build: Part Selection Using Supply Chain Data to Validate Choices
Read More
From Concept to Compliance: Managing PCB Design Data with Confidence
Open in Browser
Projects
Navigator

BRAC University
Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-01: Introduction to PCB Fundamentals

1. Objective:

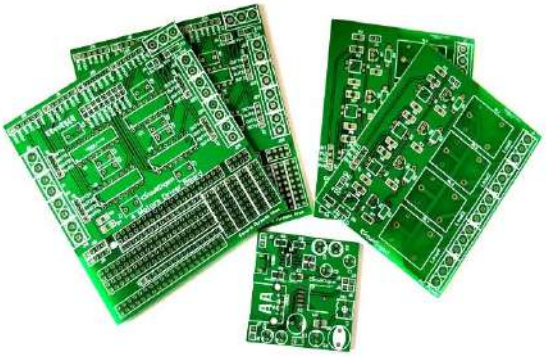
By the end of this, students will be able to:

Understand the purpose and importance of Printed Circuit Boards (PCBs) in electronic systems.
Identify the main components and structural layers of a PCB.
Distinguish between single-layer, double-layer, and multilayer PCBs.
Become familiar with common PCB materials and copper cladding.
Recognize the basic PCB features including pads, tracks, vias and solder masks.
Learn the fundamental design considerations for reliable PCB layouts.
Develop the ability to identify different PCB types and their practical applications.

2. Introduction:

PCB is a copper-laminated and non-conductive **Printed Circuit Board,** in which all electrical and electronic components are connected in one common board with physical support for all components with the base of the board. In PCB, all components are connected without wires and internally, so it reduces the complexity of the overall circuit design.

PCB is used to provide electricity and connectivity between the components, through which it functions the way it was designed. PCBs can be customized for any specifications to meet user requirements. It can be found in many electronic devices like: TV, Mobile, Digital camera, Computer parts like: Graphic cards, Motherboard, etc. It is also used in many fields like medical devices, industrial machinery, automotive industries, lighting, etc.



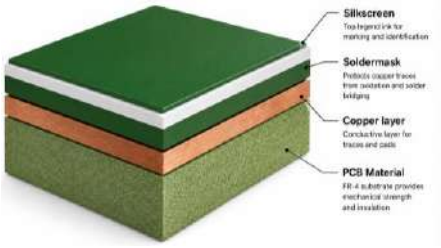
- It provides mechanical support and electrical connections for electronic components.
- It facilitates the flow of electricity between components via conductive pathways.
- It is used in a wide range of electronic devices for consumer, industrial, medical, and aerospace applications.

Feature	Traditional Wiring	PCB Board
Complexity	High - manual wire routing	Low - pre-designed traces
Reliability	Lower (connection failures)	Higher (permanent connections)
Manufacturing Time	Hours to days	Minutes (automated)
Circuit Size	Limited scalability	Highly scalable (multi-layer)
Cost (bulk production)	Expensive	Cost-effective

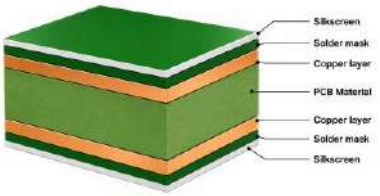
There are several types of PCB available for the circuit. Out of these types of PCB, we have to choose the appropriate type of PCB according to our application.

1. **Single-layer PCB**
2. **Double-layer PCB**
3. **Multi-layer PCB**
4. **Flexible PCB**
5. **Aluminium-backed PCB**
6. **Flex-rigid PCB**

A single-layer PCB is also known as a single-sided PCB. Single-layer PCBs are the simplest and most cost-effective option for PCB design basics for beginners. This type of PCB is simple and the most commonly used PCB because these PCBs are easy to design and manufacture. One side of this PCB is coated with a layer of any conducting material. A layer of solder mask is used to protect the PCB against oxidation followed by a silk screen to mark out all of the components on the PCB. In this type of PCB, only one side of the PCB is used to connect different types of electrical or electronic components like resistors, capacitors, inductors, ICs etc. These components are soldered. These PCBs are used in low-cost and bulk manufacturing applications like calculators, radios, printers and the solid-state drive.



A double-layer PCB is also known as a double-sided PCB. As the name suggests, in this type of PCB, a thin layer of conducting material like copper is applied to both top and bottom sides of the board. In PCB, different layers of the board consist of *via* which have two pads in corresponding positions on different layers. These are electrically connected by a *hole* through the board. More flexible, relatively lower cost and the most important advantage of this type of PCB board is its reduced size which makes the *circuit compact*. This type of PCB is mostly used in industrial controls, converters, UPS systems, HVAC applications, phones, amplifiers and power monitoring systems. This design is essential for intermediate circuit board design basics.



A multilayer PCB has more than two layers. It means that this type of PCB has at least three conductive layers of copper. For securing the board, glue is sandwiched between the layers of insulation which ensures that the excess heat will not damage any component of the circuit. This type of PCB design is very complex and used in very complicated and large electrical tasks in a very small space and compact circuits. This type of PCB is used in large applications like GPS technology, satellite systems, medical equipment, file servers and data storage. These represent advanced PCB board design basics for complex applications.

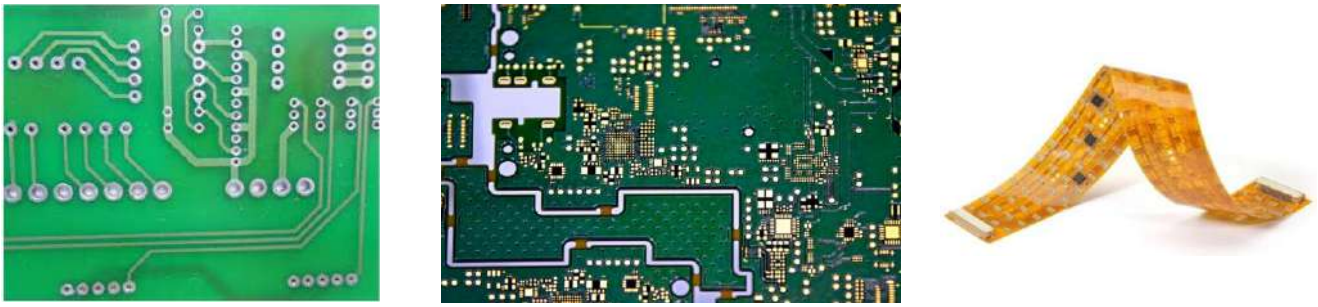


Figure: Images of single layer, Multiple Layer and Flexible PCB respectively

PCB Type	Layers	Complexity	Best Applications
Single-Layer	1 conductive layer	Simple	Calculators, LED circuits, radios
Double-Layer	2 conductive layers	Moderate	Amplifiers, UPS systems, HVAC
Multi-Layer	3+ layers (up to 50)	Complex	Computers, GPS, medical equipment
Flexible PCB	Variable	Moderate to High	Wearables, cameras, laptops
Rigid PCB	Variable	Any	Computer components (RAM, GPU)
Flex-Rigid	Mixed	Very Complex	Smartphones, aerospace, automotive

The main element is a dielectric substrate which is rigid or flexible. This dielectric substrate is used with a conducting material like copper on it. As a dielectric material, glass epoxy laminates or composite materials are used. Aluminum PCBs use a metal substrate for superior heat dissipation, essential for high-power simple circuit board design applications.

FR4: FR stands for FIRE RETARDANT (material that can self-extinguish flames). For all types of PCB manufacturing, the most common glass laminated material is FR4. Based on woven glass-epoxy compounds, FR4 is a composite material which is most useful because it provides very good mechanical strength.

FR4	Glass transition temp.	
Standard	130 °C	Most common and widely used, Cheapest
With higher glass transition temp.	170-180 °C	Compatible with the lead free reflow technology
Halogen free	--	Compatible with the lead free reflow technology

FR-1 and FR-2: This material is made from paper and phenol compounds, and this material is used for only single-layer PCBs. Both FR1 and FR2 have similar characteristics; the only difference is the glass transition temperature. FR1 has a higher glass transition temperature compared to FR2. These materials are also subdivided into standard, halogen-free free and

non-hydrophobic. FR1 and FR2 use a paper substrate with phenolic resin, offering a lower cost for simple PCB layout design projects with minimal requirements.

CEM-1: This material is made from paper and two layers of woven glass epoxy and phenol compounds, and this material is used for single-sided PCBs only. CEM-1 can be used instead of FR4, but the price of CEM-1 is higher than FR4. CEM (Composite Epoxy Material) boards blend paper core with glass fiber surfaces, balancing cost and performance in PCB board design basics.

CEM-3: This material is a white coloured, glass epoxy compound, which is mostly used in double-layer PCBs. CEM-3 has lower mechanical strength compared to FR4, but it is cheaper than FR4. So, this is a good alternative to FR4.

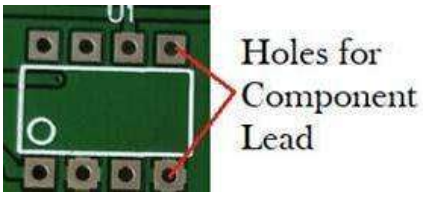
Polyimide: This material is used in a flexible PCB. This material is made from Kepton, Rogers, and DuPont. This material has good electrical properties, flexibility, a wide temperature range and high chemical resistance. Working temperature of this material is -200℃ to 300℃. Polyimide is the premier material for flexible PCBs, offering exceptional thermal and chemical resistance crucial for advanced circuit board design basics.

Prepreg: Prepreg means pre-impregnated. It is a fibreglass impregnated with resin. These resins are pre-dried, so that when it is heated, it flows, sticks and completely immerse. Prepreg has an adhesive layer which gives strength similar to FR4. There are many versions of this material according to resin content: SR- standard resin, MR- medium resin and HR- high resin. This is chosen according to the required thickness, layer structure and impedance. This material is also available in high glass transition temperature and halogen-free. Prepreg (pre-impregnated) fibreglass provides the adhesive layers in multi-layer PCB board design basics, bonding copper layers together.

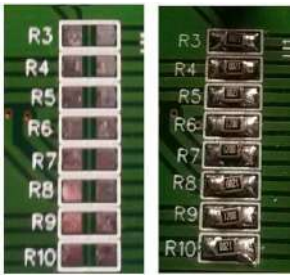
Understanding component mounting methods is crucial for simple PCB layout design.

1. Through-hole PCB
2. Surface-mounted PCB

1) Through-hole PCB: In this type of PCB, we have to make a hole using a drill on the PCB. In these holes, leads of components are mounted and soldered to pads situated on the opposite side of the PCB. This technology is most useful because it gives more mechanical support to electrical components and very reliable technology for mounting components, but drilling in PCBs makes it more expensive. In a single-layer PCB, this mounting technology is easy to implement, but in the case of double-layer and multi-layer PCBs, making holes is more difficult. This traditional method remains important for certain circuit board design basics and applications.

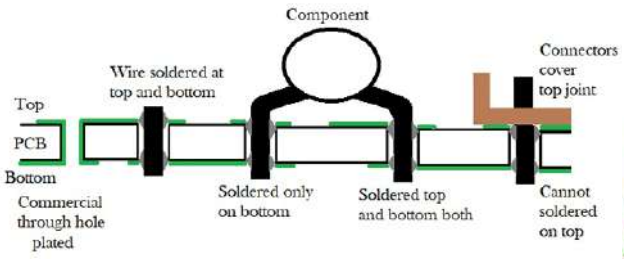


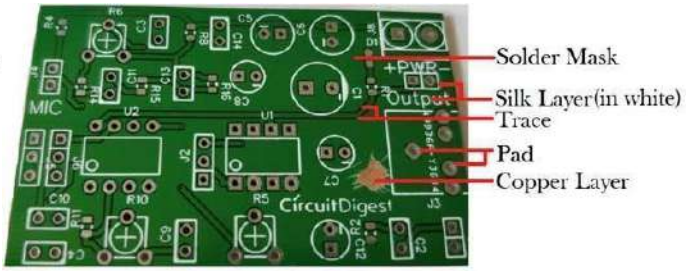
2) Surface-mounted PCB: In this type of PCB, components are small in size because these components have very small leads or no leads are required for mounting on the board. Here, in this technology, SMD components are directly mounted on the surface of the board and do not require making a hole in the board. SMT dominates modern PCB board design basics for consumer electronics due to its efficiency and compactness.



Through-Hole vs. Surface Mount Comparison

Feature	Through-Hole Technology (THT)	Surface Mount Technology (SMT)
Component Size	Larger with long leads	Compact, minimal or no leads
PCB Drilling	Required for every component	Not required
Assembly Speed	Slower (manual or semi-automated)	Fast (fully automated)
Component Density	Lower	Much higher
Mechanical Strength	Excellent	Good (but less than THT)
Cost (bulk production)	Higher due to drilling	Lower with automation
Best For	High-power, high-reliability applications	Compact, high-volume consumer electronics





Pad: A pad is nothing but a piece of copper on which leads of components are mounted and on which soldering is done. The pad provides the mechanical support to the components. In a simple PCB layout design, proper pad sizing ensures reliable electrical connections and mechanical stability.

Trace: In PCB, components are not connected with the help of wires. All components are connected with a conducting material like copper. This copper part of the PCB, which is used to connect all components that is known as a trace. Traces are thin copper pathways that replace traditional wires in modern circuit board design basics.

According to the application, cost and available space of the circuit, the user can choose the layer of PCB. The simplest in construction, easy to design and most useful in routine life is the single-layer PCB. But for very large and complex circuits, double-layer PCB or Multi-layer PCB is most preferred compared to single-layer PCB. Nowadays, in multi-layer PCBs, 10-12 layers can be connected and the most critical thing is to communicate between the components on different layers. Understanding layer stackup is essential for advanced PCB board design basics.

Silk Layers: The Silk layer is used for printing lines, text or any art on the surface of the PCB. Usually, for screen printing, epoxy ink is used. Silk layer can be used in the top and/or bottom layer of PCB according to the user's requirement, which is known as silk screen *TOP* and silk screen *BOTTOM*.

Top and bottom layer: In the Top layer of the PCB, all components are mounted in this layer of PCB. Generally, this layer is green coloured. In the bottom layer of PCB, all components are soldered through the holes and the leads of the components are known as the bottom layer of PCB. Sometimes, in the top and/or bottom layer, PCB is coated with a green colour layer, which is known as a solder mask.

Solder Mask: There is one additional layer on the top of the copper layer called Solder Mask. This layer generally has a green colour but it can be of any colour. This insulating layer is used to prevent accidental contact of pads with other conductive material on the PCB. This essential layer prevents solder bridges during assembly, a key concept in PCB design basics for beginners.

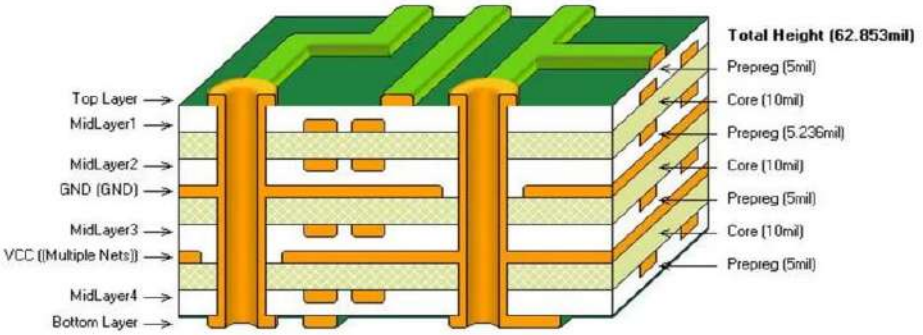
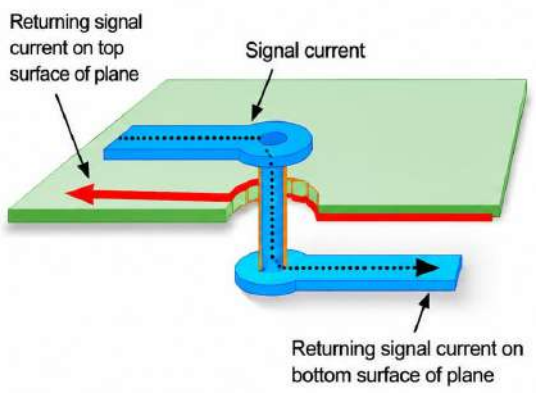
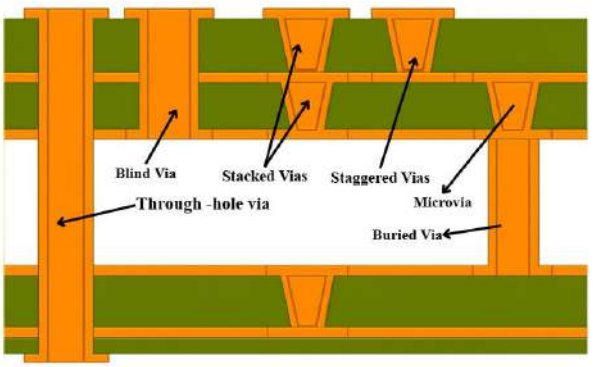


Fig: 6 Layer PCB Layer stackup

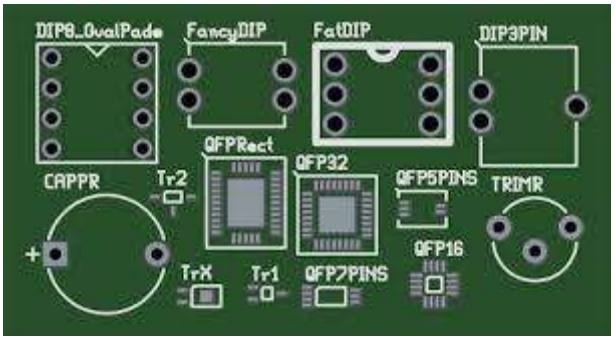
Via: Interconnection between layers is done through VIAs.

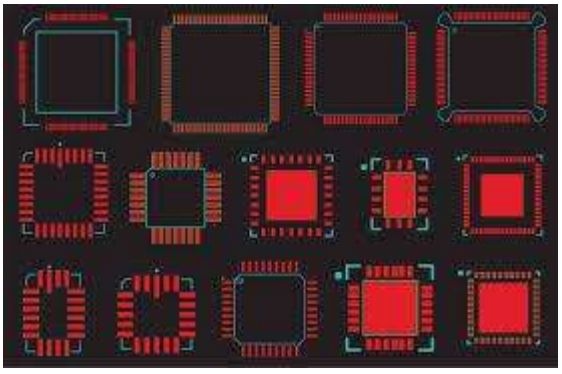




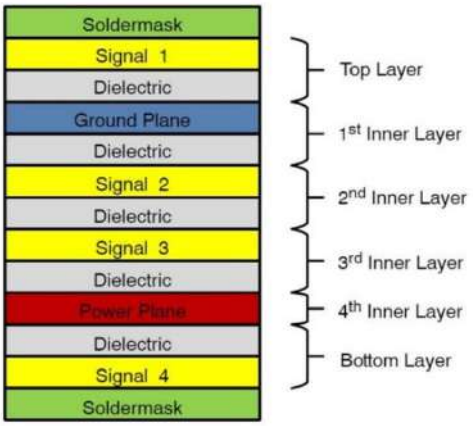
Through-Hole Via: A via that goes all the way through the PCB, from the top layer to the bottom layer
Blind Via: A via that starts on an outer layer (top or bottom) but doesn't go all the way through the board. It connects an outer layer to one or more inner layers.
Buried Via: A via that is completely inside the PCB and doesn't reach the outer layers. It connects two or more inner layers.
Micro Via: A very small via (usually less than 0.15 mm in diameter) used in high-density designs.
Stacked Via: Multiple vias (blind, buried, or micro vias) stacked on top of each other to create a connection through multiple layers.
Staggered Via: Multiple vias (blind, buried, or micro vias) placed next to each other (not directly stacked) to connect different layers.

Footprint: A PCB footprint is the physical arrangement of pads on a printed circuit board (PCB) that connects electronic components to the board.





Power and Ground Plane: Ground planes provide a low-impedance path for return currents, improving signal integrity and reducing electromagnetic interference (EMI). Power planes ensure stable voltage distribution, preventing brownout issues during sudden power spikes. The solid conductive material spreads heat, aiding in thermal management without extra cost.

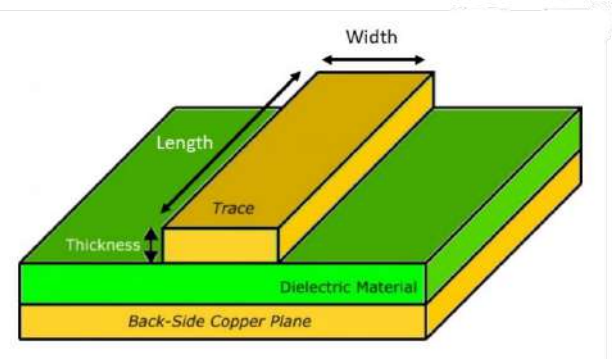


PCB layout is the physical realization of a schematic design. Component placement is arguably the most critical step. A poor placement forces suboptimal routing which degrades signal integrity, EMC performance and manufacturability. The general

placement philosophy follows signal flow: input connectors → filters → processing → output → power.

In Altium, the board outline is drawn on the Mechanical 1 layer / Board Shape via Design. The Design Rules (DRC rules) define minimum trace widths, clearances, drill sizes and via specifications. For standard FR4 double-sided PCBs (JLCPCB/PCBWay specifications): minimum trace width = 6 mil (0.15mm), minimum clearance = 6 mil, minimum drill = 0.3mm.

Decoupling capacitors (bypass caps, typically 100nF) must be placed as close as possible to IC power pins to suppress high-frequency noise on the power supply rail. This proximity minimizes the parasitic inductance of the connecting traces.



In PCB, components are not connected with the help of wires. All components are connected with a conducting material like copper. This copper part of the PCB, which is used to connect all components that is known as a trace. Traces are thin copper pathways that replace traditional wires in modern circuit board design basics.

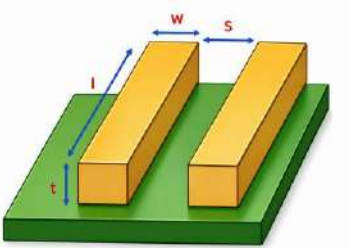
Trace width is how wide the copper lines (traces) on a PCB are. It's important because the width determines how much current (electricity) the trace can safely carry without overheating or failing. If a trace is too narrow, it can't handle the current and it might overheat, melt or even cause the PCB to fail. If the trace is wide enough, it can safely carry the current without problems, making the PCB reliable and long-lasting.

Wire Geometry : Pitch is the center-to-center spacing between two consecutive.

$$Pitch = W+s$$

$$Aspect\ Ratio\ (AR) = \frac{t}{w}$$

- In older processes AR<<1
- In modern processes AR≈ 2 (w has decreased)



Mil thickness:
A mil is a unit of thickness equal to one thousand of an inch. (0.001 inches)
Example: 2 mils = 2/1000 inches = 0.002 inches

Wire Resistance:

Resistivity, $\rho = \Omega * m$

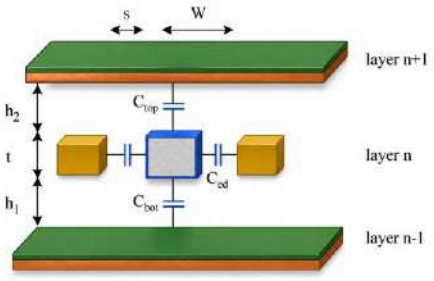
Resistance, $R = \frac{\rho l}{A} = \frac{\rho l}{tw} = R_s \frac{l}{w}$

$R = R_s$ * number of squares | here, Sheet Resistance (R_s, in Ω/sheet)

Up to 180 nm technology, most wires were Aluminum (due to low resistance). Modern processes often use copper. Copper atoms diffuse into Silicon and damage the transistor. So, it must be surrounded by a diffusion barrier.

Bulk resistivity is a measure of how much a material resists the flow of electricity through it. It tells us how easily or difficult it is for electric current to pass through a metal or substance.

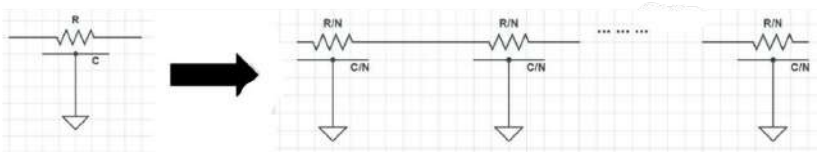
Wire Capacitance: Wire has capacitance per unit length
- With the neighbors (same layer)
- With the layers above and below



Total capacitance: $C_T = C_{top} + C_{bot} + 2C_{adj}$

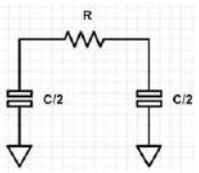
RC Delay: In printed circuit board (PCB) design, RC delay refers to the *signal propagation delay* caused by the interaction between the resistance (R) of conductive traces and the parasitic capacitance (C) between traces, to ground planes or to adjacent layers. This creates a time constant (t = RC) that slows down signal transitions, affecting rise/fall times, overall timing and signal integrity especially in high speed digital circuits.
RC delay is a lumped-element effect prominent at lower frequencies or shorter traces where inductive effects are negligible.

Lumped Elements Model (n segment consideration):
- Wires are a distributed system
- Approximate with lumped element models



The resistance is equally distributed throughout the wire. We assume the wire is divided into n segments (in reality infinite segments), if the value of n is very large it mimics the real wire.

Lumped Elements Model (π segment consideration): The 3 segment pie model is accurate to 3% in simulation. We calculate the total delay of the wire (how much delay is caused by the entire wire in the system) using this.



Trace width Calculation: Trace width is a critical parameter in PCB design that determines the amount of current a copper trace can safely carry without exceeding the allowable temperature rise. An appropriately sized trace minimizes resistive losses, reduces voltage drop and prevents excessive heating that may lead to reliability issues or PCB failure. The IPC-2221 standard provides an empirical equation for estimating the minimum trace width based on the required current, permissible temperature rise and copper thickness. The relationship is expressed as:

$$\mathbf{I = k \,.\, (\Delta T)^b \,.\, A^c}$$

Here,

I = Maximum current capacity (A)

ΔT = Allowable temperature rise above ambient in °C

A = Cross-sectional area of the trace in mils² = Trace Width × Copper Thickness

k, b and c = Constants derived from IPC experimental data:

- External Traces (Outer layers): k = 0.048, b = 0.44, c = 0.725
- Internal Traces (Inner layers): k = 0.024, b = 0.44, c = 0.725

PCB Routing Rules: The following routing rules are widely used in PCB design to ensure electrical performance, manufacturability, reliability and ease of assembly.

1. Use the shortest possible trace length to minimize resistance, inductance and signal delay.
2. Select an appropriate trace width based on the expected current.
3. Maintain adequate spacing between adjacent traces to prevent short circuits, crosstalk and electrical arcing.
4. Avoid 90° trace bends; instead use 45° bends or curved traces to improve manufacturability and signal integrity.
5. Minimize the number of vias as each via introduces additional resistance, inductance and manufacturing cost.
6. Route high-speed and sensitive signals over a continuous ground plane to provide a low-impedance return path.
7. Separate analog and digital signal traces to minimize noise coupling and electromagnetic interference (EMI).
8. Match the lengths of high-speed signal traces where timing synchronization is critical.
9. Avoid routing traces beneath high-frequency components, oscillators or switching power supplies whenever possible.
10. Maintain sufficient clearance between copper features and the PCB edge to improve mechanical reliability.
11. Use ground planes and copper pours to improve shielding, heat dissipation and return current paths.
12. Place decoupling capacitors as close as possible to the power pins of integrated circuits.
13. Avoid creating narrow copper necks that can increase resistance and reduce current-carrying capacity.
14. Keep high-current traces away from low-level analog and RF signals to reduce electromagnetic interference.
15. Follow the manufacturer's minimum design rules for trace width, spacing, via size, and solder mask clearance.
16. Ensure thermal relief connections are used where appropriate to facilitate reliable soldering.
17. Perform Design Rule Check (DRC) and Electrical Rule Check (ERC) before generating manufacturing files to verify compliance with design constraints.

BRAC University
Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-02: Introduction to Altium Designer Environment

1. Objective:

By the end of this, students will be able to:

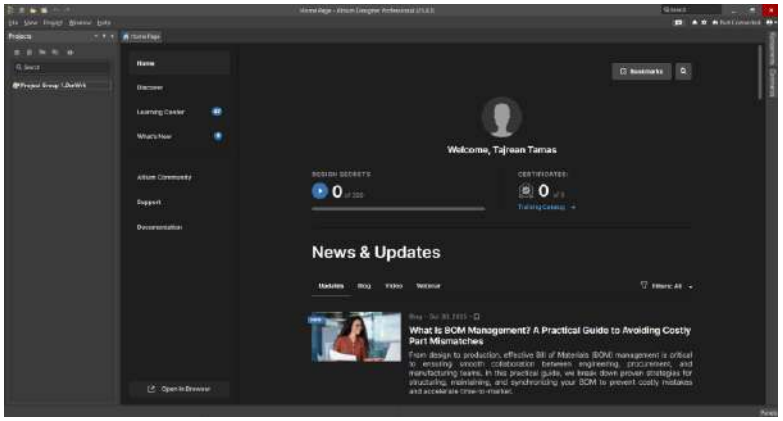
- Open, navigate and customize the Altium Designer workspace.
- Create and manage a new PCB project.
- Add schematic and PCB layout documents to a project.
- Understand file structures such as .PrjPCB, .SchDoc and .PcbDoc.
- Save and organize design files systematically.

2. Theoretical Background:

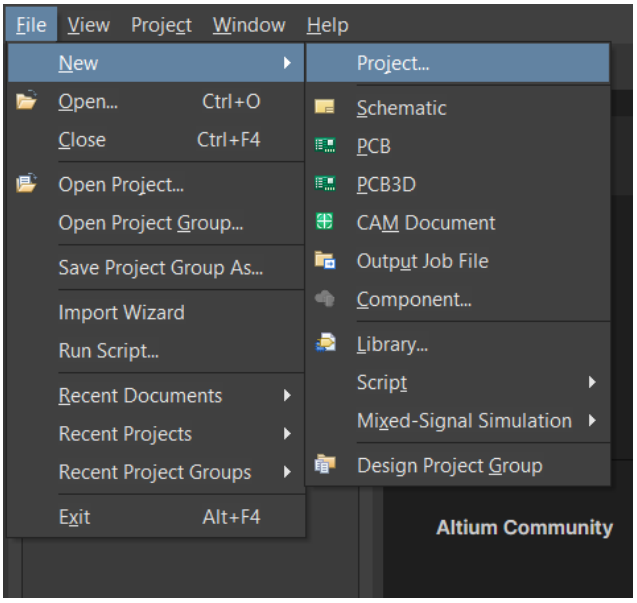
A Printed Circuit Board (PCB) is the foundation of nearly every electronic device today. It provides both the mechanical structure to support components and the electrical pathways to connect them. Altium Designer is a powerful Electronic Design Automation (EDA) tool that integrates schematic capture, PCB layout and manufacturing preparation in one environment. It maintains a synchronized connection between schematic and layout, so any circuit change automatically updates across the design.
In this part of the manual the students will get introduced with Altium's interface including panels, menu bars and project organization and also learning how to properly set up and manage a project.

3. Procedure:

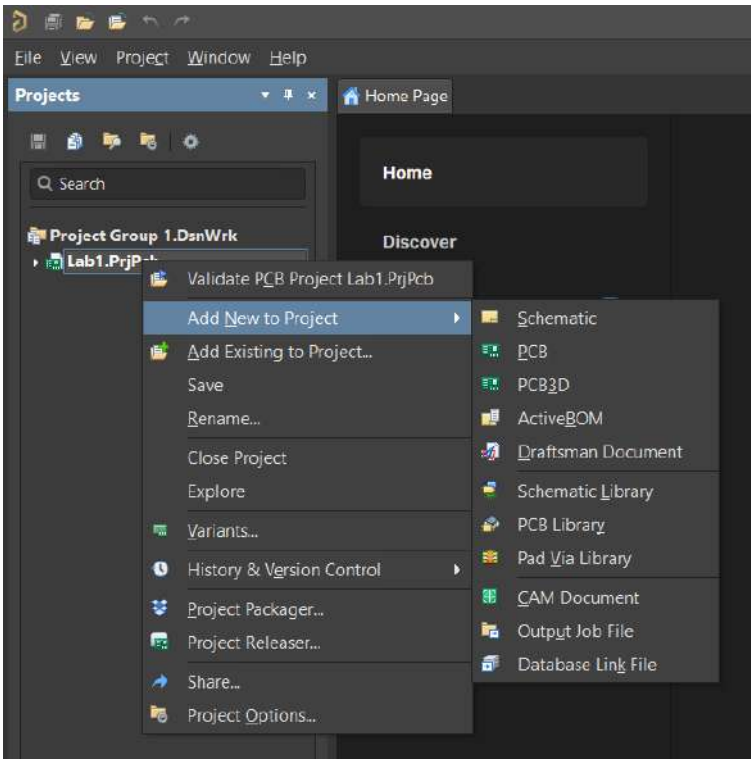
1. Once Altium Designer launches, you'll see the Start Page which acts as a control hub for all your projects and workspaces.

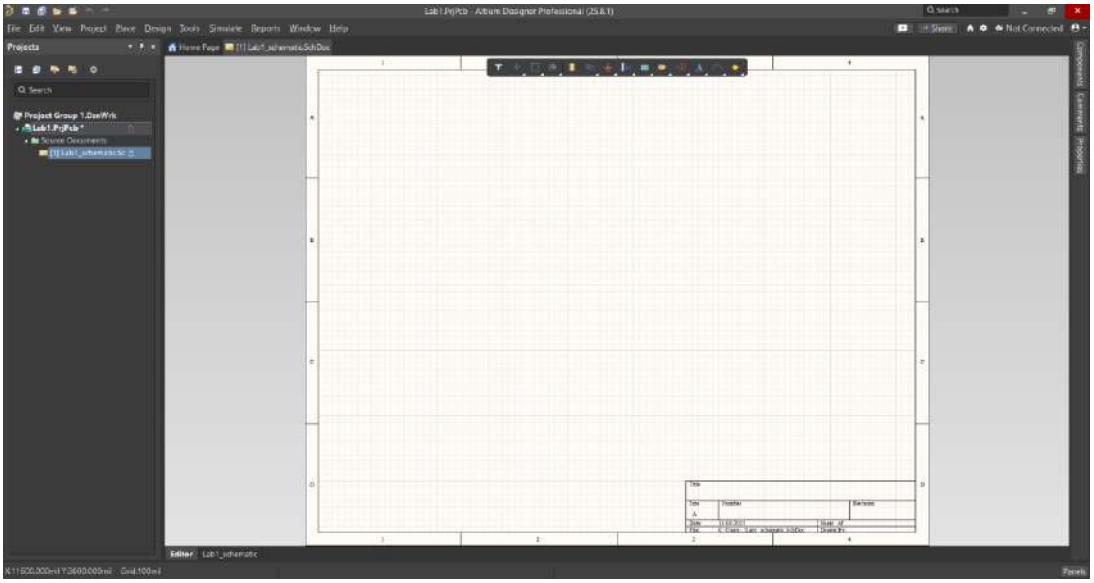


2. Create a new project by going to "File"-->"New"-->"Project", name it "Lab 01". It's important to save your projects in an organized folder structure for easy access.



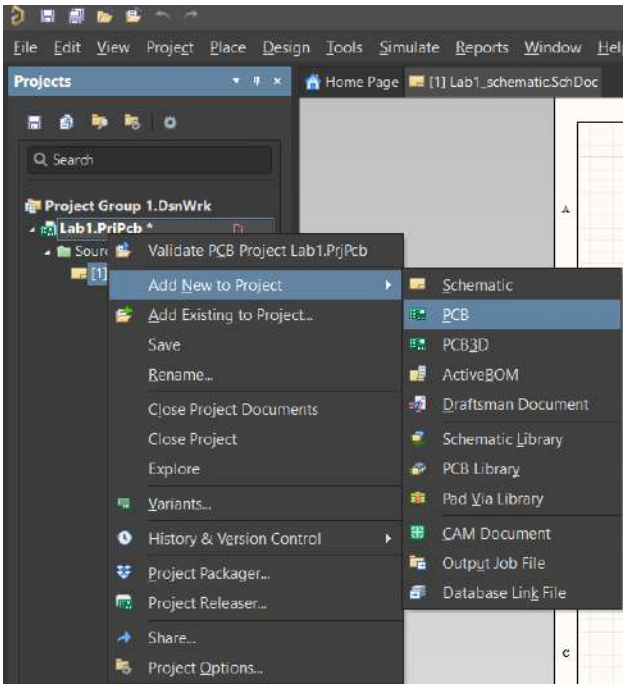
3. Right-click on the Project Name (Lab1) in the Projects Panel. From the context menu, select **Add New to Project > Schematic**. A new Schematic Sheet will appear under your project in the Projects Panel. Rename the schematic file to *Lab1_Schematic.SchDoc* by right-clicking on it and selecting **Rename**.

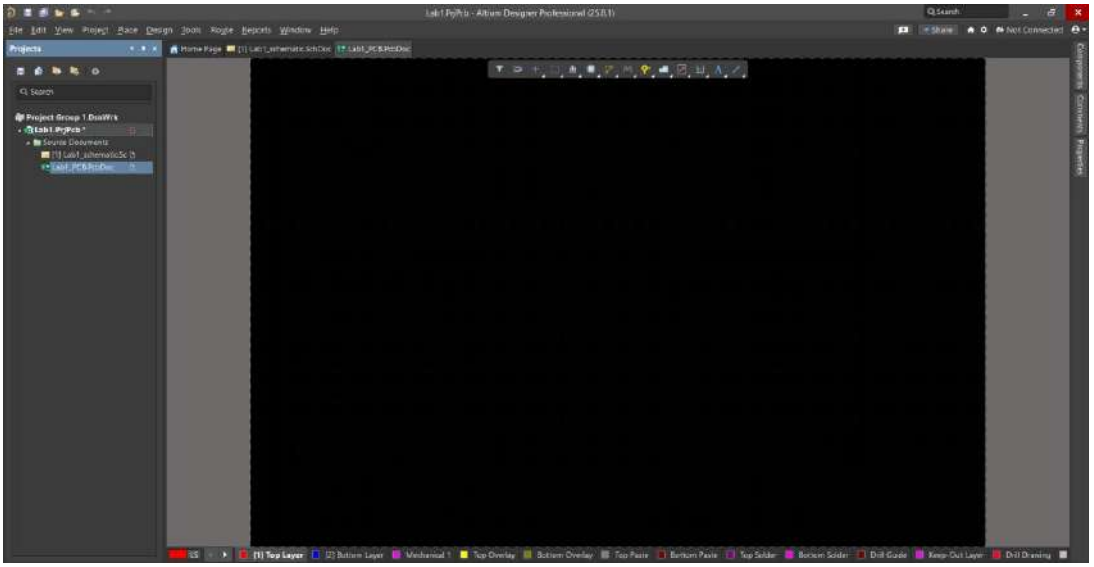




The .SchDoc file is where you'll draw the circuit schematic, it's the first step in creating a PCB design.

4. Again, right-click on the Project Name (Lab1) in the Projects Panel. Select Add New to Project > PCB to add a new PCB Layout Sheet. The new PCB layout file will appear under your project files in the Projects Panel. Rename the PCB layout file to Lab1_PCB.PcbDoc.



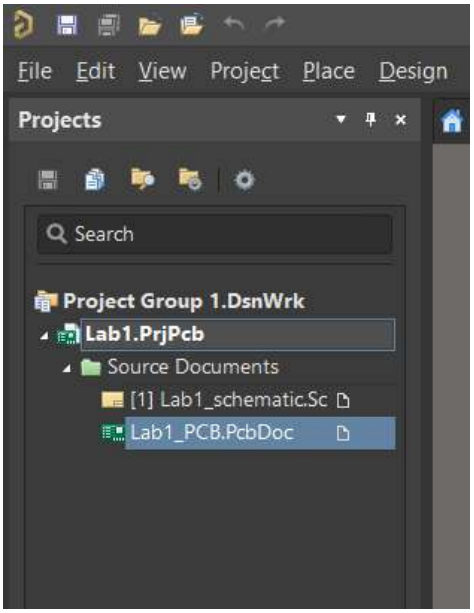


The .PcbDoc file is where you'll define the physical layout of your board, place components, and route the traces.

5. Once the schematic and PCB files are added, save the entire project.

File > Save All *or press* ***Ctrl + Shift + S.***

This step ensures that all your project files are stored in the correct folder and are ready for the next steps in your design process.



After completing the above steps, your Projects Panel should look like this.

BRAC University
Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-03: Component Selection, Sourcing, Downloading and Integration from SnapEDA and External Libraries

1. Objective:

By the end of this, students will be able to:

- Know different types of components used in electronic circuits.
- Understand the importance of component libraries in PCB design.
- Identify suitable electronic components from online sources such as SnapEDA.
- Download schematic symbols, PCB footprints and integrated component libraries from external platforms.
- Import downloaded libraries into Altium Designer correctly.

2. Theoretical Background:

Electronic circuits are made up of active-passive components, electromechanical devices, connectors and integrated circuits (ICs) that serve particular electrical and electronic functions. Each component has its own set of electrical properties that will affect the performance, reliability, efficiency and cost of the entire system. Hence, it is important to have a basic knowledge of the characteristics, applications and selection of electronic components to be able to design robust and cost-effective electronic systems. Besides the electrical characteristics and functionality of the selected components, the Electronic System Designer must also take into account the availability of the components, manufacturability, reliability and long-term maintenance. Choosing the right components contributes to a high performance system at a reduced production price and higher product reliability.

Passive components do not require an external power source to perform their intended function and do not provide signal amplification. Instead, they store, dissipate or regulate electrical energy within a circuit. Example: Resistor, inductor, capacitor. Active components require an external power source and are capable of controlling, amplifying or switching electrical signals. Example: Diode, Transistor, Op-Amp. Electromechanical components convert electrical energy into mechanical motion or vice versa which are commonly used for user interaction, control systems, power switching, and industrial automation. Example: Relays, Switches, Push buttons, Potentiometers, Buzzers, Motors, Electromagnetic actuators.

Interconnection components establish electrical connections between different sections of a circuit or between external devices. Their selection depends on current capacity, contact resistance, mechanical durability, environmental conditions and ease of assembly. However,

PCB connectors provide detachable electrical and mechanical connections between a PCB and external devices, wires or other PCBs. They simplify assembly, maintenance, testing and replacement of components. Based on their application and construction, PCB connectors can be classified into the following types.

Pin Header Connectors: Pin headers are one of the most commonly used PCB connectors. They consist of a row (or multiple rows) of metal pins soldered onto the PCB and are designed to mate with female headers or jumper wires. They are widely used in development boards such as Arduino, Raspberry Pi HATs and embedded systems. Pin headers are low cost components, easy to solder, available in male and female versions and have various pin counts and pitches (typically 2.54 mm).



JST Connectors: JST (Japan Solderless Terminal) connectors are compact wire-to-board connectors commonly used for batteries and low-current signal connections. Common series include: JST-XH, JST-PH, JST-SH. It ensures polarized connection, has compact size and secure locking mechanism. It is broadly used for battery connections and sensor interfaces.



Screw Terminal Block Connectors: Terminal blocks allow wires to be connected using screws or spring clamps. They are suitable for applications requiring frequent wiring changes or higher current handling. It has high current capacity, secured connection and easy field wiring. It is broadly used for power supplies, industrial controllers, motor drivers and relay modules.



USB Connectors: USB connectors provide power and data communication between electronic devices. Common types of USB connectors are: USB Type-A, Micro USB, USB Type-B, USB Type-C. It is broadly used for charging circuits, embedded systems,data transfer and firmware programming.



DC Barrel Jack Connectors: DC barrel connectors provide external DC power to PCBs through standard cylindrical plugs. Typical sizes are: 5.5 mm × 2.1 mm & 5.5 mm × 2.5 mm. It is widely used in power input, CCTV systems, Embedded devices, Power adapters.



RJ Connectors: RJ connectors are primarily used for networking and telecommunications. Common types are: RJ45 (Ethernet), RJ11 (Telephone). It is widely used in Ethernet communication, Network switches, Routers, Industrial communication systems.



Connector Type	Connection Type	Common Applications
Pin Header	Board-to-wire / Board-to-board	Arduino, sensors, embedded systems
Female Header	Board-to-board	Development boards
JST	Wire-to-board	Batteries, drones, robotics
Screw Terminal Block	Wire-to-board	Power distribution, industrial systems
USB	Data & Power	Embedded devices, charging
DC Barrel Jack	Power	DC-powered electronics
D-Sub	Data communication	RS-232, industrial equipment
RJ45/RJ11	Network	Ethernet and telecommunication

An Integrated Circuit (IC) is a miniaturized electronic circuit fabricated on a single semiconductor substrate which integrates numerous electronic components such as transistors, resistors, capacitors and diodes into a compact package capable of performing complex electronic functions. The use of ICs significantly reduces circuit size, power consumption, manufacturing cost and assembly complexity while improving reliability and performance. Modern electronic devices, ranging from smartphones and computers to medical instruments and industrial controllers, rely heavily on integrated circuits. According to functionality, ICs can be classified as Analog IC and Digital IC. Analog ICs process continuously varying electrical signals. Example: operational amplifiers, voltage regulators, comparators, data converters and analog filters. These ICs are widely used in sensor interfaces, audio systems, instrumentation, and power management applications. Digital ICs operate using discrete binary logic levels (0 and 1). Example: logic gates, flip-flops, counters, multiplexers, shift registers, memory devices and microprocessors.

Mixed-signal ICs combine both analog and digital circuitry within a single package. Example: Analog-to-Digital Converter (ADC), Digital-to-Analog Converters (DAC), Phase-Locked Loop (PLL) and microcontrollers with built-in analog peripherals. These ICs are extensively used in communication systems, embedded electronics and measurement instruments. Power integrated circuits (Power IC) are specifically designed for efficient power conversion, voltage regulation, battery management and motor control. Example: switching regulators, motor drivers, battery charging ICs and power management integrated circuits (PMICs). Microcontrollers integrate a processor, memory, timers, communication interfaces and input/output peripherals on a single chip, making them ideal for embedded control applications. In contrast, microprocessors primarily contain the central processing unit (CPU) and require external memory and peripheral devices, making them suitable for high-performance computing systems.

Based on the assembly process, IC packages are generally categorized into Through-Hole Packages (THT) and Surface-Mount Packages (SMT).

Through-Hole IC Packages (THT) : Through-hole packages have metal leads that pass through holes drilled in the PCB and are soldered on the opposite side of the PCB. They

provide strong mechanical support and are commonly used in educational laboratories, prototypes and high-power circuits.



The *Single In-line Package (SIP)* has a single row of pins extending from one side of the package. Because of its compact design, SIP packages occupy less PCB space than DIP packages. SIP packages are easier for manual soldering, suitable for simple circuits and usually have 4 - 20 pins. Example: Voltage regulator IC (LM78xx series), GBU810 Bridge rectifier IC.

The *Dual In-line Package (DIP)* is one of the most widely recognized IC packages which contains two parallel rows of pins that pass through the PCB. Standard pin spacing of DIP IC is 2.54 mm which is very easy to insert into sockets and simple to solder and replace in PCB. Common Pin Counts: 8, 14, 16, 20, 28, 40. Example: Operational amplifiers (LM741), Comparator ICs, Timer ICs (NE555), Logic ICs (74LS Series), ATmega328P (Arduino), EEPROMs.



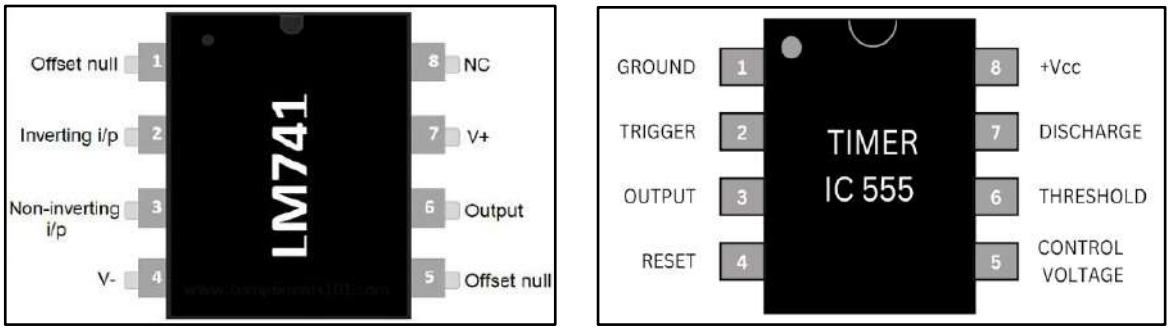


Fig: Pin layout of Op-Amp IC and 555 Timer IC respectively

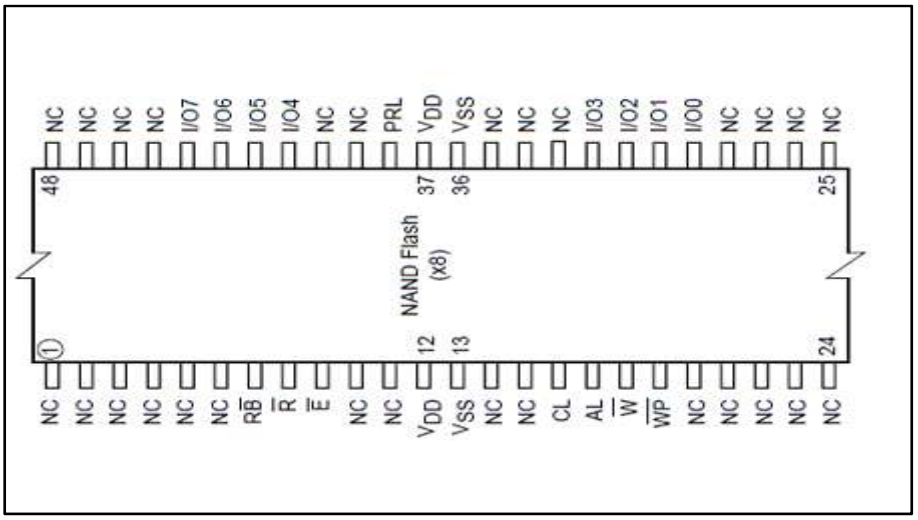


Fig: Pin layout of NAND01GW3B IC

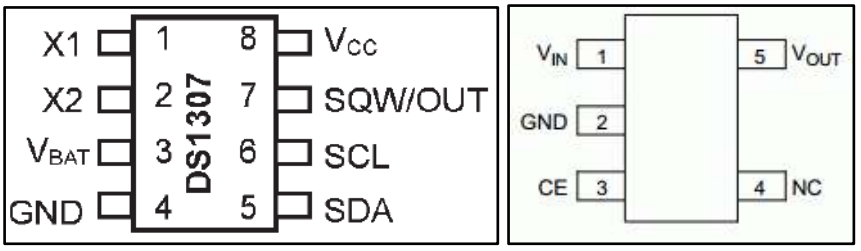


Fig: Pin layout of RTC DS1307 IC & SOT23-5 LDO IC

TO-Series (Transistor Outline) Packages are commonly used for voltage regulators, power transistors and power ICs. Example: TO-92, TO-220, TO-247, TO-3



Surface-Mount IC Packages (SMT): Surface-mount packages are soldered directly onto the PCB surface without drilling holes. They enable compact layouts, automated assembly and high-density electronic systems.



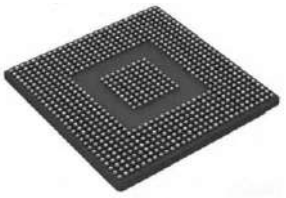
Small Outline Integrated Circuit (SOIC) is essentially the surface-mount version of the DIP package. It has Gull-wing leads, small footprint, lower profile and easy automated assembly. It has high I/O capability and is suitable for medium-to-high complexity designs but it is not suitable for high speed/frequency applications as it has larger lead inductance.



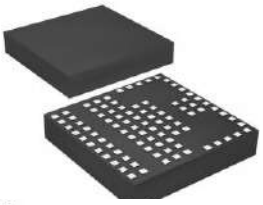
Quad Flat Package (QFP) IC has leads extending from all four sides. It has a very high pin count, thin profile and it is excellent for complex digital systems. Typical Pin Counts: 32, 44, 64, 100, 144, 208. Example: Microcontrollers, DSPs, ARM processors, FPGA devices. It has high I/O capability and is suitable for medium-to-high complexity designs. But it is not suitable for very high frequency/speed applications as it has a larger lead inductance.



Unlike QFP, the *Quad Flat No-Lead (QFN)* package has no external leads. Electrical connections are made through metal pads located underneath the package. It is very compact and has excellent thermal performance, low inductance, and high-frequency capability. QFP is widely used in RF circuits, Bluetooth modules, Wi-Fi modules, Sensors and Power management ICs.

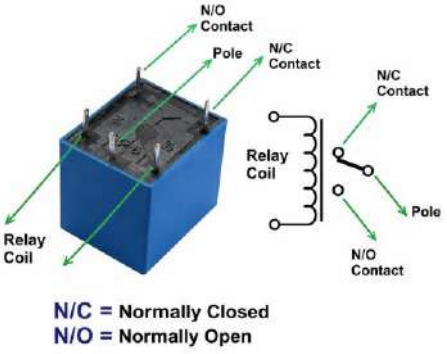


Ball Grid Array (BGA) package uses an array of solder balls underneath the package instead of peripheral leads. It has extremely high pin density, excellent heat dissipation, high-speed signal performance and requires reflow soldering and X-ray inspection. BGA is widely used in CPUs, GPUs, FPGAs, High-speed memory, Advanced microprocessors.



Land Grid Array (LGA) packages contain flat contact pads instead of solder balls or external leads. It is used in modern processors, communication ICs and high-performance computing.

A relay is an electrically operated switch that uses an electromagnetic coil or solid-state circuitry to control the opening and closing of electrical contacts. It enables a low-voltage, low-current control circuit to safely switch a high-voltage or high-current load while maintaining electrical isolation between the control and load circuits. Relays are widely used in industrial automation, motor control, protection systems, home automation and embedded applications where direct control of high-power devices is impractical or unsafe. The selection of a relay depends on



parameters such as coil voltage, contact configuration (SPST, SPDT, DPDT), switching current, voltage rating and response time.

An oscillator is an electronic circuit that generates a continuous periodic waveform without requiring an external input signal. It converts direct current (DC) power into alternating current (AC) signals at a specific frequency through positive feedback and frequency-selective components. Oscillators are fundamental building blocks in electronic systems, providing accurate timing, clock signals, carrier waves and reference frequencies for communication, computing and control applications. Depending on the required frequency stability and waveform, oscillators may produce sine, square, triangular or sawtooth waveforms.

A Global System for Mobile Communications (GSM) module is a wireless communication device that enables electronic systems and embedded controllers to communicate over a cellular network. By inserting a valid SIM card, a GSM module can transmit and receive voice calls, Short Message Service (SMS) and packet data using standard mobile communication networks. Most GSM modules communicate with microcontrollers through a Universal Asynchronous Receiver-Transmitter (UART) interface using AT (Attention) commands. GSM modules are widely used in remote monitoring, Internet of Things (IoT) systems, vehicle tracking, smart energy meters and security applications where long-distance wireless communication is required.

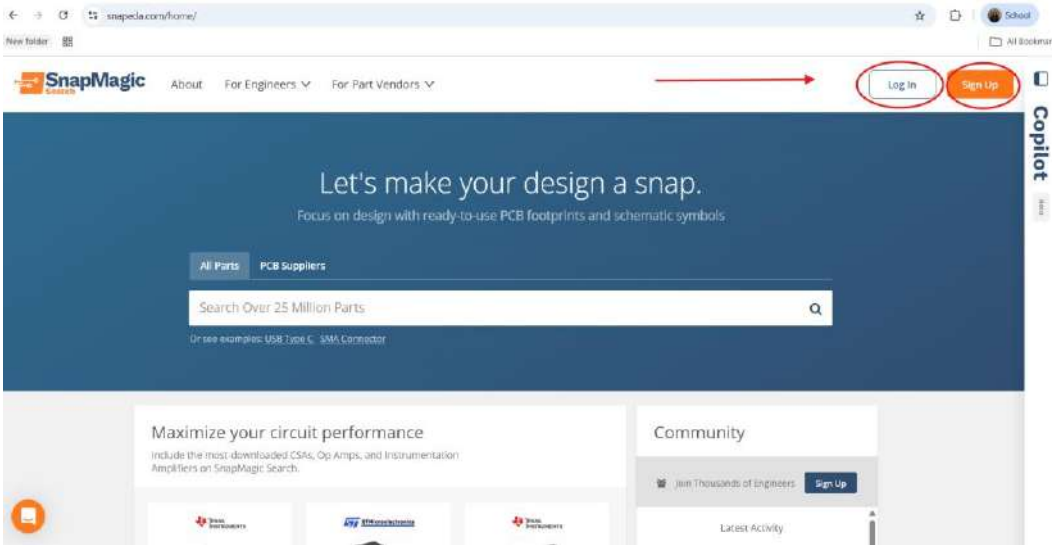
Choosing the right electronic components is an important design phase in the development of an electronic system that will ensure the proper functioning, reliability, minimal cost of manufacturing and product sustainability. When making component selection, overall performance, reliability, availability, manufacturability and life cycle cost should be considered. Electrical performance is the first factor to be considered. The components need to meet all electrical requirements such as voltage rating, current rating, power dissipation, operating frequency, tolerance, response time and the thermal properties. Choosing parts that don't meet the required critical rating could lead to failure, while choosing parts that have higher ratings than what is necessary could be an unnecessary expensive choice. Production planning has a significant impact on component availability. Using readily available parts ensures timely ordering and reduces the chances of redesign due to obsolete or discontinued parts. Availability of alternative suppliers or options should also be considered by designers to ensure supply chain resiliency. To optimize performance in cost, determine if further performance is worth the extra cost. The long-term support of the manufacturer, extensive documentation and active technical communities make troubleshooting, maintenance and product upgrades easier.

Mostly, the design of PCB is done using component libraries that include schematic symbols and PCB footprints. Such libraries enable designers to swiftly locate and insert components in a schematic and connect them to the corresponding physical components in a PCB design. Engineers will get the libraries of components from trusted online sources rather than designing all of them. The user can obtain thousands of manufacturer-verified symbols and footprints from online component databases, such as SnapEDA. This enables a reduction in design time, enhances productivity and prevents mistakes in design due to pin or component size mismatches.

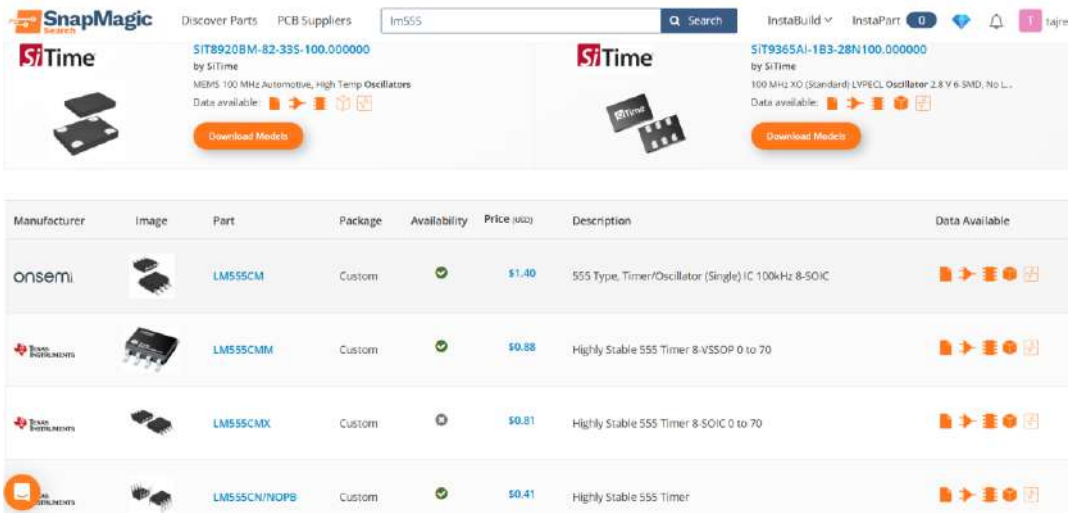
3. Procedure:

1. Open a web browser and go to www.snapeda.com.

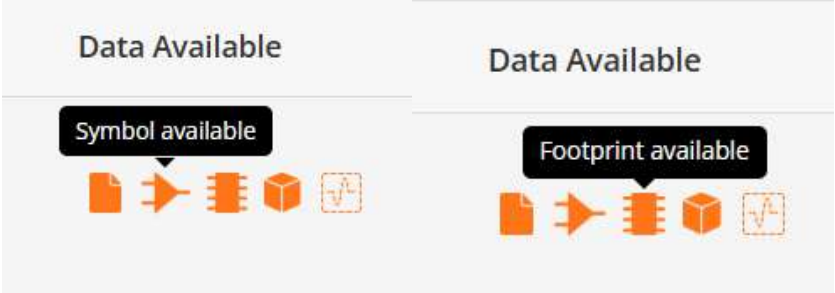
2. From the top-right corner of the webpage, select Sign Up to create a new account, or choose Log In if an account has already been created.



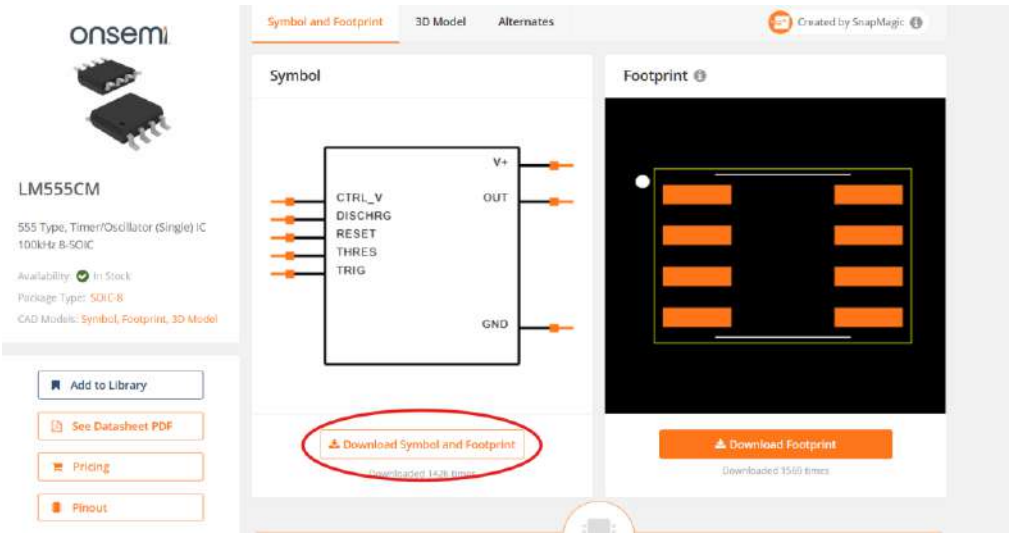
3. In the search bar, enter the name of the desired component. For example, searching for an LM555 timer will display the available matching components.



4. When selecting a component make sure, it has a symbol and footprint.

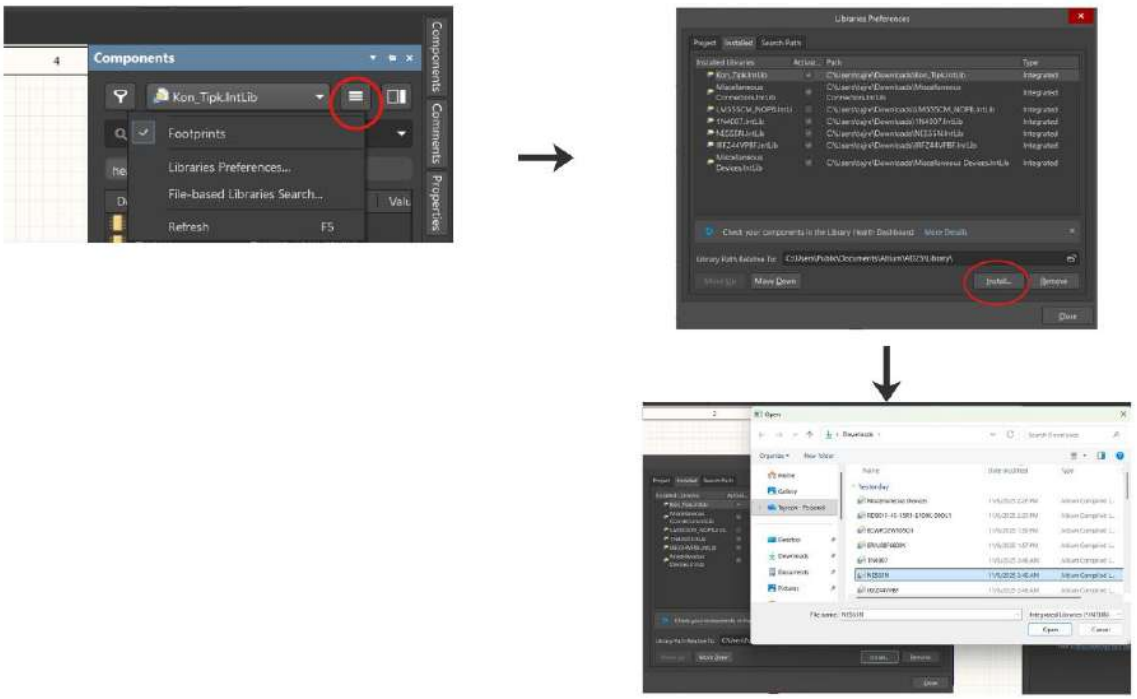


5. After selecting the desired component, click on the Download Symbol & Footprint option to download the corresponding component files to your computer.



6. To import the component in the library in Altium -

Open Components Panel > Click the ≡ (three-line menu) > Library Preferences. Then Click Add > browse to the component file. Finally Click close.



Finally, you can now use the component from the library to build your schematic.

BRAC University
Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-04: Schematic Design and Component Integration Using Altium Designer

1. Objective:

By the end of this, students will be able to:

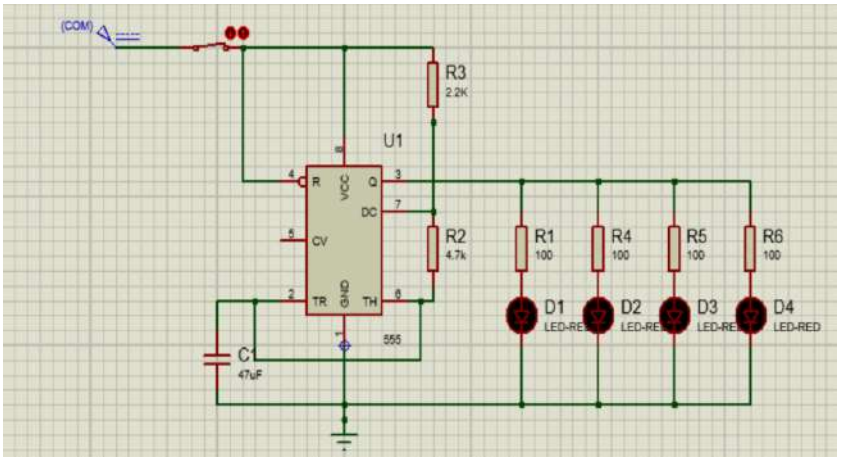
- Use Altium to create and wire a schematic circuit.
- Add and manage component libraries and footprints.
- Perform manual and automatic annotation of schematic components.
- Transfer the schematic to a PCB layout for further design.

2. Theoretical Background:

Designing a schematic is the first step in PCB development. It acts as the digital blueprint of the circuit, showing how each component connects and interacts. In Altium Designer, schematic design involves more than just drawing lines between symbols, it is about organizing, annotating and ensuring electrical accuracy before moving into physical layout. In this module, students will design an LED flasher circuit using the LM555 timer in astable mode. The schematic contains all necessary components such as resistors, capacitors, LEDs and the LM555 IC and defines how they are connected to achieve the desired blinking pattern.

An LED flasher is a simple and widely used circuit that causes an LED to blink on and off continuously. It's commonly seen in toys, indicators, and safety systems. We'll design the following circuit below using the LM555 timer IC in astable mode, where it self-oscillates without external input.

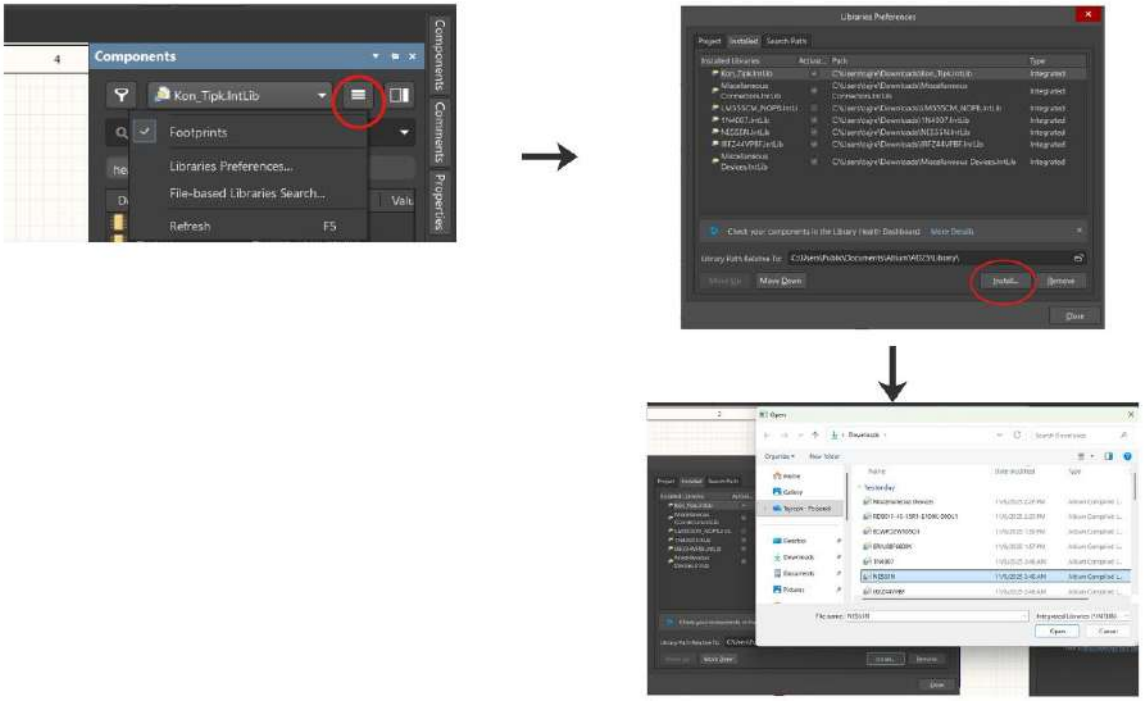
3. Circuit diagram:



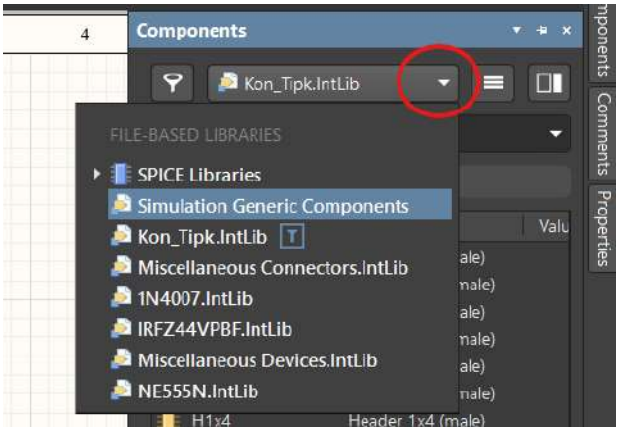
4. Procedure:

1. The first step to build the circuit schematic is to import the components from the library. We will download the required libraries. To Add Custom Libraries in Altium -

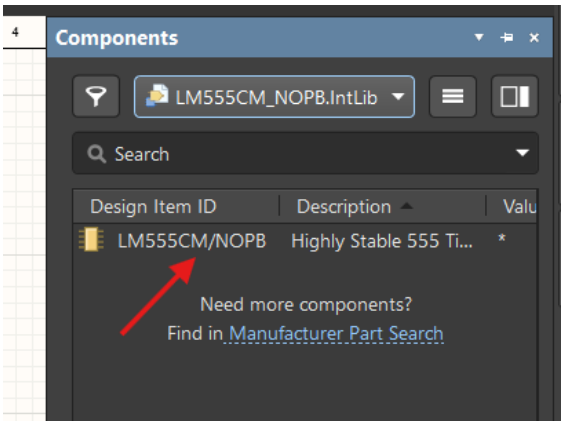
Open Components Panel > Click the ≡ (three-line menu) > Library Preferences. Then Click Add > browse to .IntLib or .SchLib files. Finally Click close.

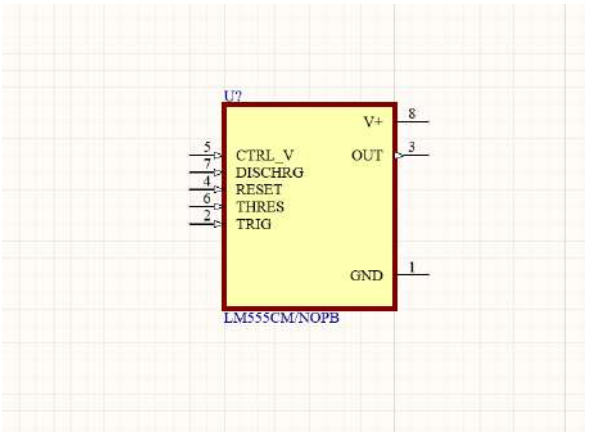


You can access the uploaded library in the following way:

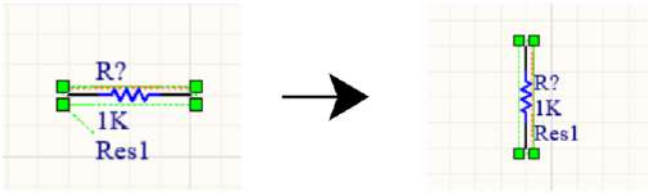


2. Select the library you want to use and double click on the component to select, add on the schematic doc by left click of mouse.

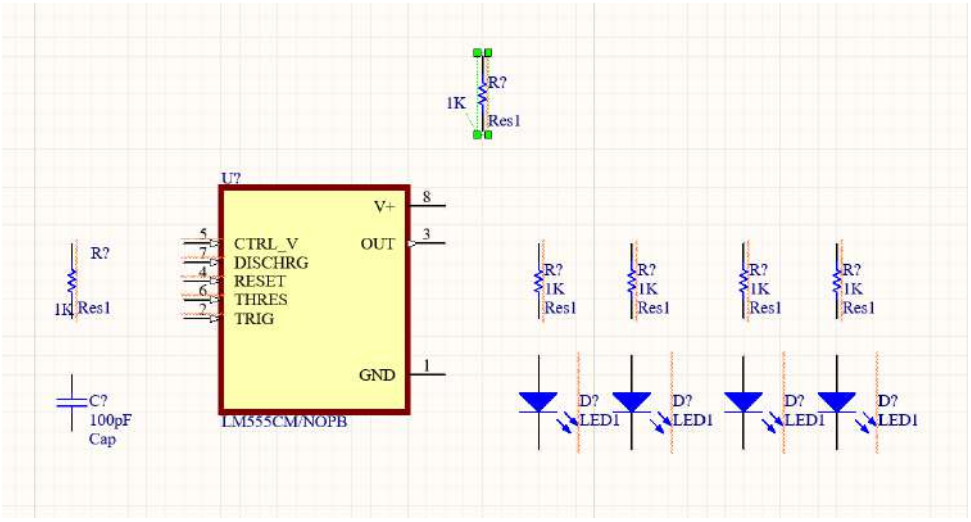




3. To rotate a component, select the component and then press the spacebar



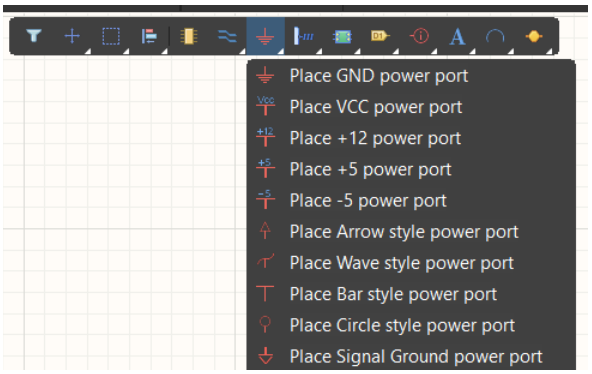
4. Then we will add all the components required in the circuit from the library



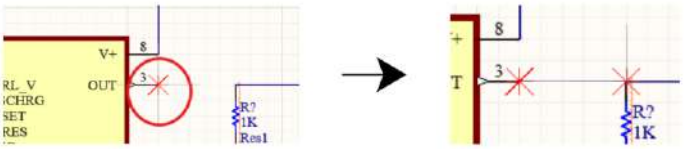
5. Then we will connect the component according to the circuit diagram by selecting wire from the panel above



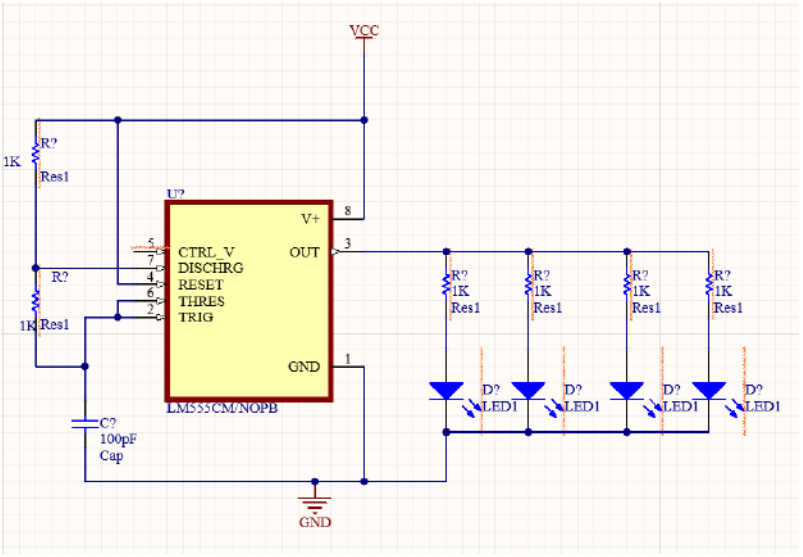
You can also press **ctrl+W** to access the wire feature easily.
The ground and Vcc can be accessed from the same panel in the following way:



6. Follow the circuit diagram and place each component in the schematic. To create connections, click on the starting node with the left mouse button and drag the wire to the destination point.



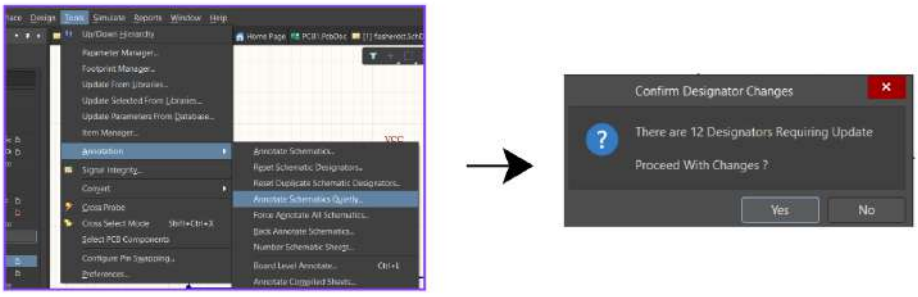
After completing the circuit connection, it will look like this:



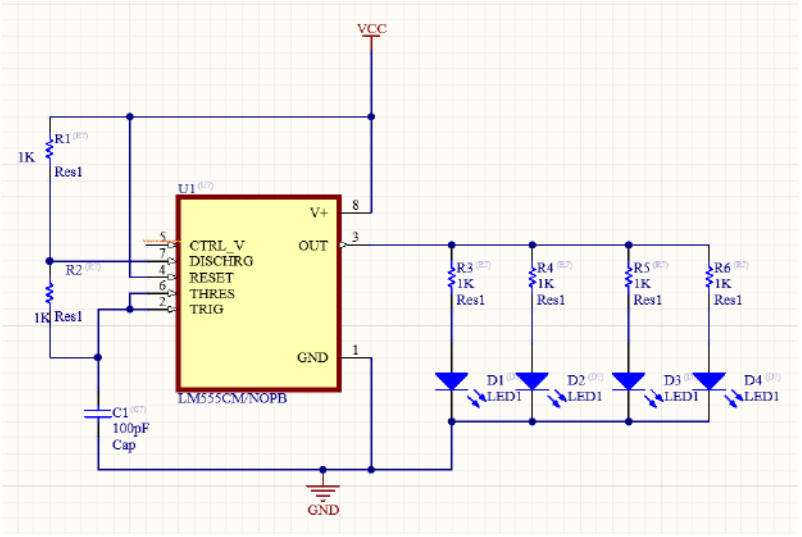
7. After completing the circuit, each component must be given a unique identifier. We can manually do that component by component but the easiest way to do all at once is to go to:

Tools → Annotation → Annotate Schematics Quietly

Click "**Yes**" to confirm the changes.

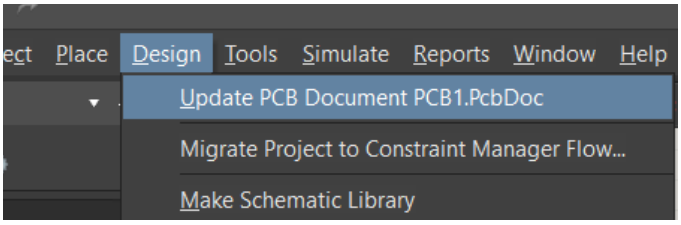


Altium will then assign unique names to all components automatically.

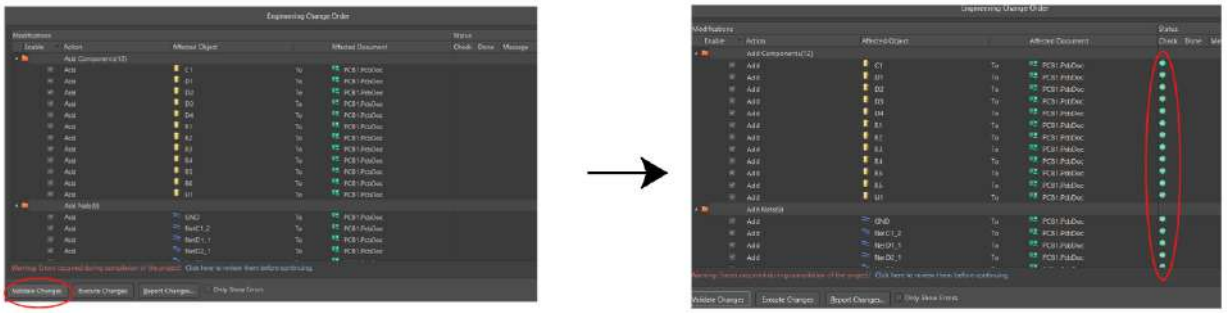


8. Finally, we need to transfer the completed schematic to the PCB layout. Go to

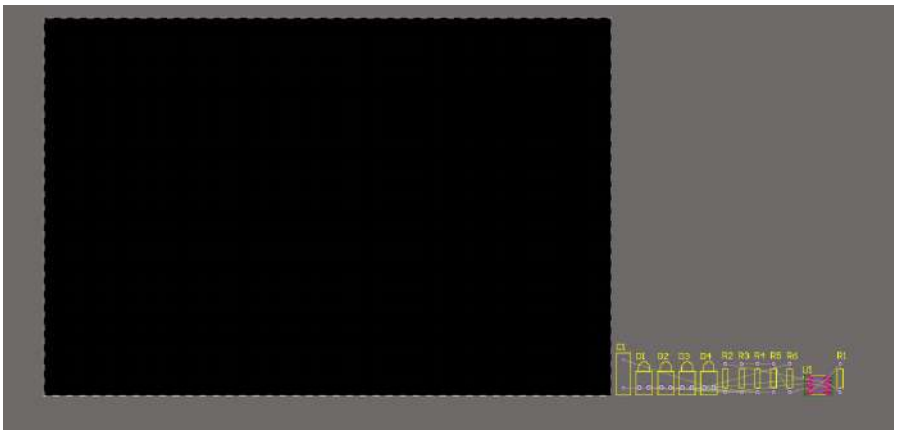
Design → Update PCB Document



Then click validate and make sure all the check status on the right are **green**.



9. Finally click Execute Changes to apply the updates. After this, all components from the schematic will appear in the PCB layout window, ready for placement and routing.



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Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-05: Board Shape Configuration, component placement and Routing Optimization for PCB Layout

1. Objective:

By the end of this, students will be able to:

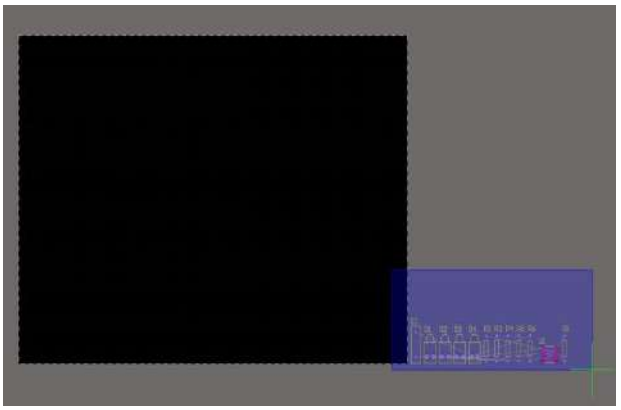
- Define the shape and size of a PCB according to a design requirement.
- Adjust the board outline using Altium's PCB editing tools.
- Place components properly before routing.
- Route signal connections using suitable trace widths and paths.

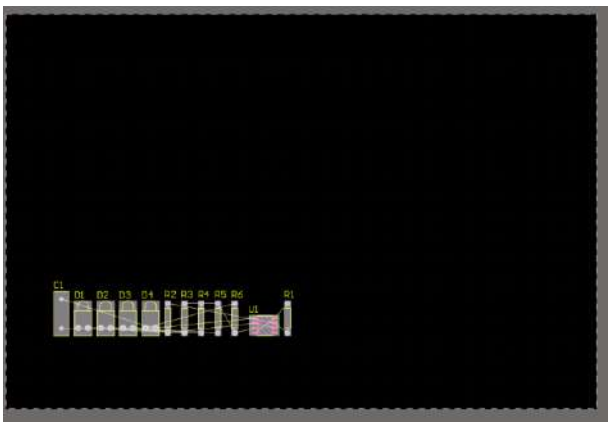
2. Theoretical Background:

Once the schematic is transferred to the PCB editor, the next steps are shaping the board and routing the connections. The board shape is important because it determines how the final circuit will fit inside an enclosure, device or prototype. After the board shape is set, the components must be arranged. Good placement makes routing easier and improves overall performance.
Routing is the process of drawing copper traces that connect all the component pins. These traces act as the "wires" on the PCB. Proper routing ensures signals flow smoothly without noise or interference. It also helps avoid issues like crossing traces, long loops or unnecessary bends.

3. Steps for shaping the board:

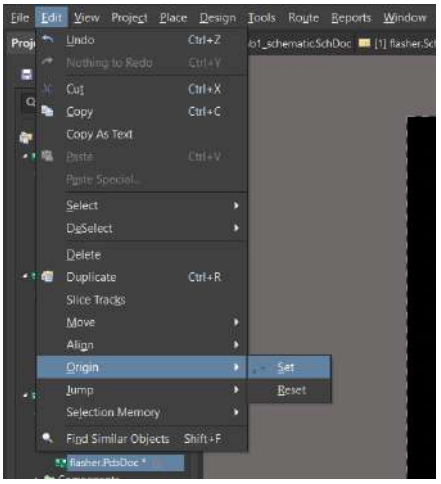
1. Select all the components and drag them to the black board.



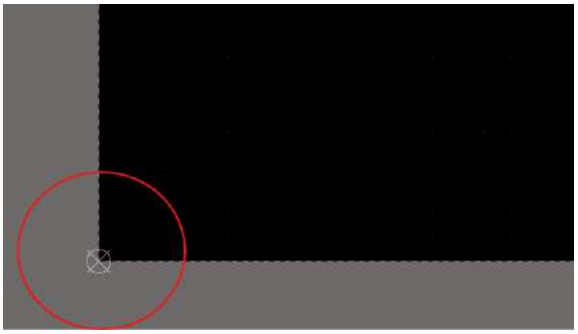


2. After that we need to shape the board. The very first step to do that is to set the origin to a convenient location so that it's easier to measure. To set the origin go to

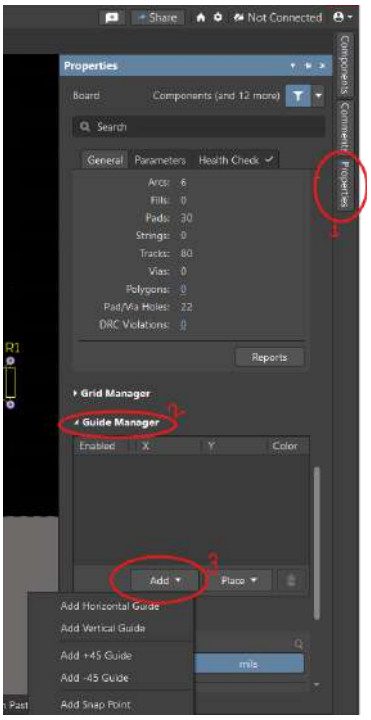
Edit → Origin → Set Origin

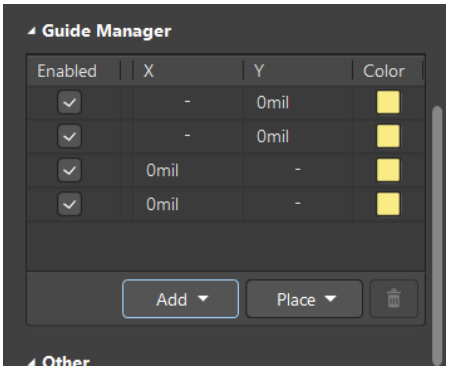


Click on any point where you want the new origin to be located (usually the lower-left corner).

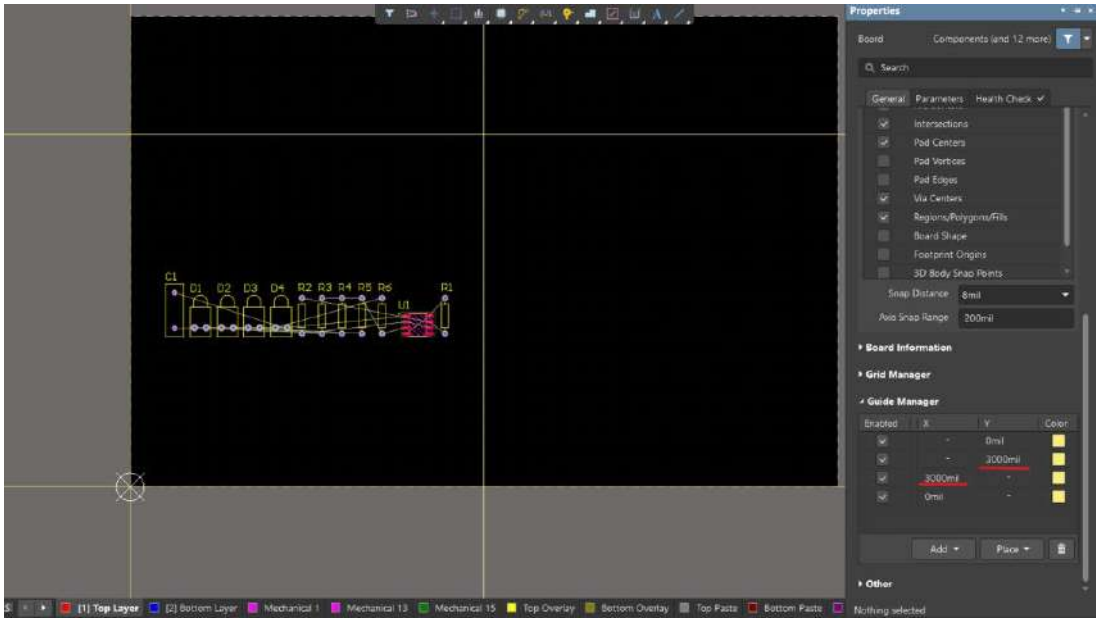


3. Now define the actual board outline. To do this, open the Guide Manager from the Properties panel. Add a pair of horizontal guides and a pair of vertical guides, These will help you mark the boundaries and draw the board shape accurately.

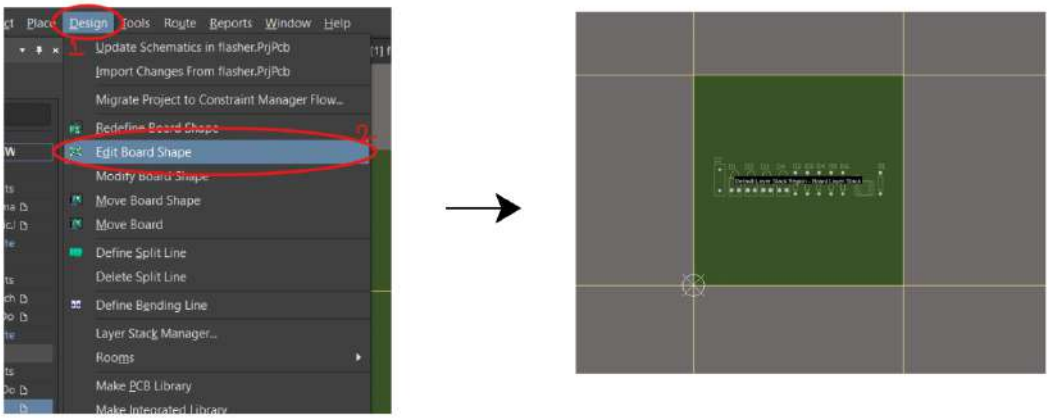




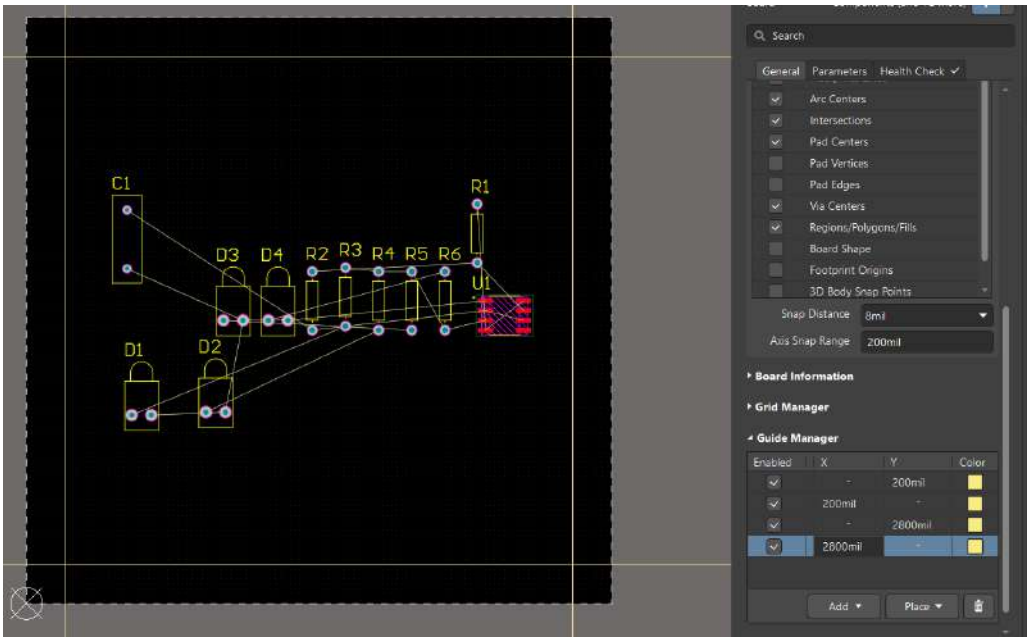
4. After adding the guide lines, use one pair of horizontal and vertical guides as your reference point and the other pair to set the required dimensions. For example, if the board size should be 3000 × 3000 mil, enter these values into one pair of guide managers. Altium will then display guide lines across the PCB workspace, helping you outline the board accurately.



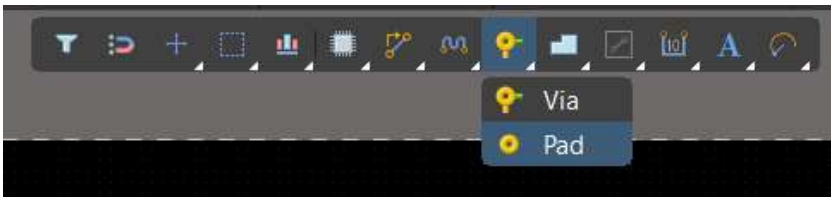
5. To crop the board to the required size, first press "1" to switch to Board Planning Mode. After that, go to Design → Edit Board Shape and adjust the outline so it aligns with the guide lines you created.



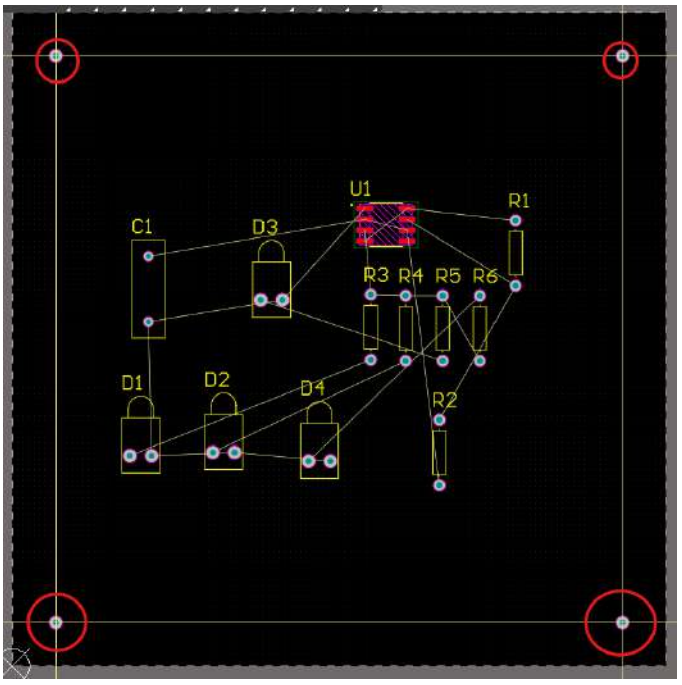
6. We will add mounting holes at the four corners of the board so screws can be attached later. To place them accurately, use the Guide Manager from properties to position guide lines. Insert horizontal and vertical guides 200 mil away from each edge side. The intersections will mark the exact spots where the mounting holes should be placed.



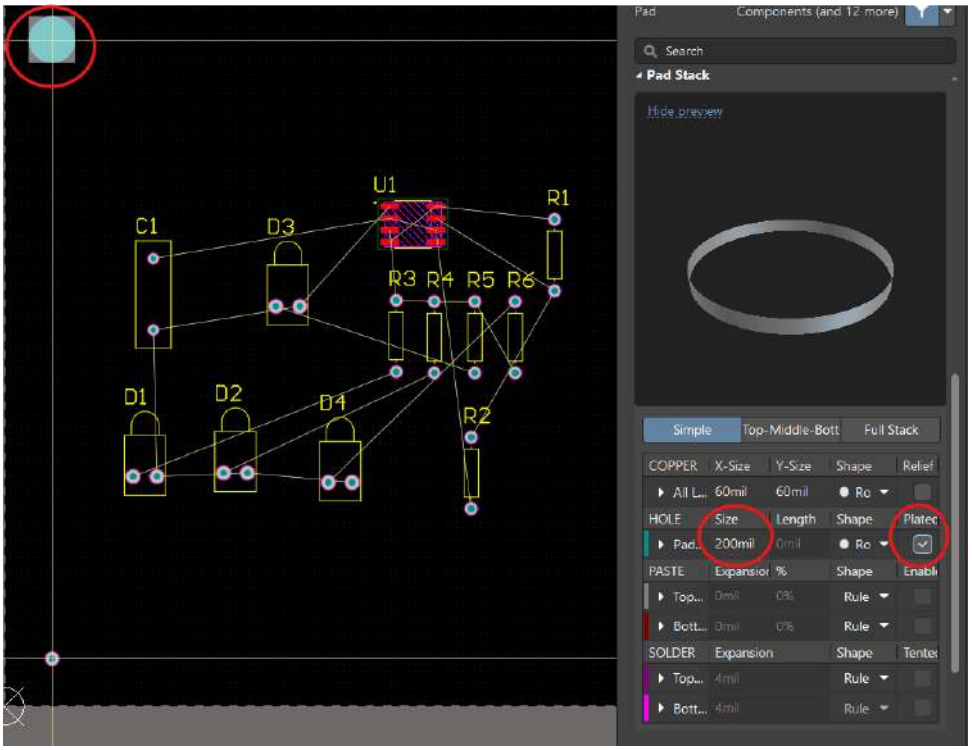
From the toolbar above select the pad option as shown in the figure below



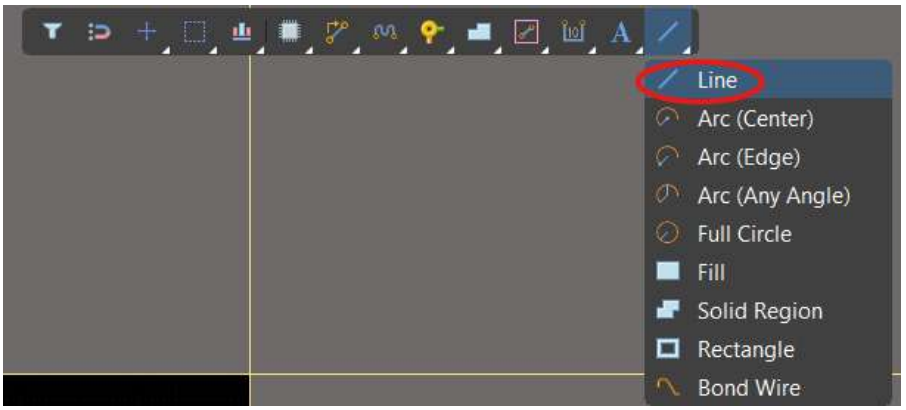
Then zoom and add the pads at the four intersections on the board.



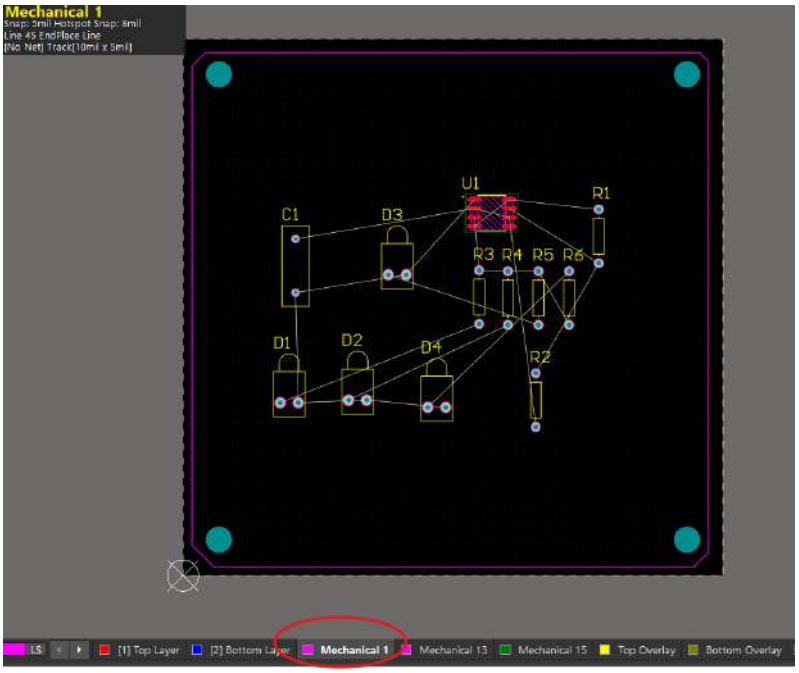
Next, set the hole diameter according to the requirement and remove the metal lining inside the hole. To do this, double-click on the mounting hole. A panel will appear on the right side of the screen. Scroll down to find the hole size settings and enter the desired diameter (200 mil in this case). To make the hole non-plated (no metal lining), simply uncheck the ‘Plated’ option. Repeat the process for the other 3 holes.



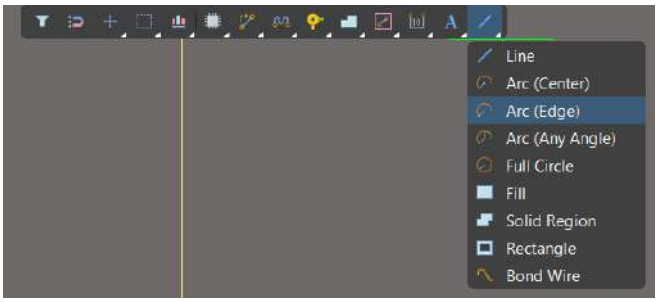
7. Now we will have to define the board shape to round the edges. To do that first press "2" to switch the display to the 2D Layout Mode. Then from the layer options at the bottom, select the "Mechanical Layer". Now from the top bar select 'line' and draw the outline of the board using it.



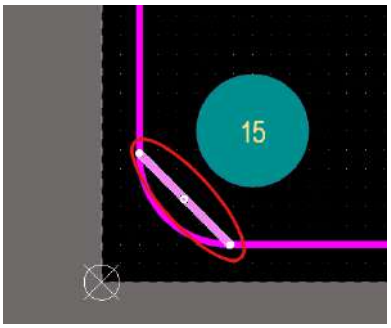
Double-click at the end of the outline to close the shape completely and ensure there are no gaps. Then right-click to exit the Line tool.Then right click to remove the line option. The board will look like the following after that:



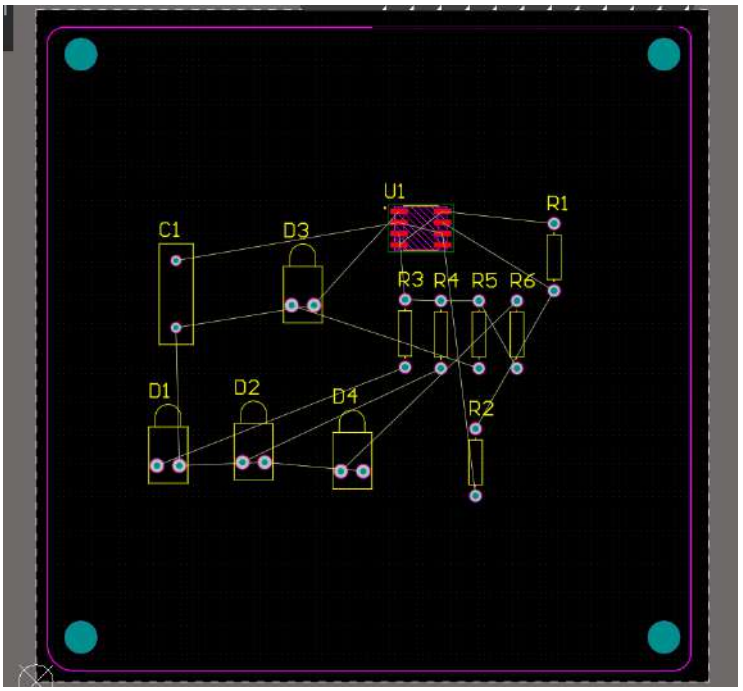
To smooth or round the corners, go to the top toolbar and select the ‘Arc (Edge)’ tool. Then click on the corner points where you want the rounded edge to appear. It’s best to select the points in an anti-clockwise direction for a cleaner arc.



After rounding the edges, delete the excess lines which were drawn previously

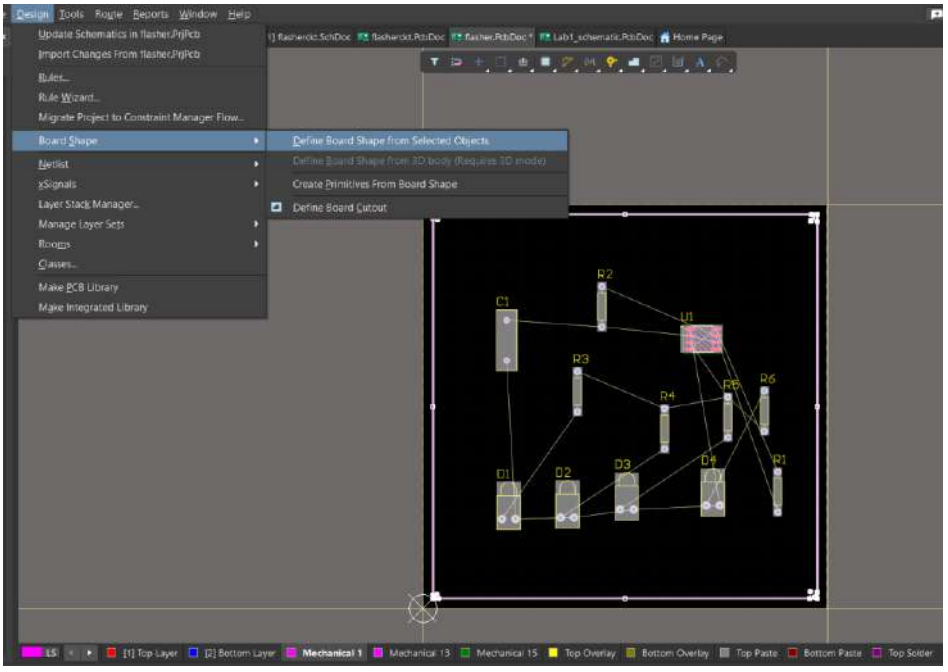


Then it will look like this:



8. Next, convert your drawn outline into the actual board shape. Left-click on the outline you created and make sure it is fully selected.
   Then go to:

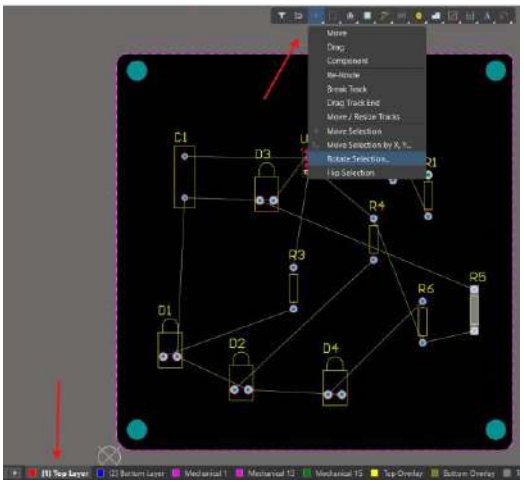
   Design → Board Shape → Define from Selected Objects



Altium will replace the previous outline with the new shape you just drew, including rounded edges.

4. Steps for routing:

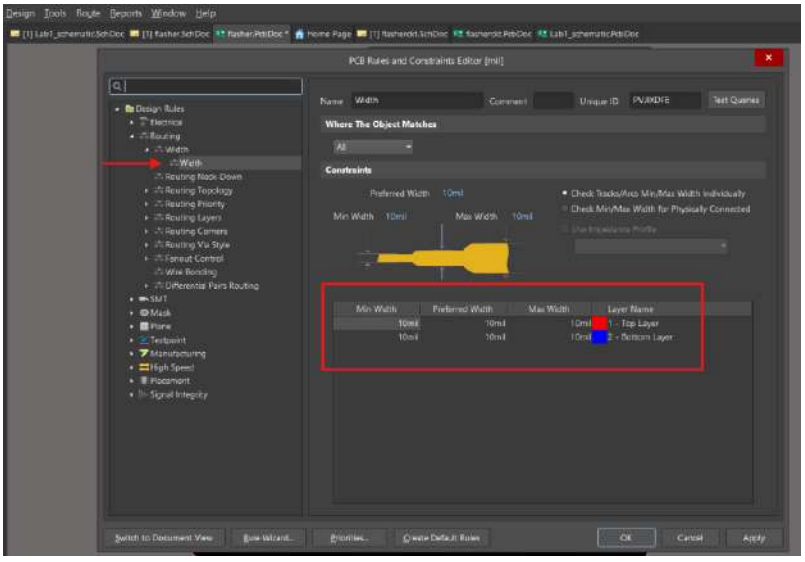
1. To begin routing, first select the ‘Top Layer’ from the layer options at the bottom. After that, arrange the components on the board in a way that makes the connections easy and logical. Good placement will make routing simpler. You can also rotate/flip the components if required. Here’s a smoother version: To quickly rotate a component, select it with a left-click and simply press the Spacebar.



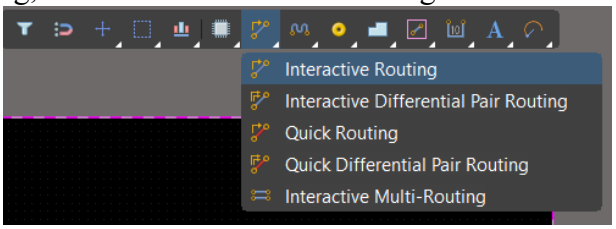
2. Next, set the required routing width. For example, if the width requirement is 10 mil, you need to update the routing rules so Altium uses that width automatically. To set the width, go to:

Design → Rules → Design Rules → routing → Width

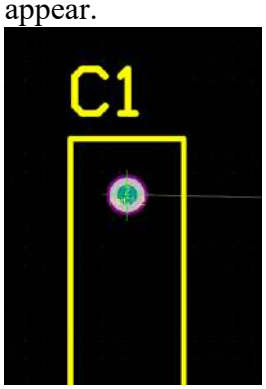
Double-click on Width to open the settings window. Here you can enter the minimum, maximum, and preferred trace width (e.g., 10 mil). Once this is set, all new traces will follow the correct width.

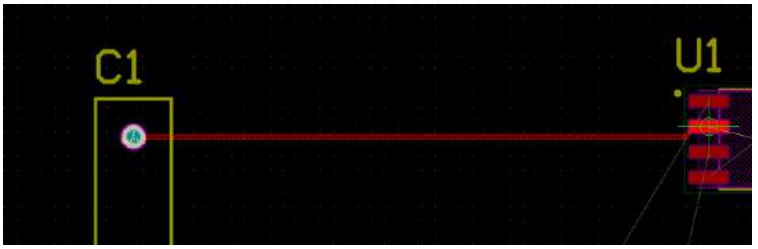


3. To begin routing, select the “Interactive Routing” tool from the top toolbar.

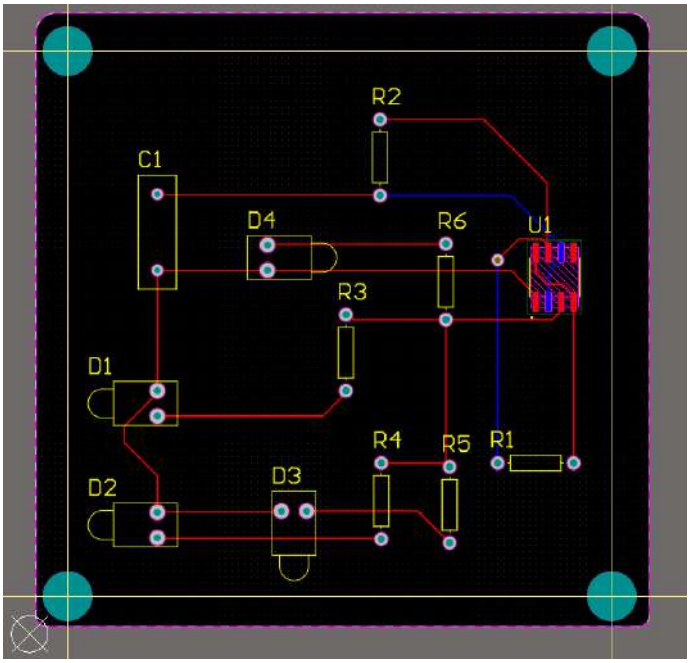


Then **place** it on a pad of the component where you want to start. Altium will highlight it with a **green circle**. **Left-click** to begin the routing, and follow the guideline (airwire) to complete the connection by clicking on the destination pad, where another **green circle** will appear.





In the same way, complete the routing of all the components. After completing routing the board will look like this:



BRAC University
Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-06: Implementation of Copper Planes, Design Finalization and Manufacturing Output Generation

1. Objective:

By the end of this, students will be able to:

- Understand the purpose of polygon pours in PCB design.
- Apply a polygon pour to create a ground or power plane.
- Rebuild or repour polygon.
- Write on silkscreen
- Perform final checks before generating manufacturing files.

2. Theoretical Background:

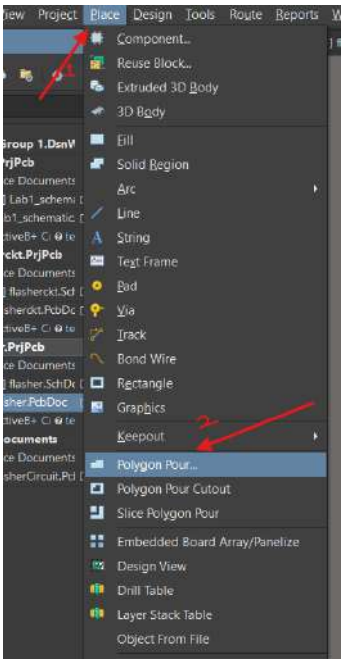
A polygon pour is a large copper-filled area placed on a PCB, usually used to create a ground plane (GND) or a power plane. It helps improve signal quality, reduce noise and provide better current return paths throughout the board. Polygon pours also help with heat spreading and make routing easier by reducing the number of individual ground traces.

In Altium Designer, polygon pours cover selected areas of a board and automatically connect to specific nets (like GND). After placing one, the software clears space around other traces according to design rules, ensuring the board remains manufacturable. Once the polygon pour is added, the design is almost complete. The final steps include checking for errors, cleaning up the layout, reviewing all connections and preparing the board for fabrication. These final checks ensure that the PCB is reliable and ready for manufacturing without issues.

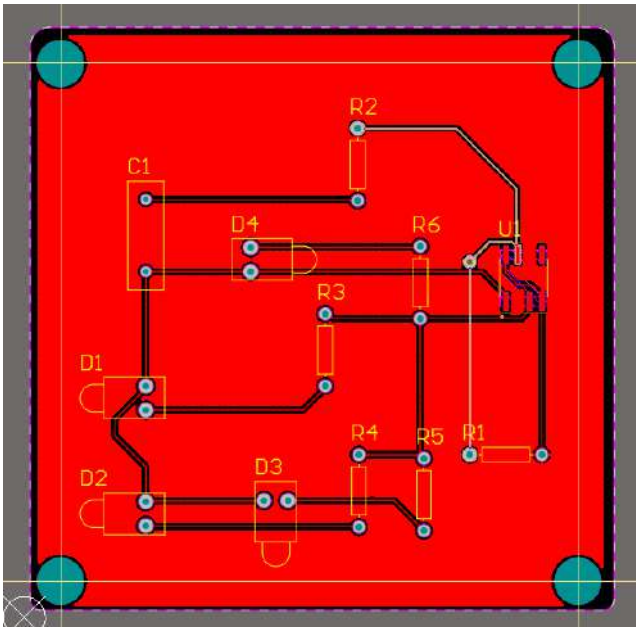
3. Steps for polygon pour:

1. For this design, we are working with two layers: Top Layer and Bottom Layer. In this case, the Top Layer will be used for the VCC plane, and the Bottom Layer will be used for GND.

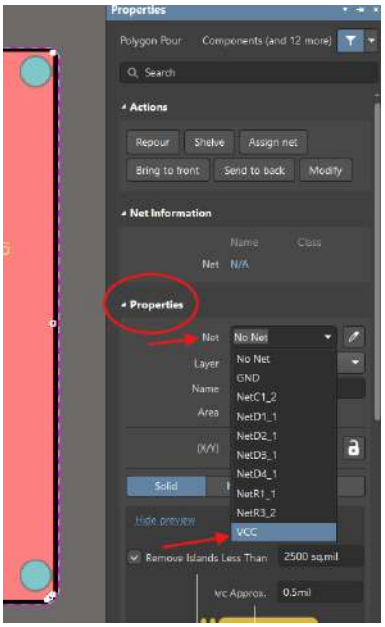
   To begin, select the Top Layer from the layer options at the bottom of the screen. Next, go to **Place → Polygon Pour.**



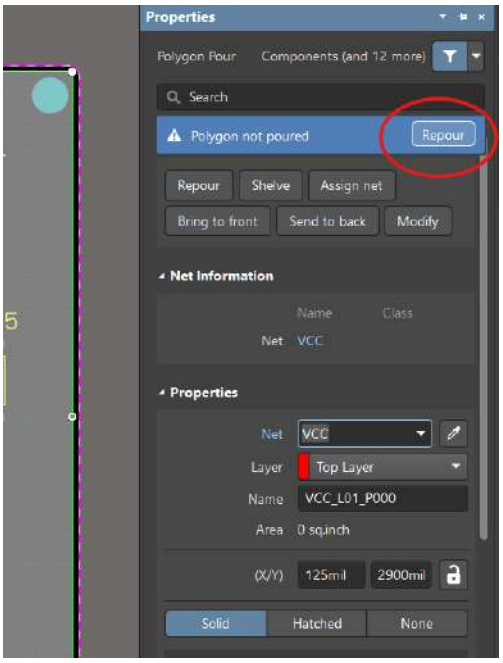
Now draw the area where you want the copper to be poured, usually the entire board outline. Make sure you start and finish at the same point so the polygon closes properly without any gaps. After that board will look like this:



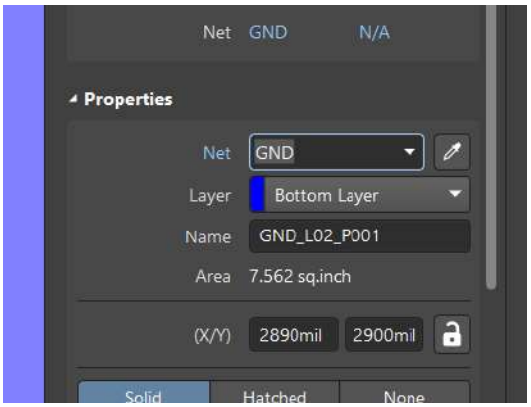
2. **Double-click** on the polygon pour to assign the correct net. A panel will open on the right side of the screen. Under the Properties section, change the Net to VCC.



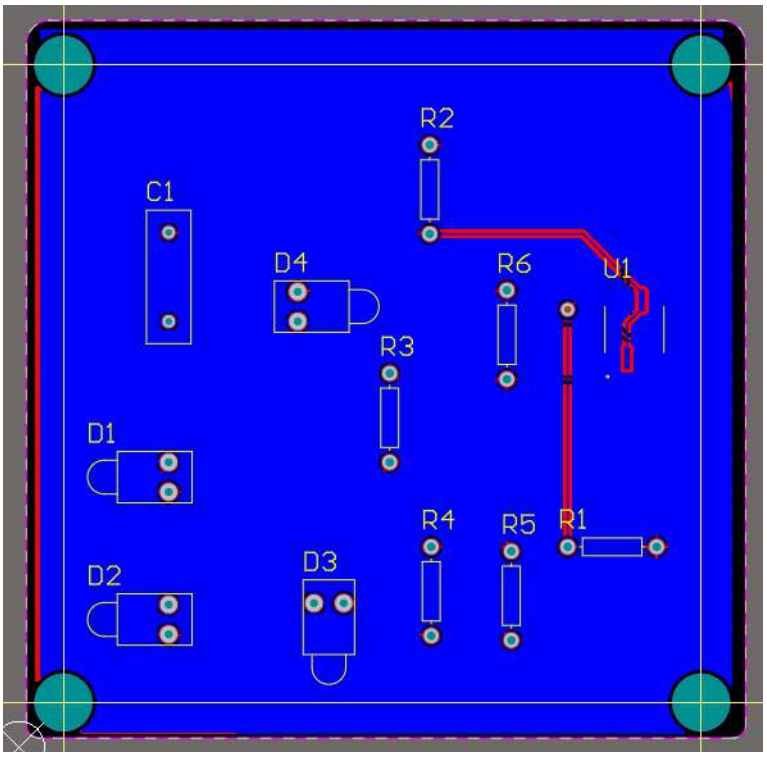
After selecting the net, click **Repour** to update the polygon so it fills the board using the VCC connection.



3. Now go to the bottom layer and repeat the process. This time **change the net to GND click the Repour** option that appears to update the polygon with the correct connection.



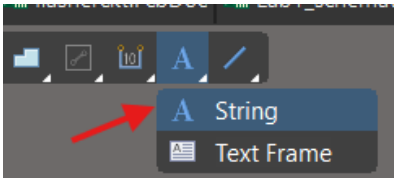
And it will look like the board below:



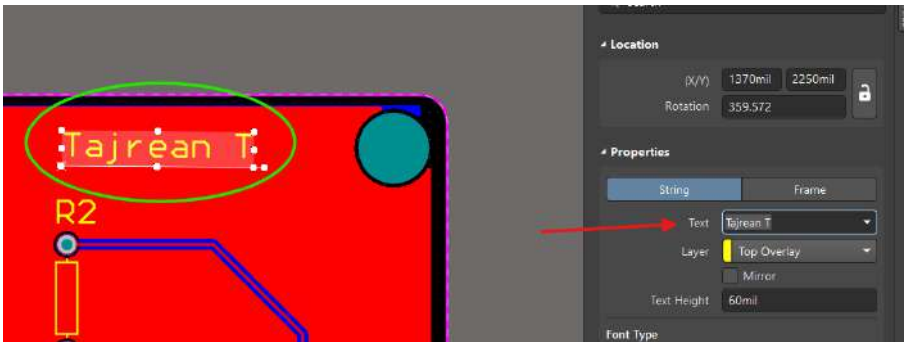
4. Steps for Adding Text on the PCB:

1. To place labels, names or any text on the PCB, first select the Top Overlay layer from the layer options at the bottom. This ensures the text appears on the silkscreen of the manufactured board.

Now select String from the top toolbar. Click anywhere on the board where you want the text to appear.



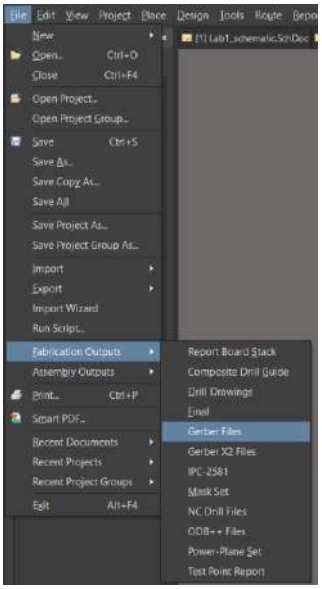
A text box will be placed, double-click on it to edit the content from the panel on the right. You can also adjust the font size, orientation, and style as needed.



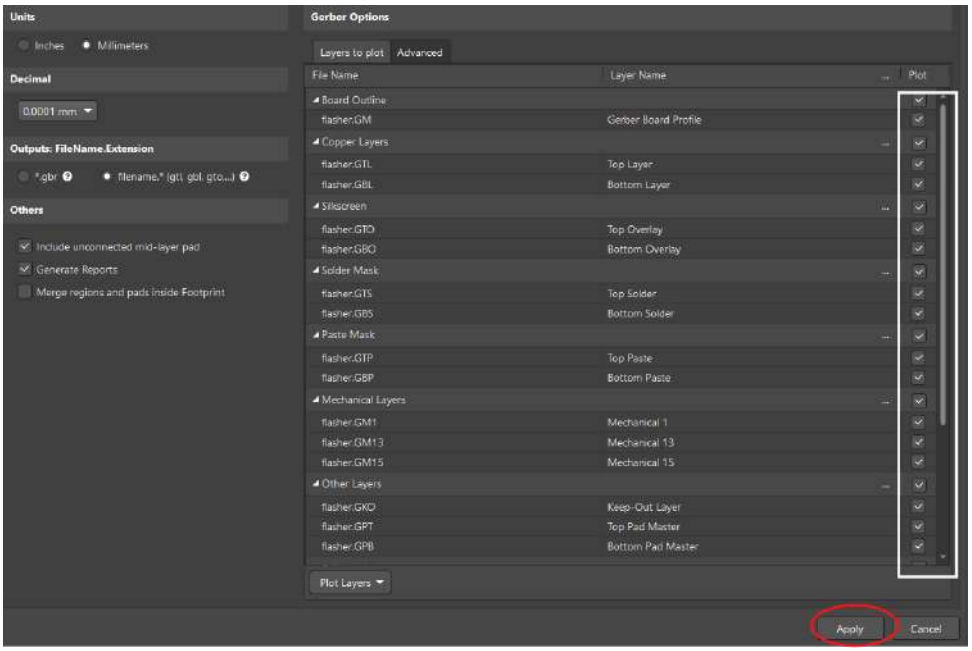
5. Steps for Generating Gerber Files:

1. To prepare the board for fabrication, we first generate the Gerber files. Go to:

File → Fabrication Outputs → Gerber Files



2. After checking all required layers, click apply.

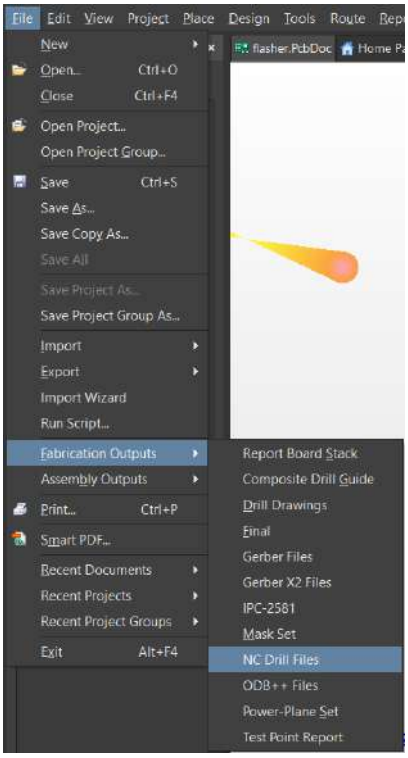


And then you will have your gerber file ready for fabrication

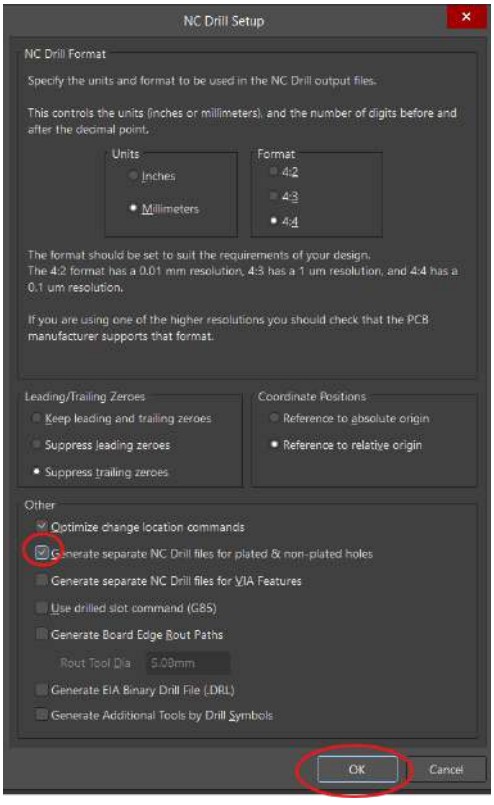
6. Steps for Generating NC drill:

1. To prepare the board for fabrication, we first generate the Gerber files. Go to:

File → Fabrication Outputs → NC drill files



2. After completing all the requirements, click OKAY.



BRAC University
Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-07: Custom Library Making

1. Objective:

By the end of this, students will be able to:

Differentiate between schematic symbols and PCB footprints.
Understand package dimensions using component datasheets.
Design custom schematic symbols and PCB footprints.
Associate symbols with footprints for PCB design.
Verify library accuracy using Design Rule Check (DRC) and visual inspection.

2. Theoretical Background:

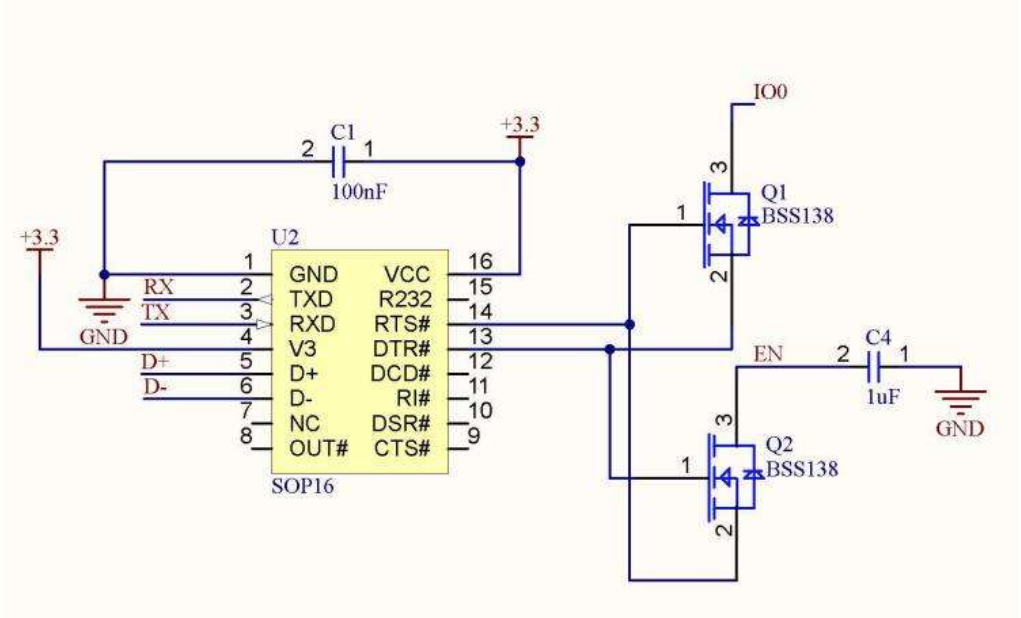
Electronic design automation (ECAD) software uses component libraries to model electronic devices in the schematic design and PCB layout. A library is a database that contains graphical symbols, PCB footprints, electrical parameters, package information and metadata, all of which are needed during the design process. Although most PCB design software provides extensive built-in libraries, there are some components that are unavailable or require manufacturer-specific footprints. Under these circumstances, designers' own libraries have to be developed to ensure that the schematic will be drawn properly, the PCB will be assembled correctly and the electrical performance will be reliable.

Creating a custom library consists of these steps: converting data from the component's datasheet into a schematic symbol and then a PCB footprint. This allows for the PCB to be manufactured correctly and the physical dimensions, pin assignments and electrical connections to be properly represented. A well-designed custom library helps to reduce assembly errors, design reliability and manufacturing defects.

Altium Designer organizes library data into three distinct file types, and a student must understand the purpose of each before beginning construction:

File Type	Extension	Purpose	Primary Editor
Schematic Library	.SchLib	Stores schematic symbols: graphical outline, pins, electrical type and default designator.	Schematic Library Editor (SCH)
PCB Library	.PcbLib	Stores footprints: pads, courtyard, silkscreen outline, assembly and 3D body.	PCB Library Editor (PCB)
Integrated Library	.IntLib	Compiled, locked-down combination of schematic and PCB libraries with model links resolved.	Compiled from .SchLib + .PcbLib via a Library Package (.LibPkg)

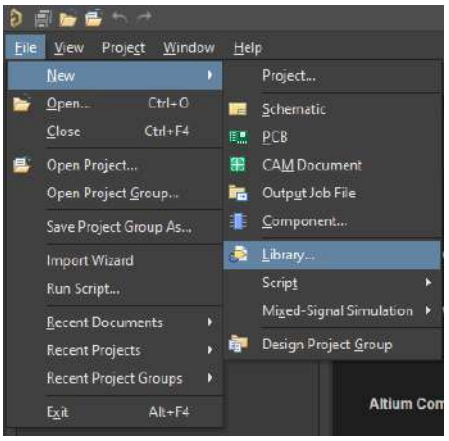
3. Circuit Diagram:



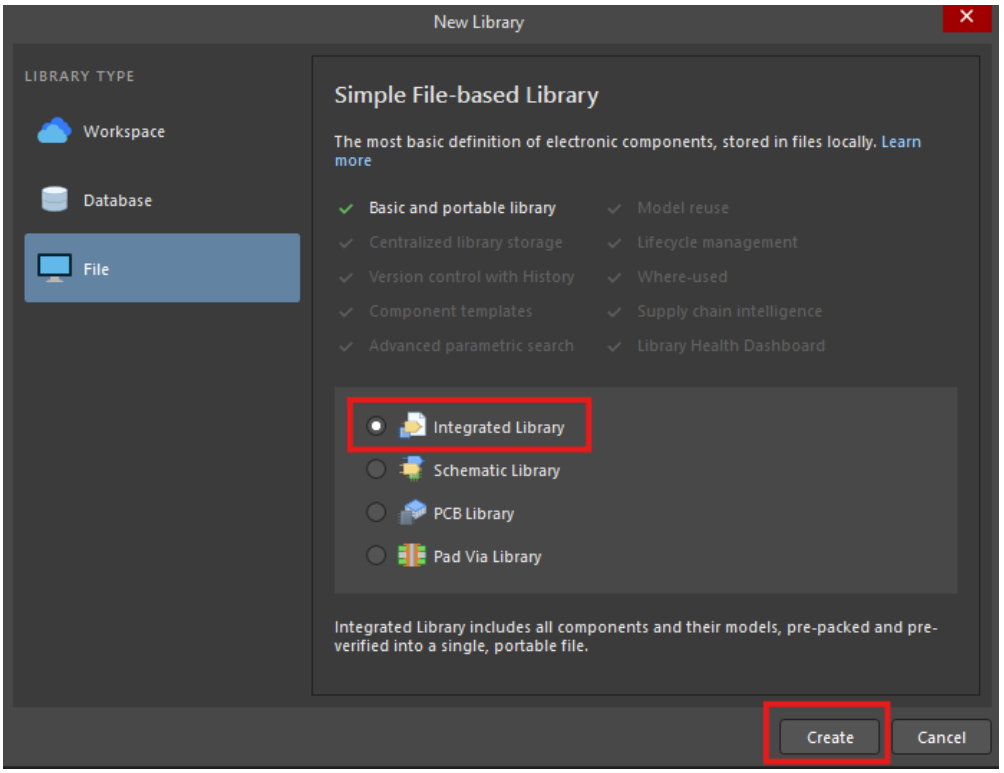
We have to build a custom library for the CH340C IC to build the circuit.

4. Procedure:

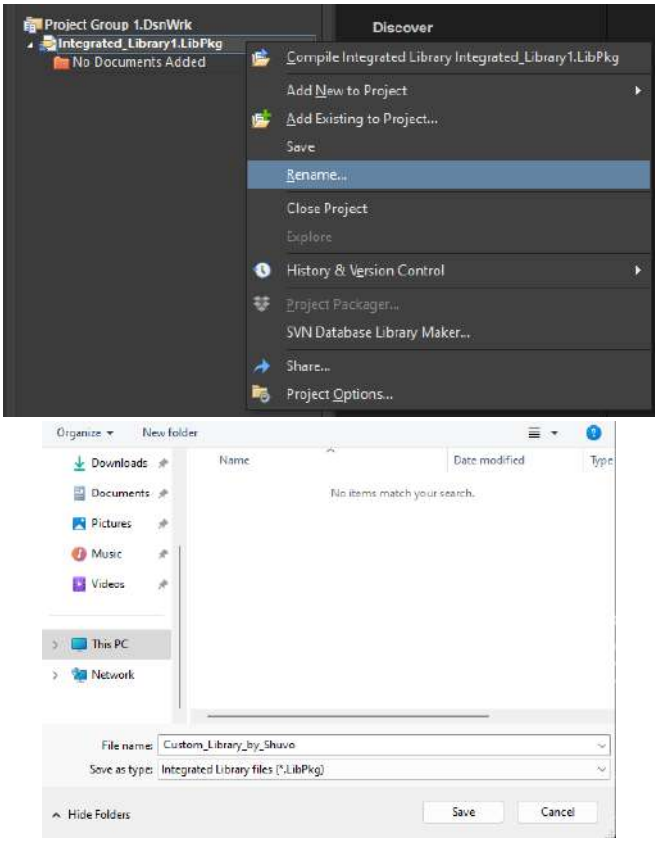
1. Click here to download the datasheet of CH340C IC.
2. Open the *Altium Designer* app.
3. Go to: File > New > Library.



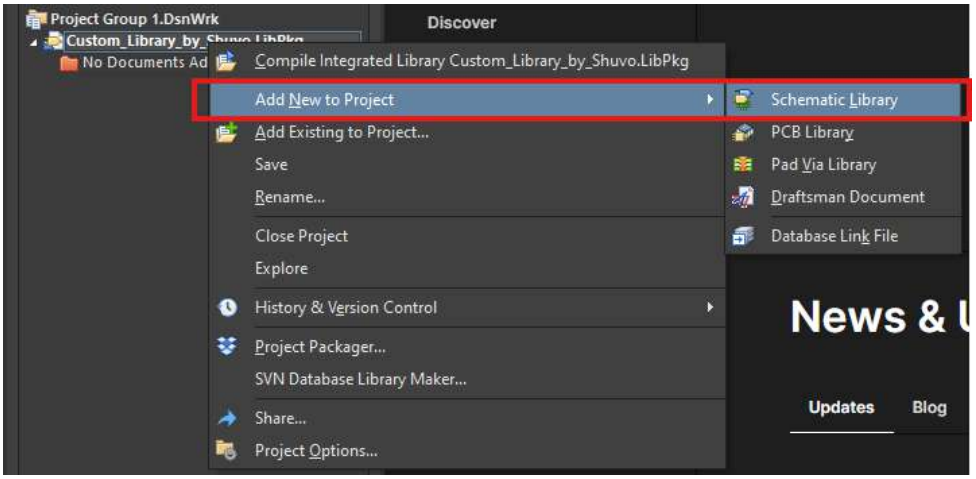
4. Select File > Integrated Library > Create.



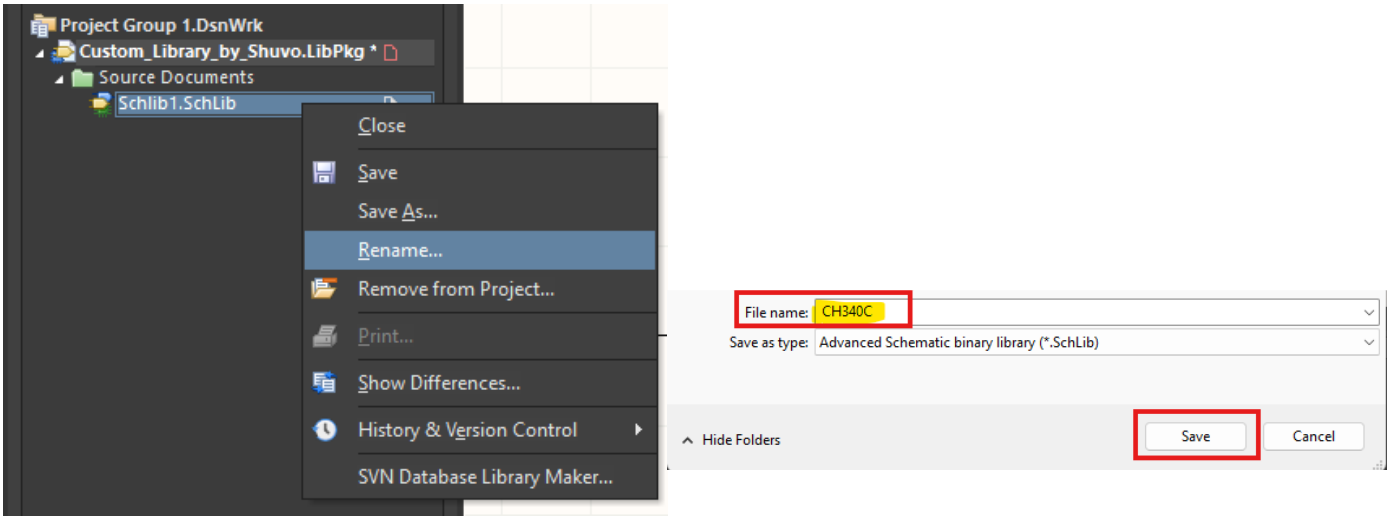
5. An integrated library file will be created. Right click on the *library name* and select Rename. Rename it ensuring the naming convention and hit *save*.



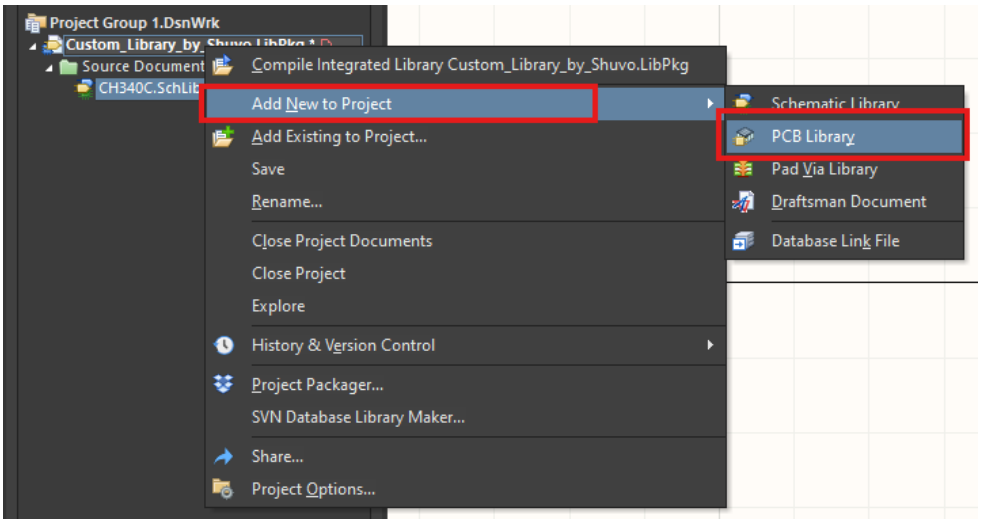
6. Right click on the library name and select *Add New to Project* > *Schematic Library*.



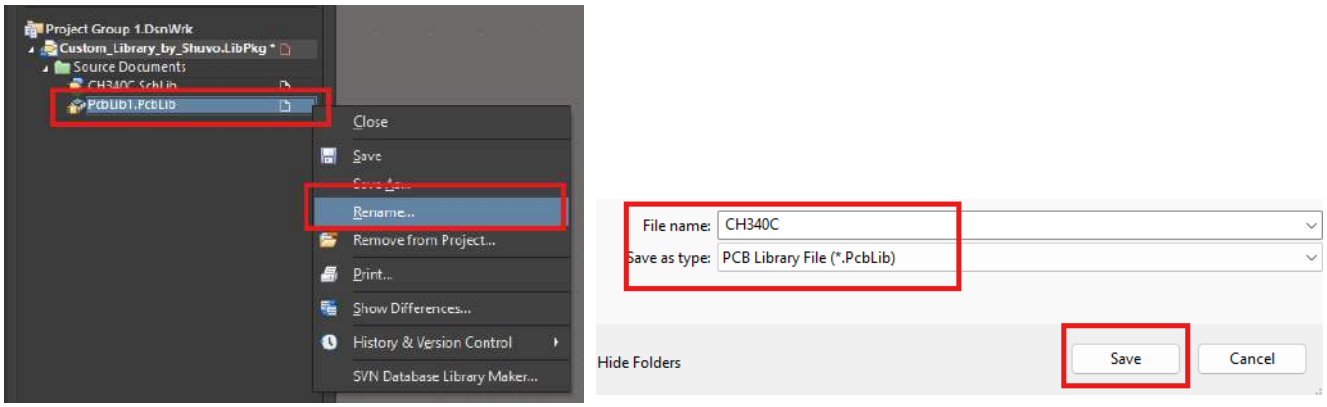
7. A new schematic window will be opened. Go to Project > Right click on the schematic file > select Rename > Rename the file according to the component name > Save.



8. Again, right-click on the library name and select *Add New to Project* > *PCB Library*.

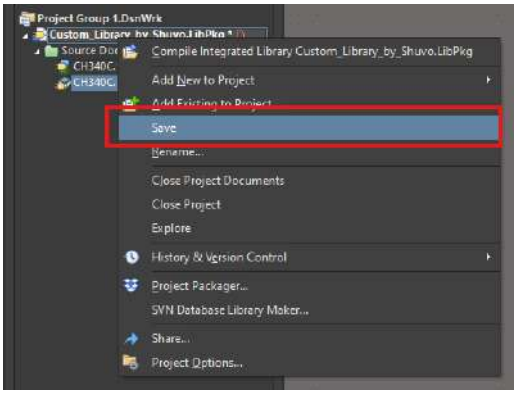


9. A new PCB window will open. Go to *Project* > Right click on the PCB file > select *Rename* > Rename the file according to the component name > *Save*.

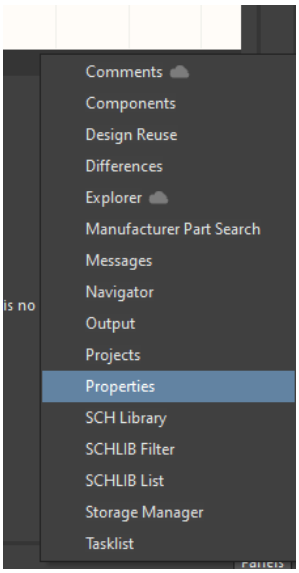


The file name for the schematic and PCB file should be the same but the file name extension should be different.

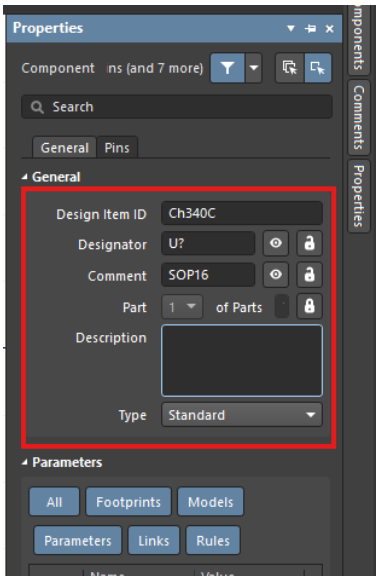
10. Go to *Project* > Right click on the *Library* file > *Save*.



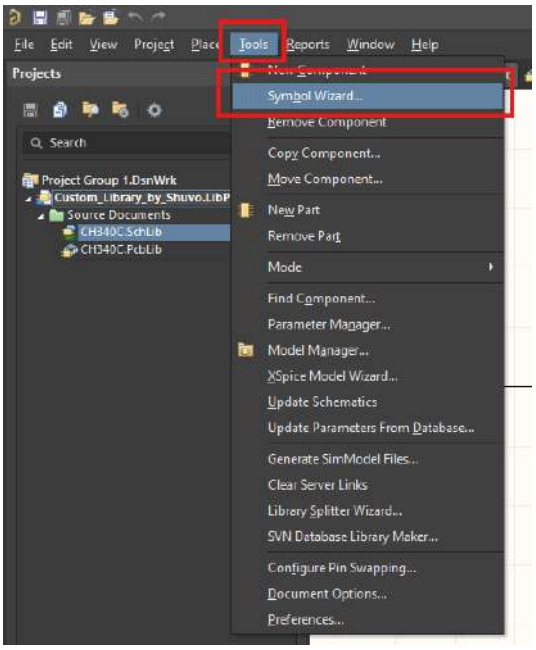
11. Open the schematic file > Click on panels (at the right bottom corner) > Select Properties >Go to the general section > Change the placeholders in the dialogue box of the general section as mentioned below.



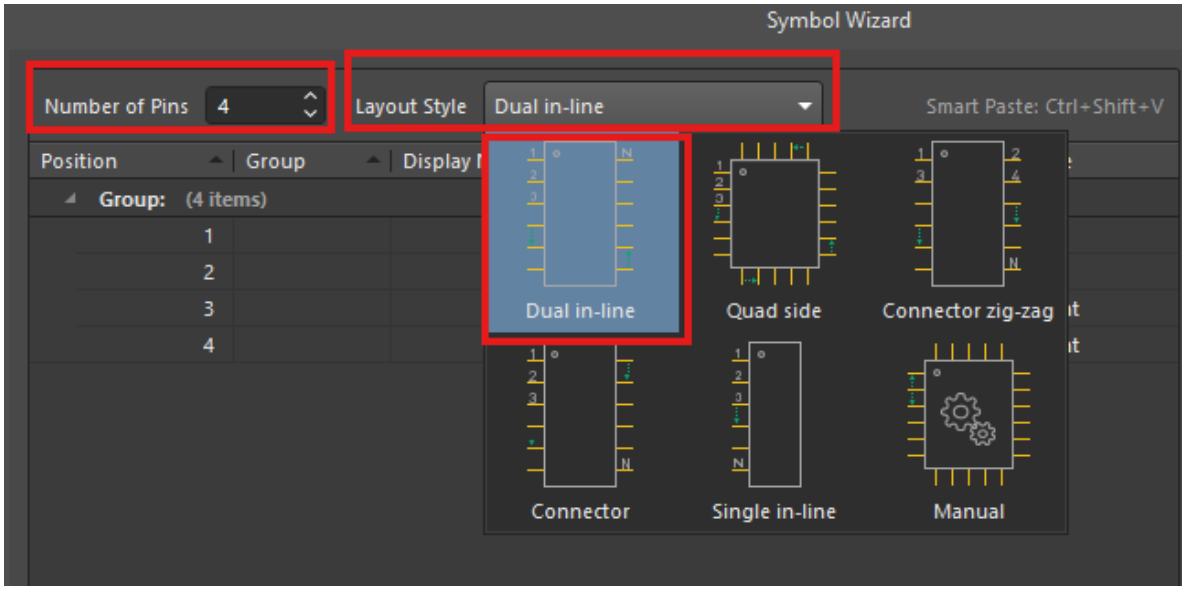
>



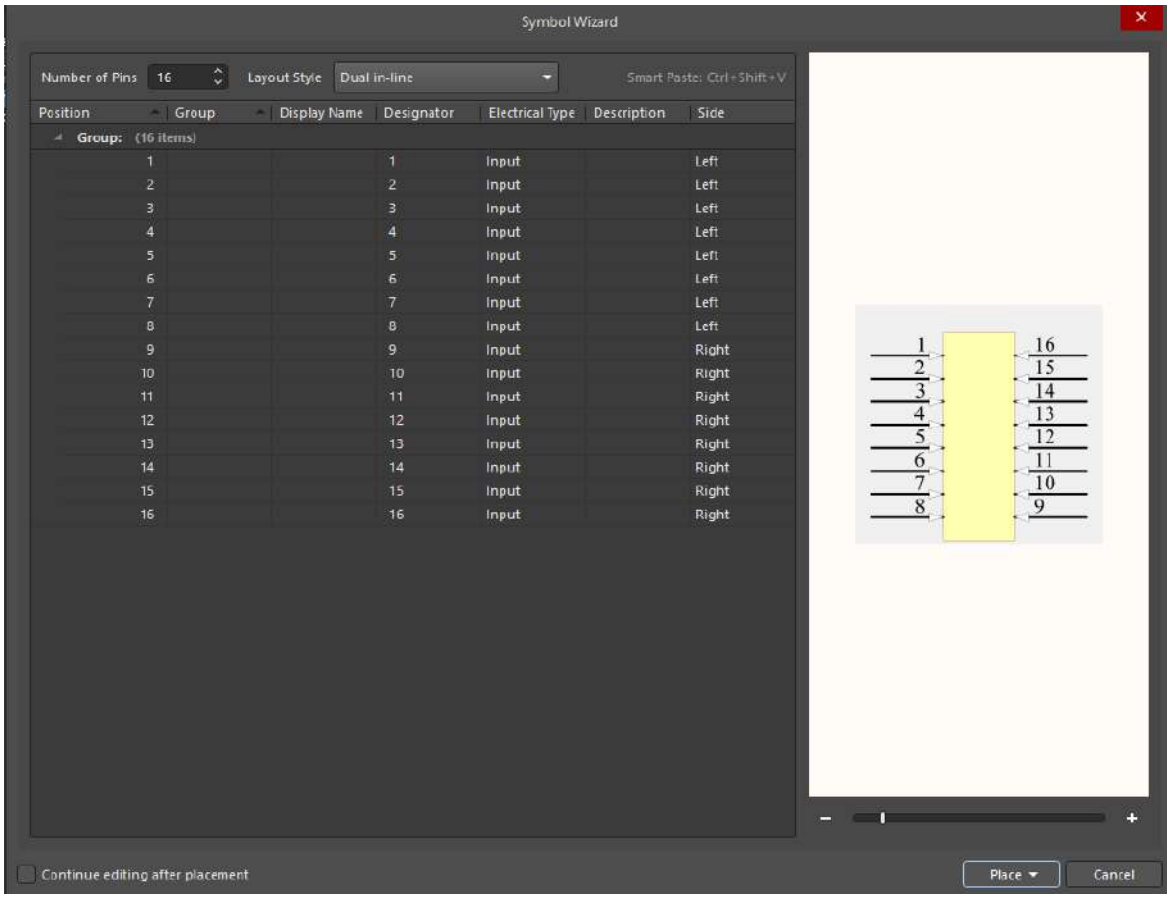
12. Go to the Tools menu from the top navigation bar > Select Symbol wizard.



13. Select Layout Style as *Dual in Line* > Enter pin number (here, 16). Click Place.



14. The window will get updated with the new input and will look like the following image:



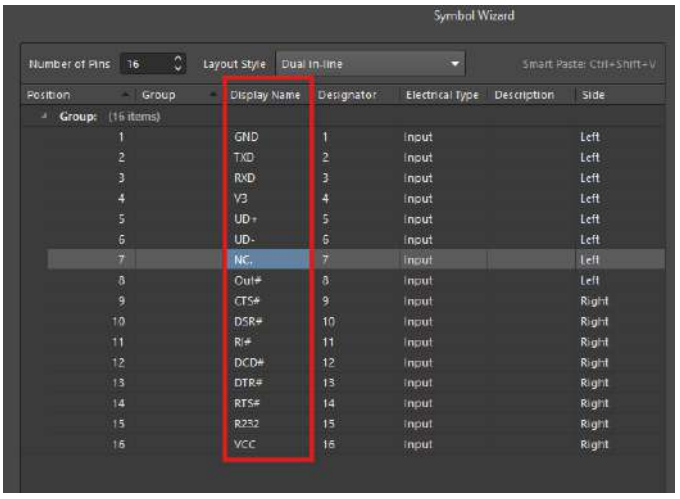
15. Change the Display name according to the *Datasheet Pinout*.



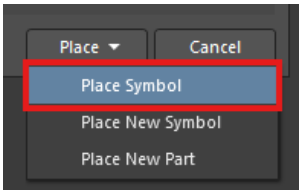
CH340 Datasheet 3

4. Pin Out

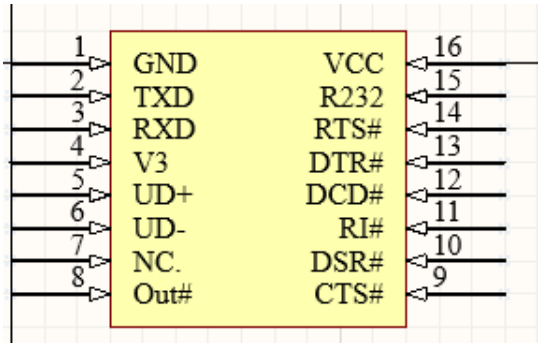
SSOP20 Pin No.	SOP16 Pin No.	ESSOP10 Pin No.	SOP8 Pin No.	Pin Name	Pin Type	Pin Description (description in bracket is only about CH340R)
19	16	7	5	VCC	POWER	Power supply voltage input, requires an external 0.1uF decoupling capacitor
8	1	3, 0	3	GND	POWER	Ground
5	4	10	8	V3	POWER	Connect to VCC when VCC is 3V3, connect to 0.1uF decoupling capacitor when VCC is 5V
9	7	NONE	NONE	XI	IN	CH340T/R/G: Input of crystal oscillator, connect to 12MHz crystal and capacitor
				NC.	NONE	CH340C: No Connection, must be suspended
				RST#	IN	CH340B: Input of external reset, active low, integrated pull-up resistor
10	8	NONE	NONE	XO	OUT	CH340T/R/G: Output of crystal oscillator, connect to crystal and capacitor
				OUT#	OUT	CH340C: MODEM output IO, software controlled, active low
				NC.	NONE	CH340B: No Connection, must be suspended
6	5	1	1	UD+	USB signal	Connect to USB D+ Signal directly
7	6	2	2	UD-	USB signal	Connect to USB D- Signal directly
20	NONE	NONE	NONE	NOS#	IN	Forbid USB device suspending, active low, integrated pull-up resistor
3	2	8	6	TXD	OUT	Transmit asynchronous data output(reverse output for CH340R)
4	3	9	7	RXD	IN	Receive asynchronous data input, integrated configurable pull-up and pull-down resistor
11	9	5	NONE	CTS#	IN	MODEM input signal, clear to send, active low(high)
12	10	NONE	NONE	DSR#	IN	MODEM input signal, data set ready, active low(high)
13	11	NONE	NONE	RI#	IN	MODEM input signal, ring indicator , active low(high)
14	12	NONE	NONE	DCD#	IN	MODEM input signal, data carrier detect, active low(high)
15	13	4	NONE	DTR#	OUT	MODEM output signal, data terminal ready, active low(high)
16	14	6	4	RTS#	OUT	MODEM output signal, request to send, active low(high)
2	NONE	NONE	NONE	ACT#	OUT	USB configuration completed state output, active low
18	15	NONE	NONE	R232	IN	CH340T/R/G/C: Assistant RS232 enable,



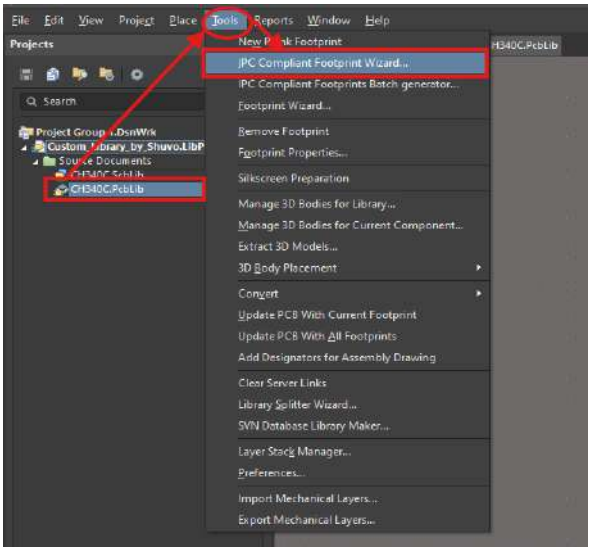
16. Click on Place > Place Symbol.

1. 

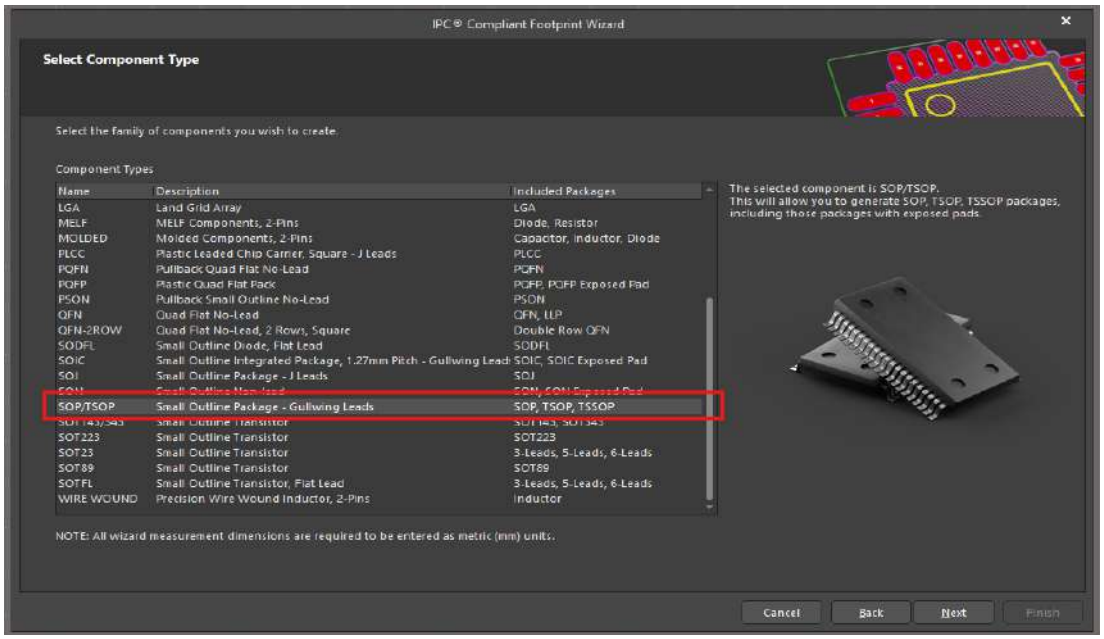
17. The final schematic will look like:



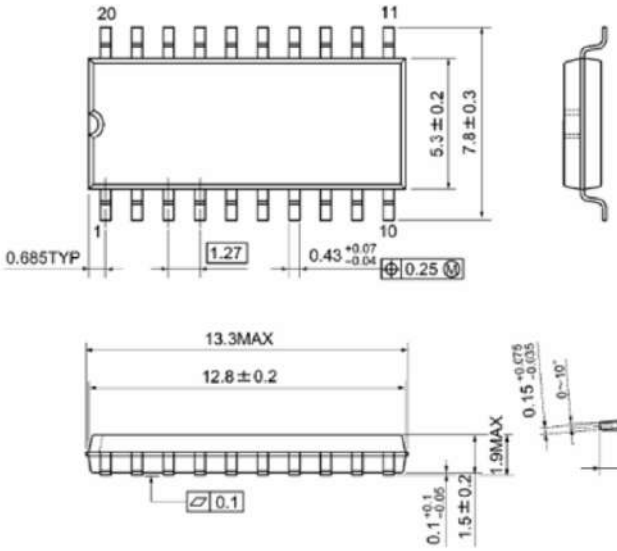
18. Go to Project > Right-click on the Schematic file > Save.
19. Go to Project > Select *PCB* file > Click on Tools > Select IPC Compliant Footprint Wizard (IPC is a global trade association that develops internationally recognized standards for the design, manufacture, assembly, inspection and testing of PCBs and electronic assemblies. These standards ensure consistency, quality, reliability and manufacturability across the electronics industry).

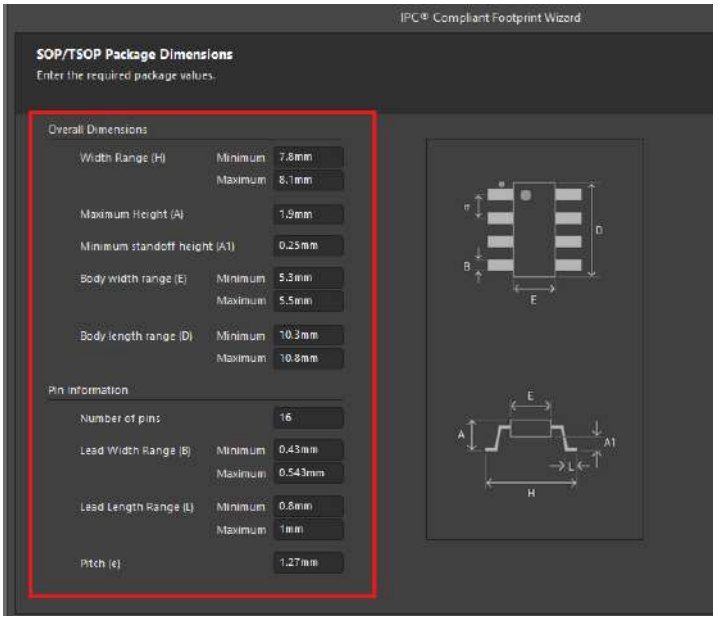


20. A new window will be opened > Click Next > Select SOP/TSOP as Component type > Next.

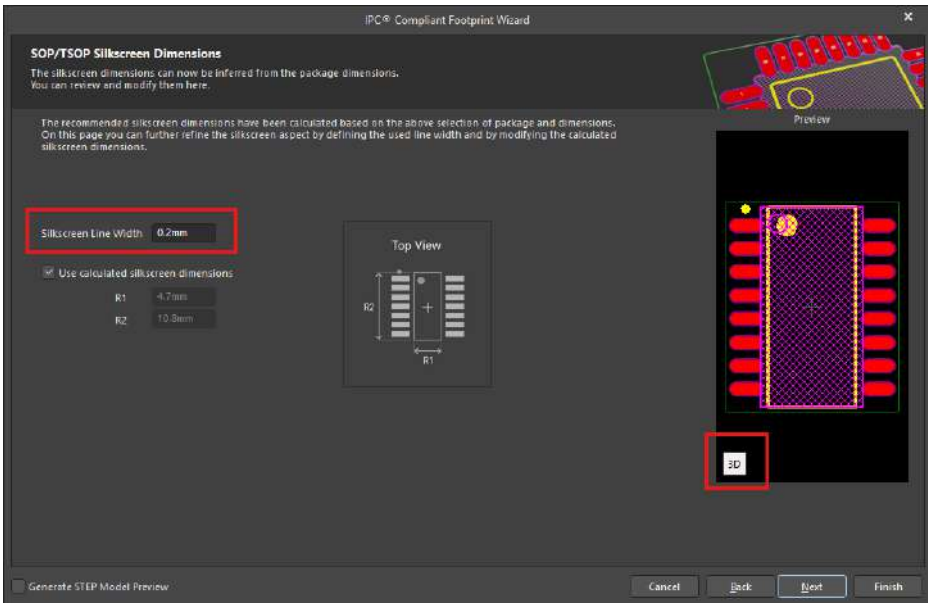


21. A new window will be opened. *Fill up* the dialogue boxes based on the mechanical measurement mentioned in the datasheet.

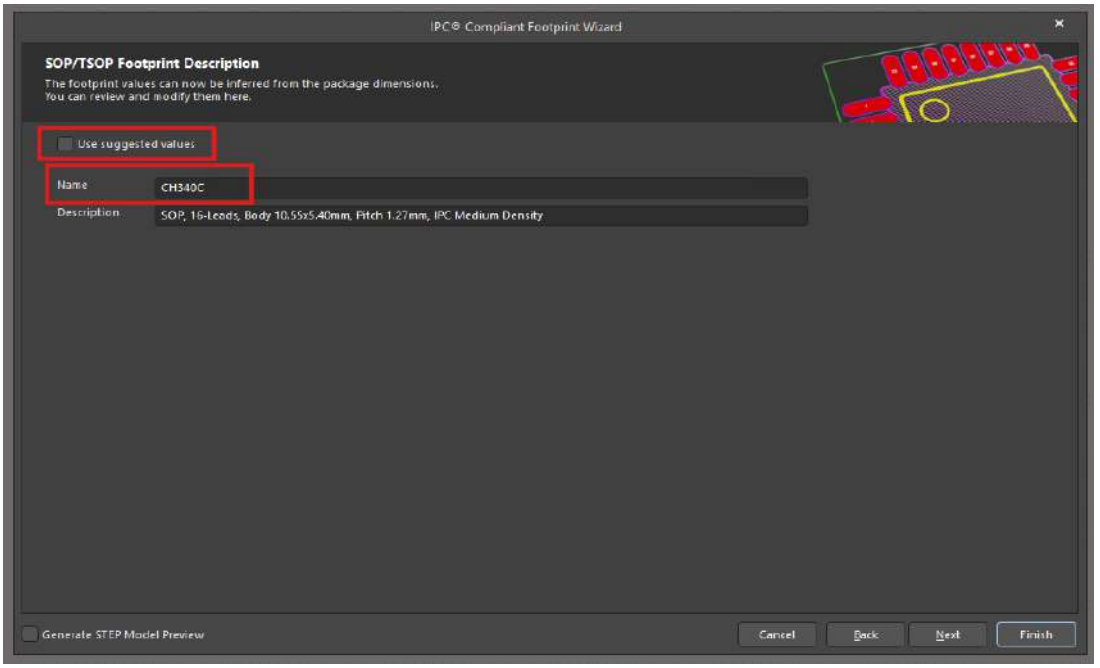




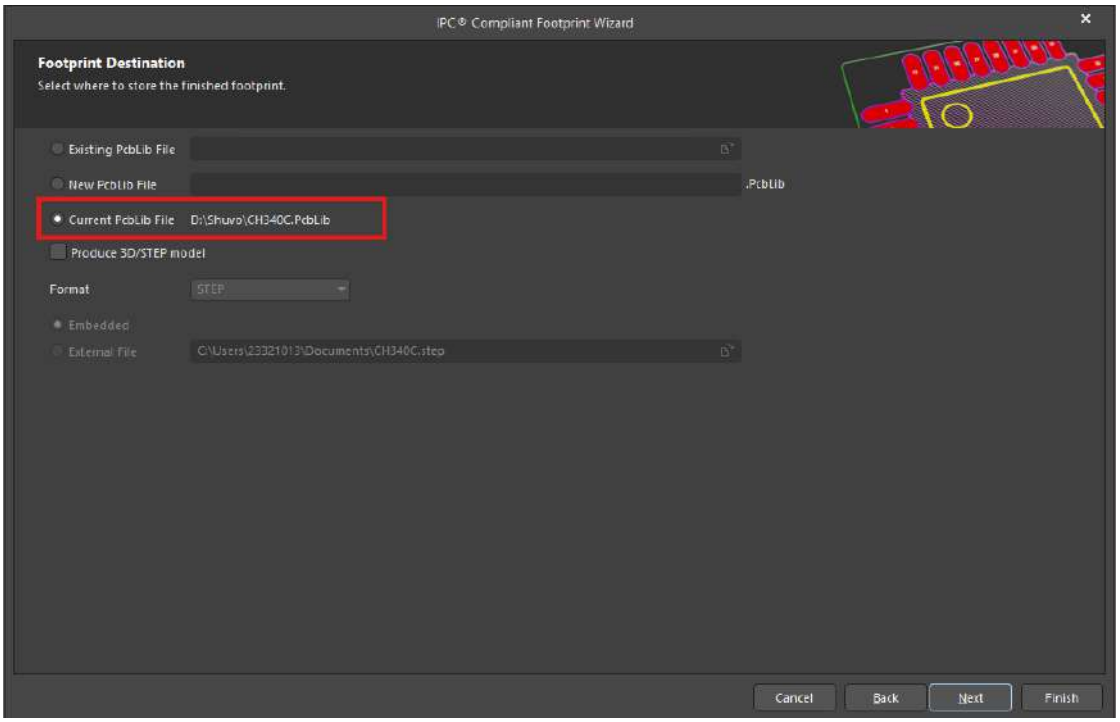
22. After filling up the dialogue boxes, it will look like the following image. Click *next*.
23. According to the datasheet, no thermal pad should be used. So click next until you reach the SOP/TSOP Silkscreen Dimension window.
24. In the SOP/TSOP Silkscreen Dimension window, set the *silkscreen line width* as 0.2 mm . You may view the *silkscreen line width* in the final PCB file by clicking the 3D button at the corner. Click next.



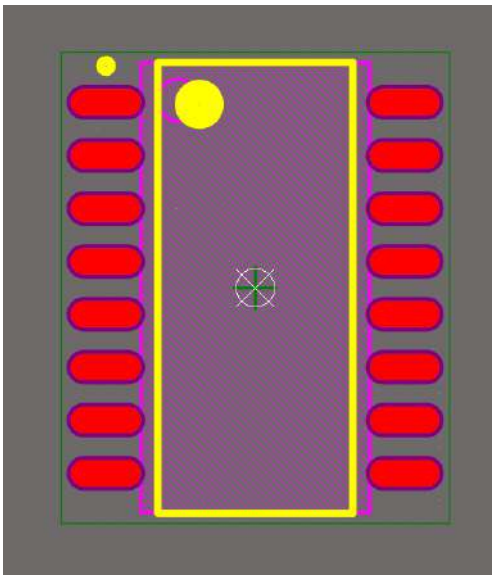
25. A new SOP/TSOP Footprint window will be opened. Uncheck the *use suggested values* option and rename the file as given before.

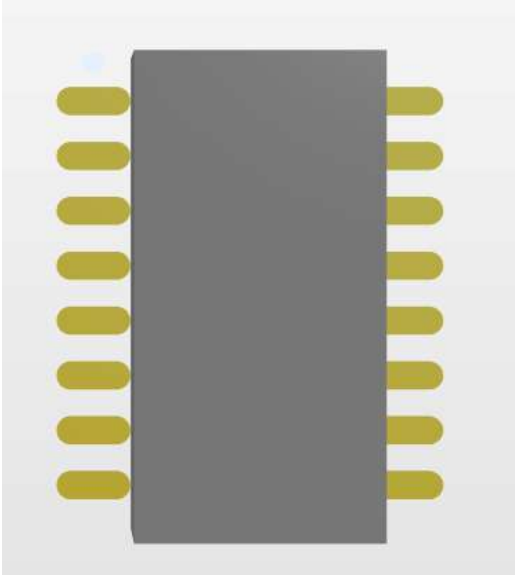


26. A Footprint destination window will be opened. Select *Current PCB File* > Click *next* > *Finish*.

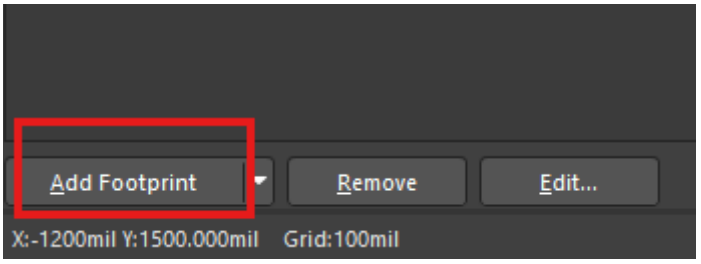


27. Save (ctrl + s) the PCB file. The final PCB footprint will be created.

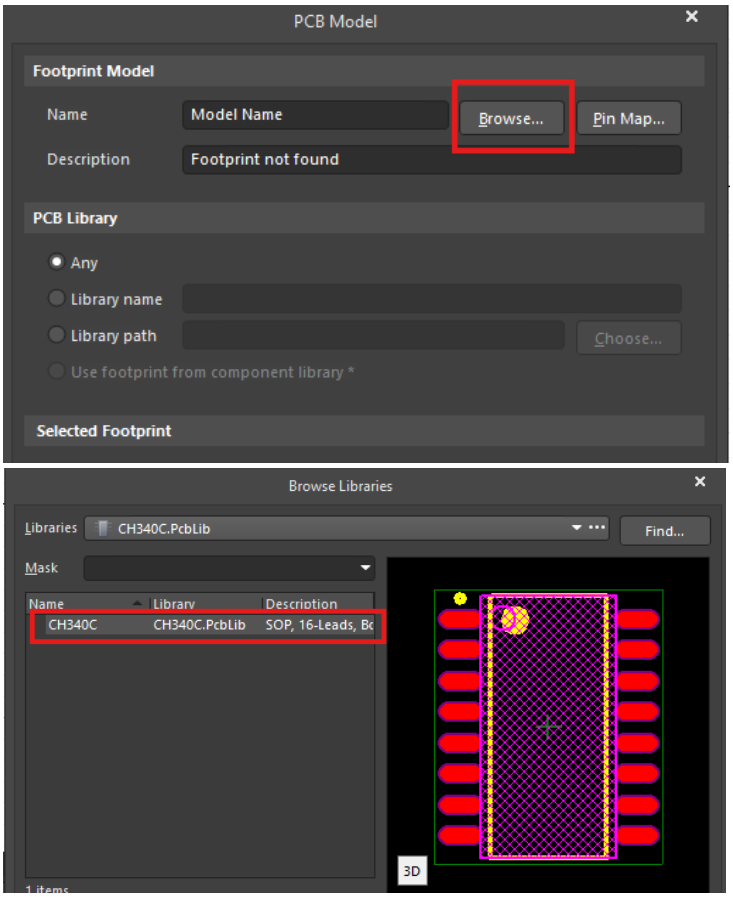




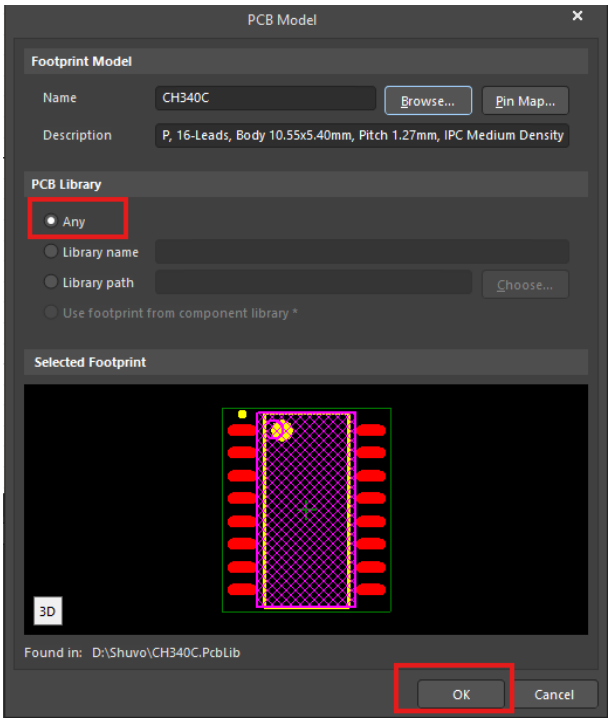
28. Go to *Schematic file* > Click *Add Footprint*.



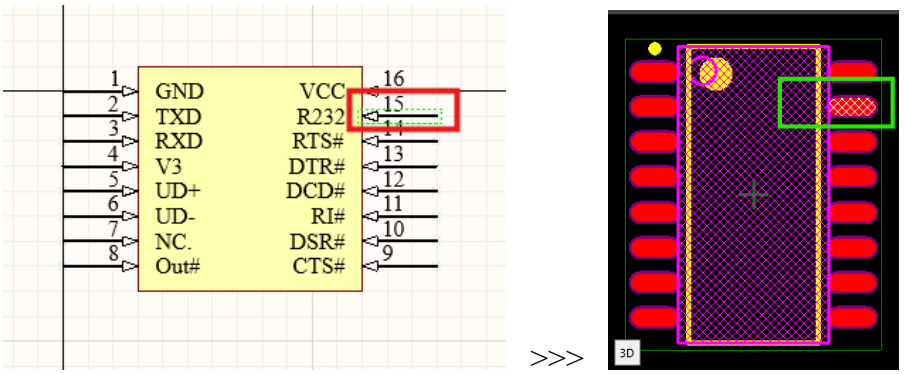
29. Click *Browse* > Select *CH340C PCB* file.



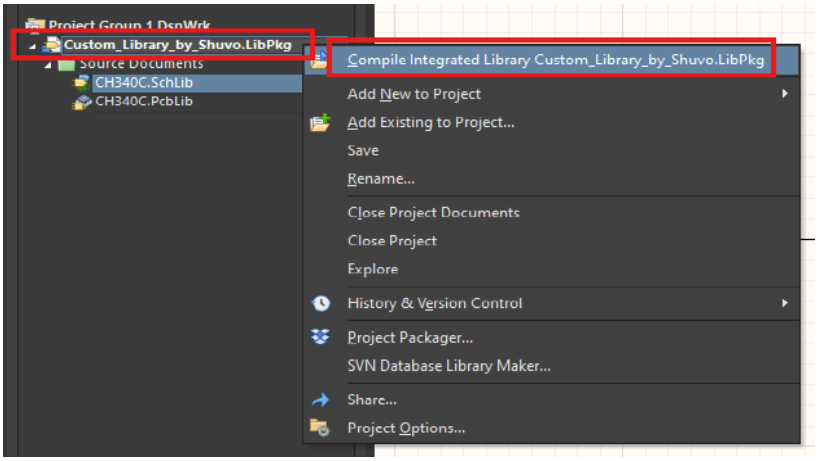
30. Select *any* as PCB Library > click ok.



31. Check that both the files are linked or not by clicking any pin in the schematic file (*red marked*). You will see that the dedicated pin will be shaded in the PCB file (*green marked*). Save (ctrl + s) the *Schematic* file.



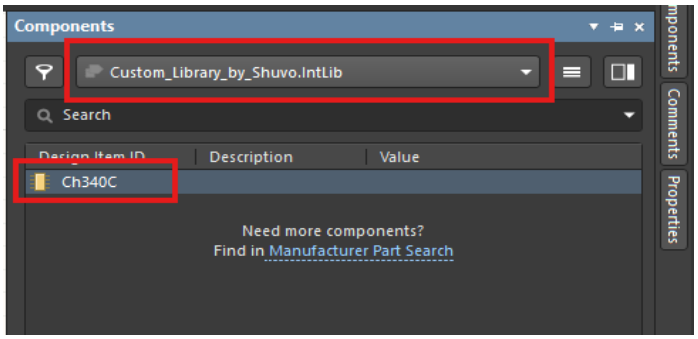
32. Go to *project* > select Compile Integrated Library Custom_Library_by_Shuvo.LibPkg



Your custom library file is ready to use.

33. Verification:

Go to Components > Select the library file named *Custom_Library_by_Shuvo* > You will get the new component named *Ch340C*.



BRAC University
Department of Electrical and Electronic Engineering
EEE383
Electronic System Design
Experiment-08: Soldering Basics

1. Objective:

By the end of this, students will be able to:

- Understand the basic principles and importance of soldering in electronic circuit assembly.
- Identify the tools, materials and safety precautions required for soldering.
- Develop practical skills in soldering electronic components onto a PCB.
- Produce reliable and high-quality solder joints using proper soldering techniques.
- Inspect solder joints and identify common soldering defects.

2. Theoretical Background:

Soldering is one of the most fundamental manufacturing and assembly processes in electronics. It is a joining process in which two or more metallic surfaces (typically electronic component leads and copper conductor pads) are permanently bonded together through the application of a low-melting-point filler metal known as solder. Unlike welding, soldering does not melt the base metals themselves; instead, the solder is heated above its melting point and allowed to flow into the interface between the surfaces where it wets the metal, forms an intermetallic bond upon cooling and creates a mechanically strong and electrically conductive joint. Soldering serves two simultaneous purposes: it provides a durable mechanical fixture that holds a component in place on a board or wire and it establishes a low-resistance electrical pathway between the component lead and the copper trace or pad.

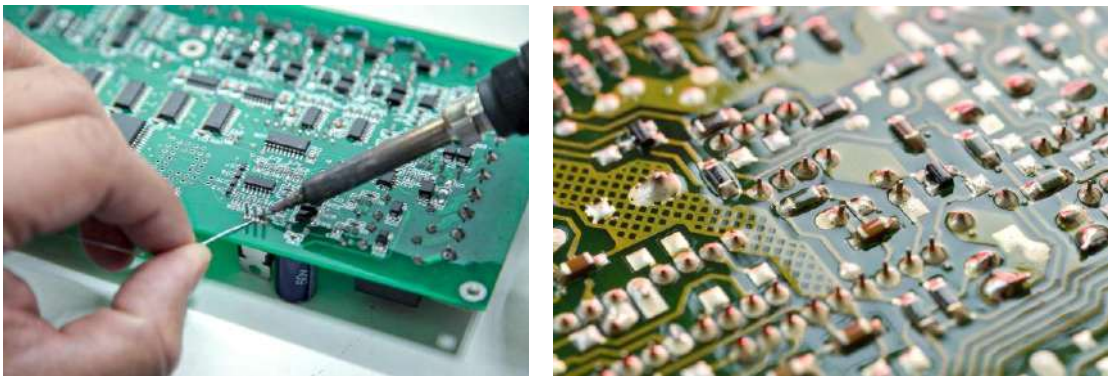


Fig: Soldering using an electric soldering iron and the final product, a PCB with solder joints

In modern electronic product development, soldering is essential for assembling prototypes, repairing faulty circuits, manufacturing commercial products and conducting laboratory experiments. Proper soldering techniques ensure reliable electrical conductivity, mechanical stability and long-term durability of electronic systems. Conversely, poor soldering practices may result in open circuits, short circuits, excessive electrical resistance, intermittent faults or complete system failure.

Soldering techniques can be broadly classified according to the operating temperature of the process and according to the method by which heat and solder are applied to the joint.

Soft Soldering: Soft soldering employs filler alloys with melting points typically below 450°C (most commonly in the range of 180 to 250°C). Tin-lead (Sn-Pb) and lead-free tin-copper-silver (Sn-Cu-Ag) alloys are the most widely used solders for this category. Soft soldering is the standard technique used in nearly all electronic assembly work including PCB soldering because the relatively low temperature minimises the risk of thermal damage to semiconductor devices, plastic connectors and laminate substrates.

Hard Soldering: Hard soldering includes silver soldering and brazing, uses filler alloys that melt above 450°C. This technique produces substantially stronger joints than soft soldering and is used primarily in mechanical, plumbing and jewelry applications rather than in electronic assembly, because the elevated temperatures would damage most electronic components.

Components required for soldering:

- **Soldering Iron /Soldering Station**: The soldering iron is the primary heat-delivery tool. A basic iron consists of a heating element enclosed in a metal barrel, terminating in a replaceable metal tip that conducts heat to the joint. For laboratory and professional use, a temperature-controlled soldering station is strongly preferred over a fixed-wattage iron as it allows the operator to select and maintain a precise tip temperature, typically between 300 and 370°C for lead-free solder, which improves joint consistency and reduces the risk of thermal damage to components.



- **Solder Wire:** Solder wire is the filler metal consumable, typically supplied as a spool of wire with a diameter between 0.5 and 1.0 millimetre for electronics work. Modern electronic solder is most commonly either a leaded alloy (Sn63/Pb37, made of 63% tin and 37% lead) which offers a low, well-defined melting point and excellent wetting characteristics. Most electronic solder wire is manufactured with a core of flux running through its centre, so that flux is deposited automatically as the wire is fed into the joint.

- **Flux:** Flux is a chemically active substance that removes oxide layers from the metal surfaces being joined and prevents further oxidation during the heating process, thereby allowing the molten solder to wet and adhere to the base metal properly. Flux may be supplied as the core of solder wire, as a separate liquid or paste applied with a brush or pen or as a paste combined with powdered solder for reflow processes. Without adequate flux activity, solder tends to bead up on an oxidized surface rather than flowing smoothly, resulting in a poor-quality joint.

- **Soldering Iron Stand and Sponge/Brass Wool:** A stand holds the hot iron securely when it is not in active use, preventing burns and accidental damage to the

workbench. Most stands incorporate a cleaning medium, either a moistened cellulose sponge or a coil of brass wire wool, against which the hot tip is wiped periodically to remove oxidized solder and residual flux which preserves good heat transfer and prolongs tip life.

- **Desoldering Tools:** A desoldering pump (Solder sucker) and desoldering braid, a fine copper mesh impregnated with flux, are used to remove unwanted or excess solder or when correcting a mistake or removing a defective component. These tools are considered standard equipment in any soldering kit intended for rework or repair.

- **Wire Cutters and Pliers:** Flush-cut wire cutters are used to trim excess component lead length after a joint has been soldered while fine-nosed pliers assist in bending leads to the correct shape and in holding small components in position during assembly.

3. PCB soldering:

A successful solder joint depends on wetting in which molten solder spreads evenly across a clean metal surface rather than forming a spherical bead. Wetting occurs when the molten solder reacts with the base metal to form a thin intermetallic compound layer at the interface which is the true source of the joint's mechanical and electrical continuity. Adequate surface cleanliness, sufficient temperature and active flux are the three prerequisites for good wetting; the absence of any one of these factors typically results in a poor joint.



Good Joint



Cold Joint



Too Much Solder



Insufficient Solder

Heat must be transferred efficiently from the iron's tip to both surfaces being joined, not solely to the solder wire itself. Melting solder directly on the iron's tip and then carrying the molten droplet to the joint produces a joint with poor wetting because the base metal itself was never brought to the correct temperature. Solder alloys transition from solid to fully liquid over a temperature range at which melting begins (solidus temperature) and at which the alloy is completely molten (liquidus temperature). Sn63/Pb37 have a single melting point at which the transition from solid to liquid is essentially instantaneous, producing joints that solidify quickly and with a bright, smooth finish. Once sufficient solder has been applied and the joint has reached the desired shape, the heat source is withdrawn and the joint must be allowed to cool and solidify without any movement of the component or the board.

In through-hole assembly, component leads are inserted through drilled holes in the PCB and are then soldered on the opposite side of the board. Through-hole joints are mechanically robust and relatively easy to inspect and rework by hand. In surface-mount assembly,

components are placed directly onto pads on the surface of the board without any drilled holes. Surface-mount components are considerably smaller than their through-hole counterparts and are soldered either by hand using a fine-tipped iron and considerable care or by automated reflow soldering in industrial settings. Hand soldering of surface-mount components requires a steadier hand, finer tools and greater attention to preventing solder bridges between closely spaced pins.

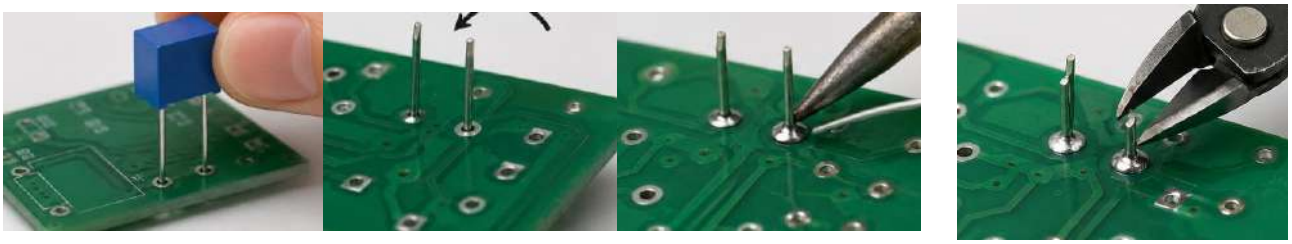


Figure: Through-hole PCB assembly Process

Defect	Typical Cause
Cold joint (dull, grainy surface)	Insufficient heat, or movement during solidification
Solder bridge	Excess solder between adjacent pads or pins
Dry joint / insufficient wetting	Inadequate flux, oxidised surface, or low temperature
Lifted pad	Excessive heat or mechanical force applied to the pad
Tombstoning (SMT)	Uneven heating of the two pads of a small SMT component

Table-01: Common PCB solder joint defects and their typical causes

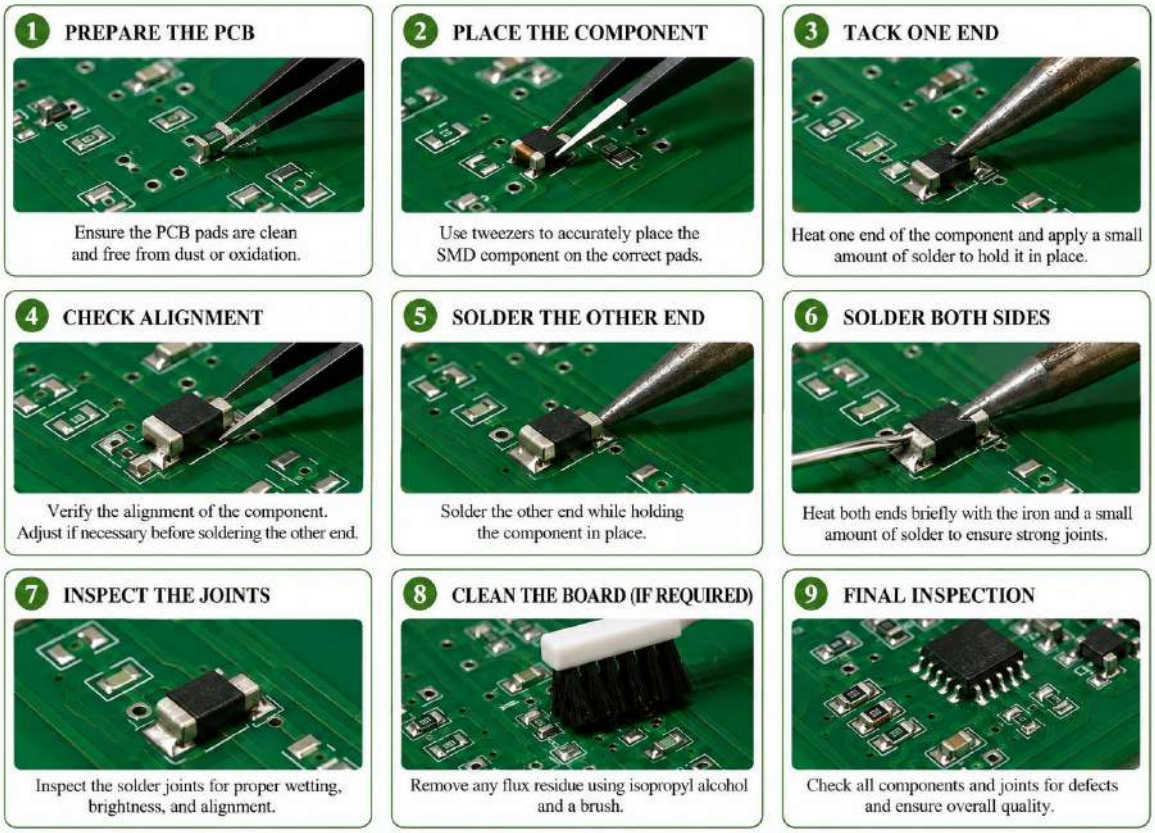


Figure: surface-mount PCB assembly Process

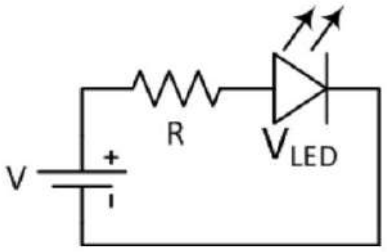
PCB Soldering Procedure:

- PCB layout is made on PCB design software (Altium). Once the PCB design is completed fully for production, then export the design into Gerber file format.
- After the completion of the design output, the second step is to paste the files onto the Printed Circuit boards. Manufacturers use a special printer known as a plotter that makes the photo films of the PCB to print the PC boards.
- When the PCB files are pasted fully on the PC boards, then ensure the work area is clean, well-lit and adequately ventilated, ideally beneath a fume extractor. Place the soldering iron in its stand and connect it to a suitable power supply. Moisten the cleaning sponge or verify that the brass-wool cleaner is in place. Set the temperature of the soldering station to the appropriate value for the solder being used, (320 to 350℃ for lead-free solder or 300 to 320℃ for leaded solder) and allow the tip to reach thermal equilibrium.
- Identify the correct orientation and placement of the component on the PCB using the silkscreen legend and the circuit diagram. Insert the component leads through the corresponding holes from the top (component) side of the board. Bend the leads slightly outward on the underside of the board to hold the component temporarily in place and secure the board in a vice or holding fixture.
- Wipe the hot tip across the damp sponge or brass wool to remove any oxidized solder residue. Apply a very thin, fresh layer of solder to the cleaned tip (tinning) which improves thermal contact for the subsequent step.

- Position the iron so that its tip makes simultaneous contact with both the component lead and the copper pad, forming a small thermal bridge between the two surfaces. Hold the tip in this position for approximately one to two seconds to allow both the lead and the pad to reach soldering temperature. Feed the solder wire into the joint from the side opposite the iron tip, touching the wire to the heated lead and pad rather than to the iron tip itself and allow the solder to melt and flow freely around the joint. Continue feeding solder until a smooth fillet has formed that fully covers the junction between the lead and the pad, then withdraw the solder wire. Remove the iron tip from the joint immediately after the solder wire is withdrawn, taking care not to move the component or the board. Allow the joint to cool and solidify undisturbed for several seconds without applying any mechanical stress.
- Inspect the joint visually and trim the excess lead protruding beyond the joint using flush-cut wire cutters, leaving a small, standard-length stub above the solder fillet. Repeat it for each remaining joint on the board. Once all joints are complete, clean the board of any flux residue if required and return the iron to its stand. Perform a continuity and visual inspection of the completed board using a multimeter before applying power to the circuit.

4. Experimentation:

1. Construct the circuit in the figure.



2. Properly solder the components according to the instructions given above.
3. Turn on the power supply and observe the LED glow.